

Assignment 1 – FIT3152 Data Analytics

Student ID	33986010
Focus Country	South Korea (KOR)

Question 1 Descriptive Analysis

Dataset Overview

The dataset contains 100,000 rows (observations) and 36 columns (variables), including 35 numeric variables and 1 character variable (Country). Although many variables are numerically encoded, 32 represent categorical responses, many of which are ordinal and are treated as numerical for analysis. *(Detailed table of data types variables is provided in Appendix 1-Q1a)*

Variable Classification

- **Identifier Variables:** Wave, Country, and Year [contextual information]
- **Outcome Variables:** Variables prefixed with C* (e.g. CArmedForces) [respondents' confidence / target outcome variables]
- **Predictor Variables:** Demographics, attitudes, and behavioural (e.g. FutureRespect)

Data Cleaning

Handling Invalid Responses

Negative survey responses were recoded to missing (NA) since they represent non-substantive responses (-1 = Don't know, -2 = No answer/refused, -3 = Not applicable, and -5 = Missing / not applicable for other reasons).

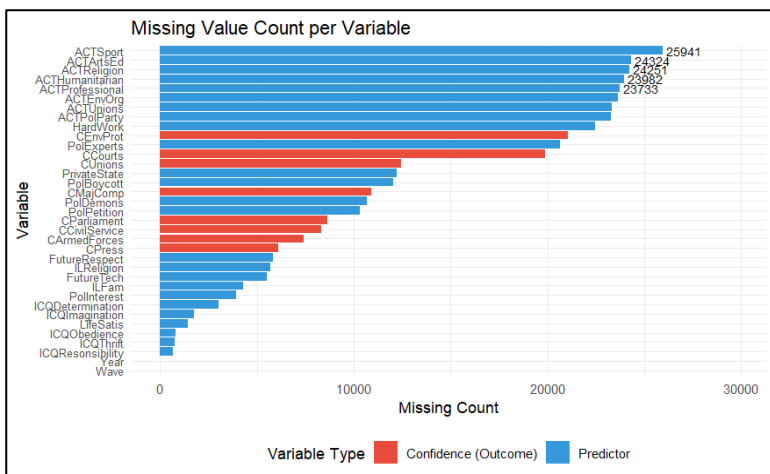
Variable Group	Valid Range	Is 0 = NA?	Notes
Confidence Variables	1–4	Yes	-
Sex	1 = Male, 2 = Female	Yes	Binary categorical
Political Action (Pol*)	1–3 / 1–4 / 1–10	Yes	
Political System Variables (PolLeader etc.)	1–4	Yes	
Life Satisfaction	1–10	Yes	-
Demographics (e.g., Age)	Numeric	No	-
Future Change (Future*)	1–3	Yes	-
Membership (ACT*)	0–2	No	0 = Don't belong
Important Child Qualities (IQR*)	1–2	Yes	Binary selection
Important in Life (IL*)	1–4	Yes	-
Happiness	1–4	Yes	-
Health / Trusted	1–5	Yes	-
PrivateState / GovtRespons / Competition / HardWork	1–10	Yes	-

Handling Zero Values

Zero values were also recoded to NA for numeric variables where 0 was assumed invalid, except for:

- **ACT variables:** retained as valid (0 = Don't belong, 1 = Inactive member, 2 = Active member).
- **ICQ variables:** retained as valid although the survey documentation specifies 1=Mentioned, 2 = Not Mentioned, the dataset uses 1 = Mentioned and 0 = Not Mentioned.

Missing Data



After cleaning, missingness varies:

ACT variables: highest (~24–26%)

Confidence variables: moderate

(6-21%), with CEnvProt and

CCourts having the highest

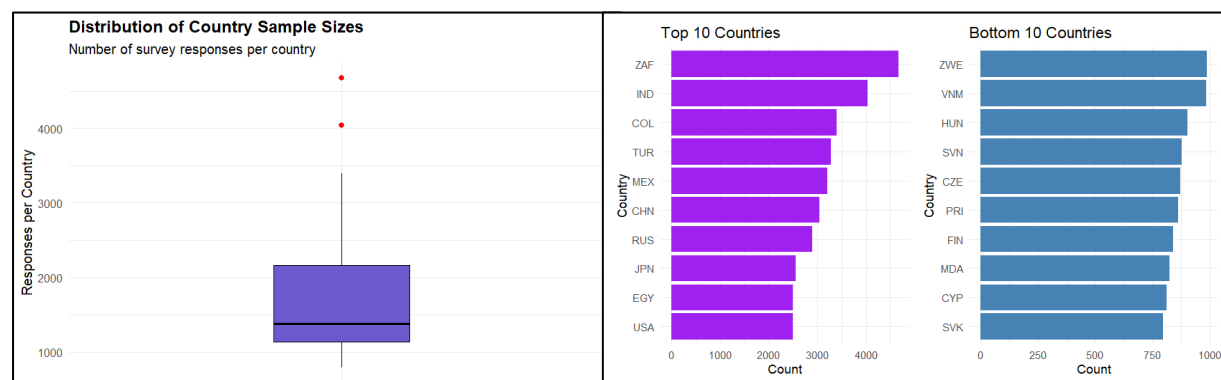
missingness.

(A detailed summary table of the

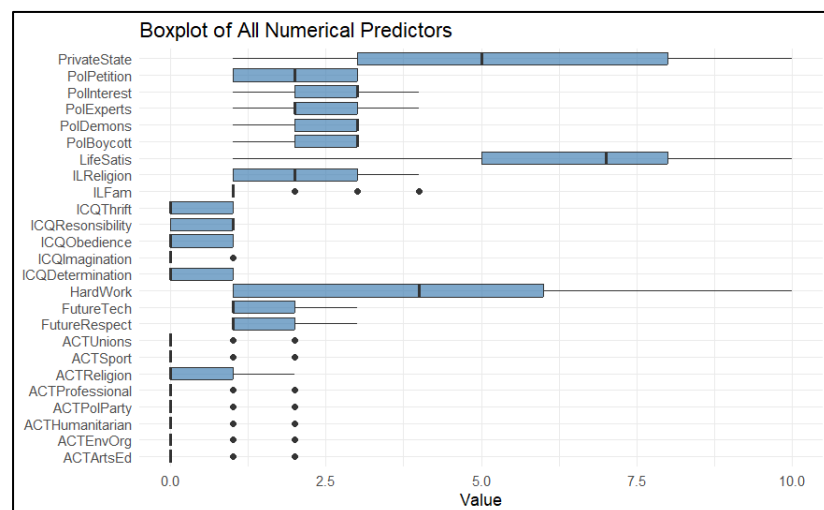
numerical variables is provided in

Appendix 1 -Q1a)

Distributional Analysis



The dataset covers 58 countries. Most countries have around 1,000-2,000 responses (median $\approx$ 1,300), showing moderate variability. A few countries (e.g., ZAF, IND, COL) have much larger samples (3,000-4,500), while others (e.g., ZWE, VNM, HUN) have smaller samples (700-1,000).



The boxplot shows

differences in scale and

variability among predictors.

Variables like Life Satisfaction

(LifeSatis) and PrivateState

have wide spreads, showing

more variation, while ICO and

ACT are closely grouped due

to limited ranges. Most

variables are low skewed.

variables are low skewed.

Question 2: Focus Country vs Other Countries (Independent of Time)

Question 2a

The dataset includes 2,339 respondents from South Korea and 97,661 respondents from all other countries combined.

Screening: Multicollinearity Check

All predictors were retained because the Pearson correlation matrix showed no strong correlations ($|r| \geq 0.70$), indicating no multicollinearity. Cohen's d was used for effect size, and Welch's t-test was used due to unequal variances and sample size

Hypothesis Testing

A significance level of $\alpha = 0.05$ was applied.

- **Null hypothesis (H_0):** The mean response for South Korea is equal to that of other countries

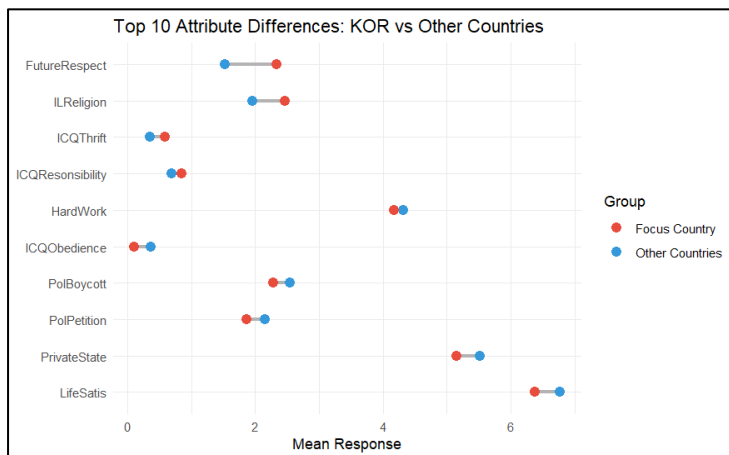
$$H_0: \mu_{KOR} = \mu_{Other}$$

- **Alternative hypothesis (H_1):** The mean response for South Korea is different from that of other countries

$$H_1: \mu_{KOR} \neq \mu_{Other}$$

Welch's t-tests show most variables are significant ($p < 0.05$), leading to rejection of the null hypothesis and showing that South Korea differs from other countries.

Comparison of Means



Dumbbell plot highlights the largest differences between South Korea and other countries, while Table reports the corresponding mean differences and significance levels. South Korea scores higher on

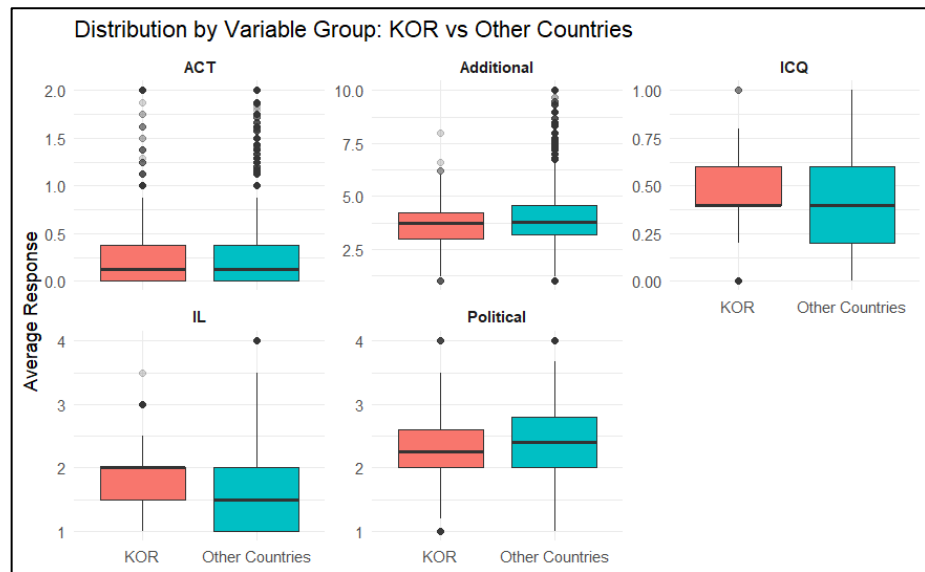
FutureRespect and **ILReligion**, denoting a greater emphasis on authority and religion. In contrast, it scores lower on **LifeSatis** and political participation (**PolPetition**,

PolBoycott), showing lower life satisfaction and civic engagement.

There are a few differences in **HardWork** and **ICQResponsibility**, suggesting that South Korea and other nations respond similarly. Overall, these patterns are consistent with cultural forces that emphasise hierarchy and established social standards.

Variable	Mean_Diff	P_Value	Direction
FutureRespect	0.812	<0.001	Focus Higher
ILReligion	0.511	<0.001	Focus Higher
LifeSatis	-0.401	<0.001	Focus Lower
PrivateState	-0.362	<0.001	Focus Lower
PolPetition	-0.282	<0.001	Focus Lower
PolBoycott	-0.259	<0.001	Focus Lower
ICQObedience	-0.259	<0.001	Focus Lower
ICQThrift	0.242	<0.001	Focus Higher
ICQResponsibility	0.159	<0.001	Focus Higher
HardWork	-0.14	0.022	Focus Lower

While political involvement and ACT variables remain low across both groups, with the majority of responses clustered near zero, South Korea often shows higher values on IL and ICQ variables. Overall differences are minimal at the group level.



Question 2b

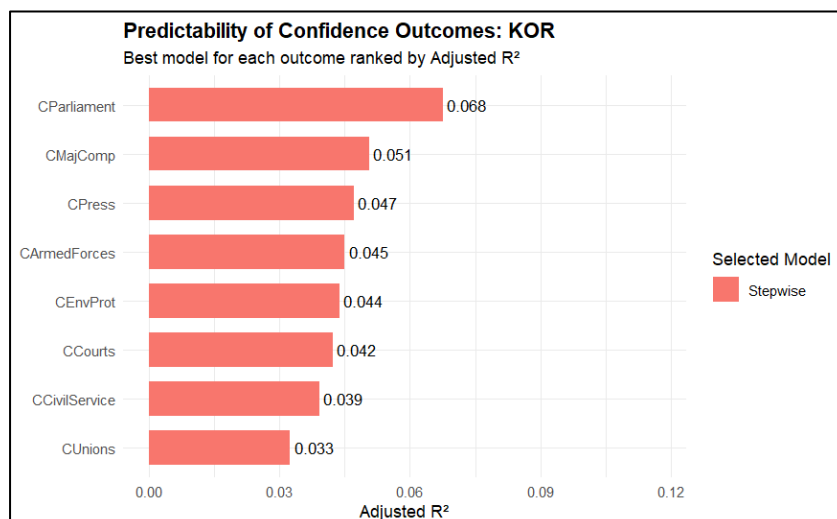
Stepwise and forward regression were used for variable selection to identify the most relevant predictors of institutional confidence while reducing model complexity. These methods iteratively add or remove variables based on statistical significance and model fit (adjusted R^2), allowing efficient identification of key predictors. However, results are interpreted with caution, as stepwise methods may lead to overfitting and instability in variable selection, particularly in large datasets with many predictors.

Hypothesis

A significance level of $\alpha = 0.05$ was applied

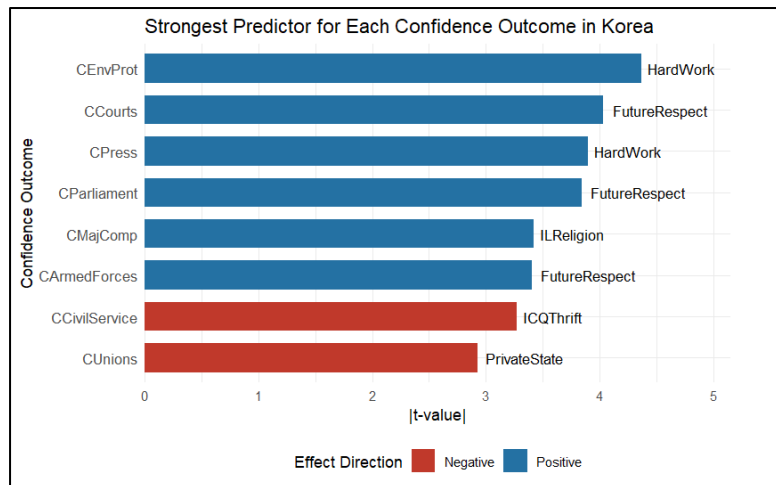
- **Null hypothesis (H_0):**
Predictor variables have no significant effect on confidence in social organisations (all coefficients $\beta = 0$).
- **Alternative hypothesis (H_1):**
At least one predictor variable has a significant effect on confidence in social organisations (at least one $\beta \neq 0$).

Overall Model Performance



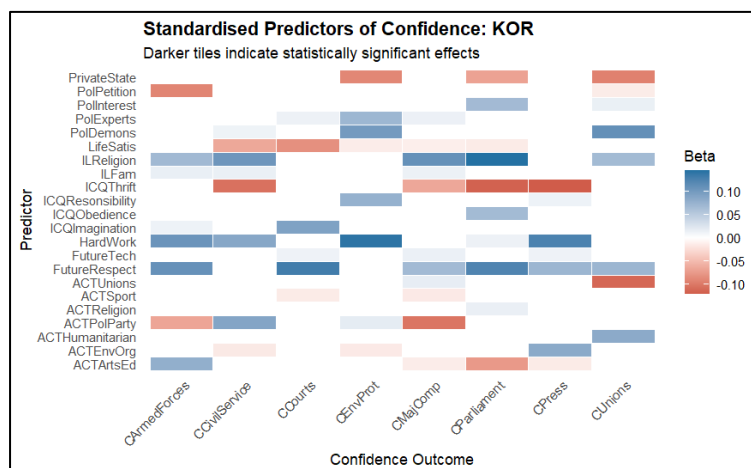
All confidence outcomes are statistically significant ($p < 0.05$) but show limited explanatory power. The most reliably predicted outcome is confidence in **Parliament** (Adj $R^2 \approx 0.068$), followed by major companies, the press, and the armed forces, while the civil service and unions are the least predictable. Overall, the models explain

less than 7% of the variance, showing weak predictive ability. (Detailed statistics of the model results for confidence outcomes in the focus country are provided in Appendix 2- Q2b)

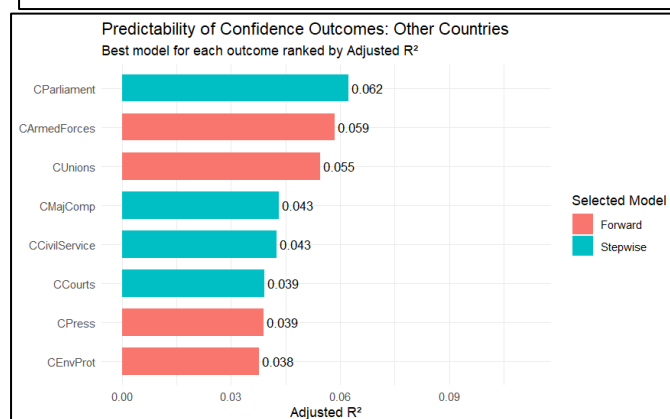


The t-value identifies the strongest predictor by showing both effect size and reliability. Most effects are positive, with **FutureRespect** predicting higher confidence in courts, parliament, and the armed forces, and **HardWork** predicting higher confidence in environmental protection and the press. **ILReligion** predicts major companies, while

ICQThrift and **PrivateState** have negative effects on civil service and unions. Overall, cultural values are key drivers of institutional confidence. (Detailed statistics, including t-values and coefficients for the strongest predictors in the focus country, are provided in Appendix 2-Q2b)



The heatmap shows standardised effects; significant associations are shown by darker colours. Most effects are weak, and few predictors are significant, implying confidence in institutions is not strongly explained by the variables. (Detailed statistics of Standardised coefficients from best models for other countries are provided in Appendix 2- Q2b)

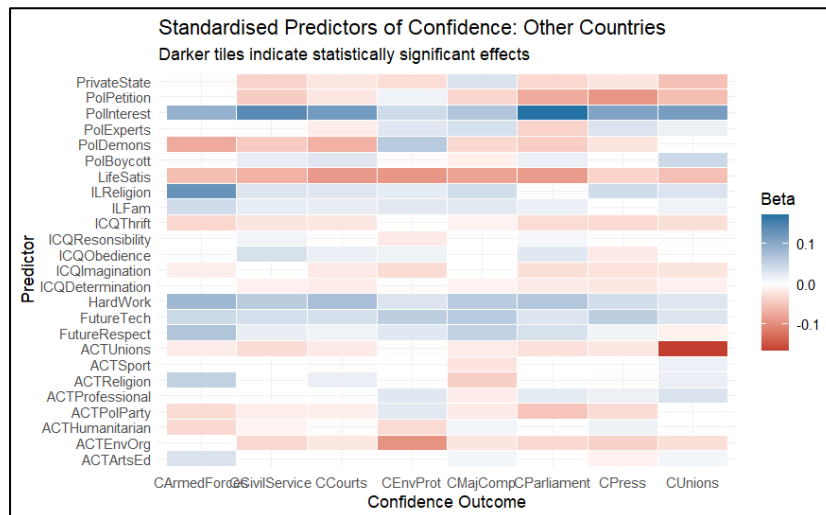


Question 2c

This section compares the findings with South Korea and evaluates how participant attributes predict confidence in social organisations in other countries. Overall, predictive results are weak, with all adjusted R² values below 0.07.

Confidence in **parliament** (Adj R² ≈ 0.062) and **armed forces** (Adj R² ≈ 0.059) are the most predictable outcomes, while courts,

press, and **environmental protection** show lower predictability (Adj R² ≈ 0.038 - 0.039). This suggests that confidence in institutions is only partially explained by the available predictors.



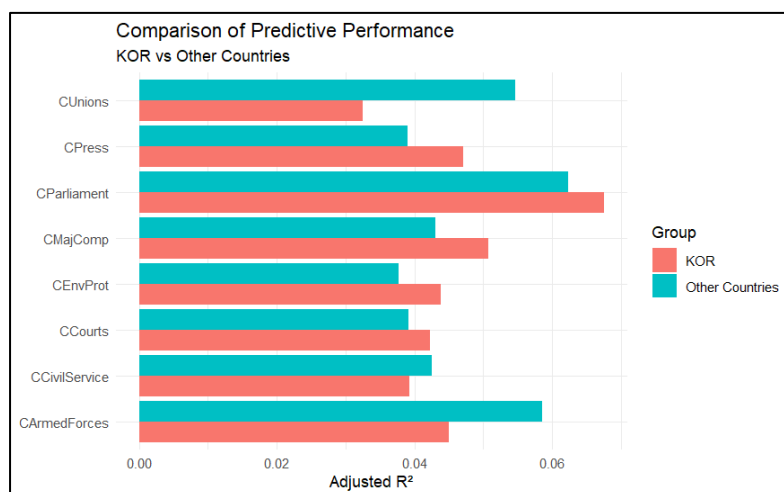
Predictors vary across outcomes, but steady patterns emerge. PolInterest and HardWork show positive effects on confidence, while variables such as LifeSatis and ACTUnions show negative effects, suggesting that higher life satisfaction and union participation are associated with lower confidence in some institutions.

(Detailed statistics of Standardised coefficients from best models for other countries are provided in Appendix 2- Q2c)

Outcome	Focus_Predictor	Other_Predictor
CArmedForces	FutureRespect	ILReligion
CCivilService	ICQThrft	PolInterest
CCourts	FutureRespect	PolInterest
CEnvProt	HardWork	LifeSatis
CMajComp	ILReligion	LifeSatis
CParliament	FutureRespect	PolInterest
CPress	HardWork	PolInterest
CUnions	PrivateState	ACTUnions

Table: Strongest Predictors for Each Outcome Variable (Focus Country vs Other Countries)

(Detailed statistics, including t-values and coefficients for the strongest predictors in other countries, are provided in Appendix 2- Q2c)



Compared to South Korea, other countries show slightly higher predictability for institutions such as armed forces and civil service, while South Korea shows greater predictability for parliament and major companies. However, both groups have generally low explanatory power, showing that institutional confidence is weakly predicted. This implies

that other nations display variability across institutions, whereas South Korea has more consistent predictive patterns.

Question 3: Focus Country vs Other Countries Over Time

Question 3a

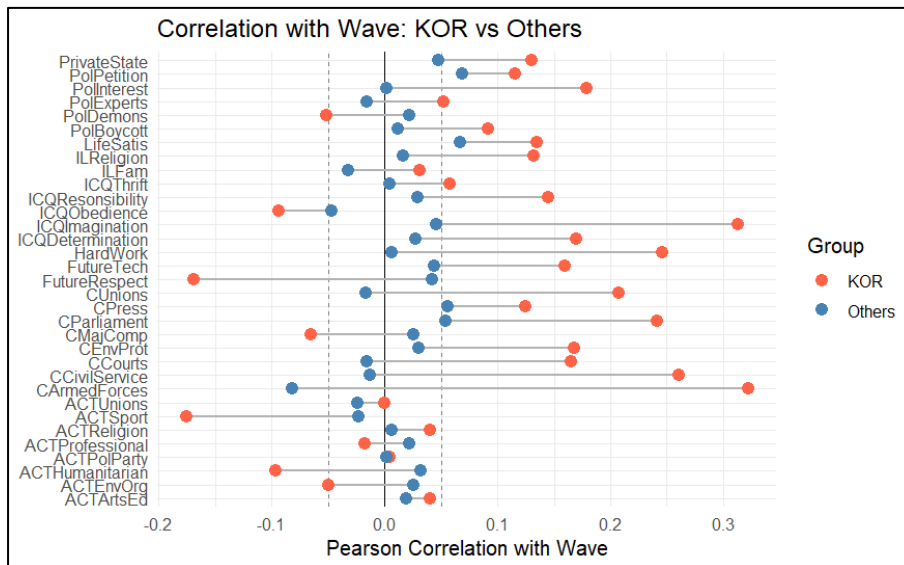
Box plots indicate variation across waves, with Korea showing greater changes over time. ANOVA was used to test differences in mean responses across waves, as it allows comparison across more than two groups.

An interaction regression model including Wave, Group, and their interaction (Wave $\times$ Group) was used to assess whether trends over time differ between South Korea and other countries. The interaction term tests whether the effect of time varies between groups. Significant interaction effects ($p < 0.05$) indicate that trends differ between South Korea and other countries. Results show that South Korea exhibits more dynamic changes over time, while other countries remain relatively stable.

Korea shows stronger increases in **HardWork**, **CArmedForces**, and **CCivilService**, but declines in **FutureRespect** and **ACTSport**, indicating more dynamic changes over time compared to the global pattern. *(More detailed table of data types variables is provided in Appendix 3-Q3a)*

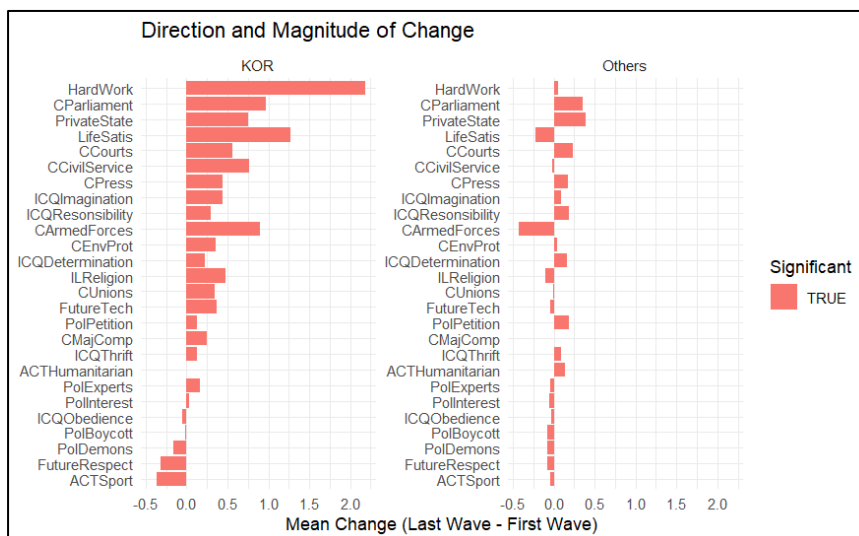
Table: Interaction Regression Results (Wave $\times$ Group Effects on Temporal Trends)

Variable	Interaction_Effect	Interaction_P	Interaction_Significant	Trend_Comparison
HardWork	0.323503	1.25E-17	Yes	KOR changes more positively over time
CArmedForces	0.177989	2.53E-71	Yes	KOR changes more positively over time
CCivilService	0.105582	4.22E-29	Yes	KOR changes more positively over time
PrivateState	0.095515	0.010073	Yes	KOR changes more positively over time
CUnions	0.086453	3.38E-19	Yes	KOR changes more positively over time
FutureRespect	-0.08413	2.38E-27	Yes	KOR changes more negatively over time
CParliament	0.072616	4.00E-13	Yes	KOR changes more positively over time
ACTSport	-0.07111	2.37E-08	Yes	KOR changes more negatively over time
PollInterest	0.070883	1.18E-11	Yes	KOR changes more positively over time
CCourts	0.067188	1.39E-11	Yes	KOR changes more positively over time

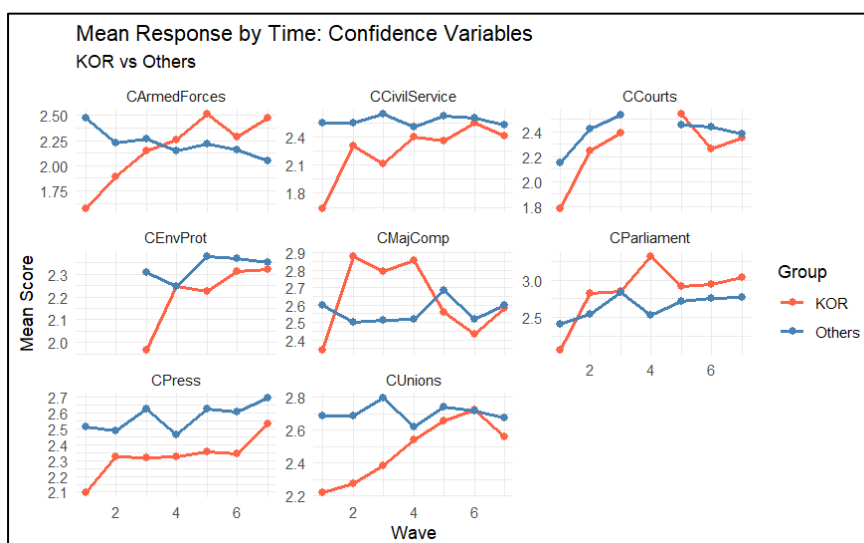


South Korea shows stronger correlations with time, indicating clearer trends, while other countries remain near zero. Confidence in institutions and FutureTech increases in South Korea, whereas FutureRespect and some ACT variables decline. Overall,

changes are more pronounced in South Korea and minimal elsewhere.

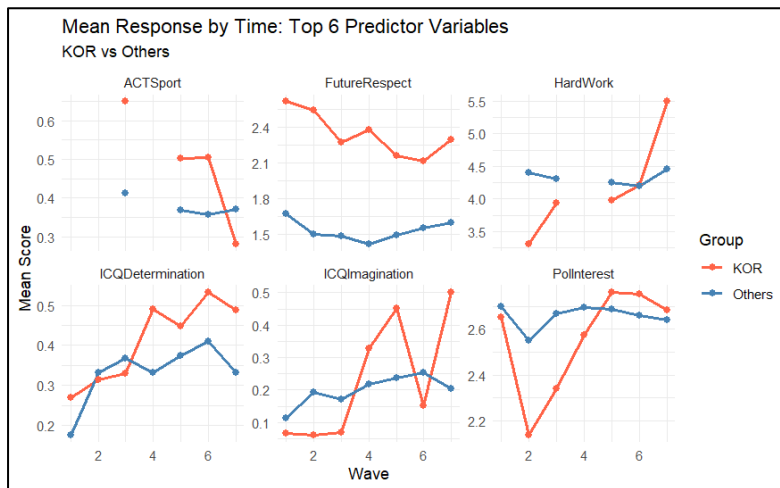


South Korea shows larger changes between the first and last wave, with **increases in HardWork, LifeSatis, and institutional confidence.** Some variables, such as **FutureRespect and ACTSport, decline.** In contrast, other countries show minimal change, showing more stable attitudes over time.

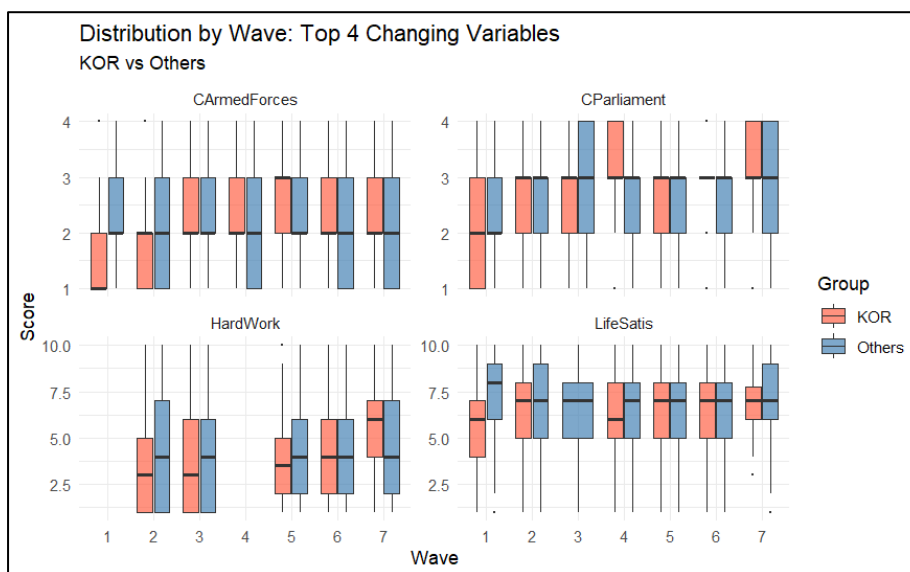


South Korea shows greater fluctuation and overall upward trends in institutional confidence, especially among the armed forces, the civil service, the courts, and parliament. Some variables peak and then slightly decline, indicating variability. Other countries remain relatively stable with minor changes. South

Korea starts lower but gradually converges toward other countries.



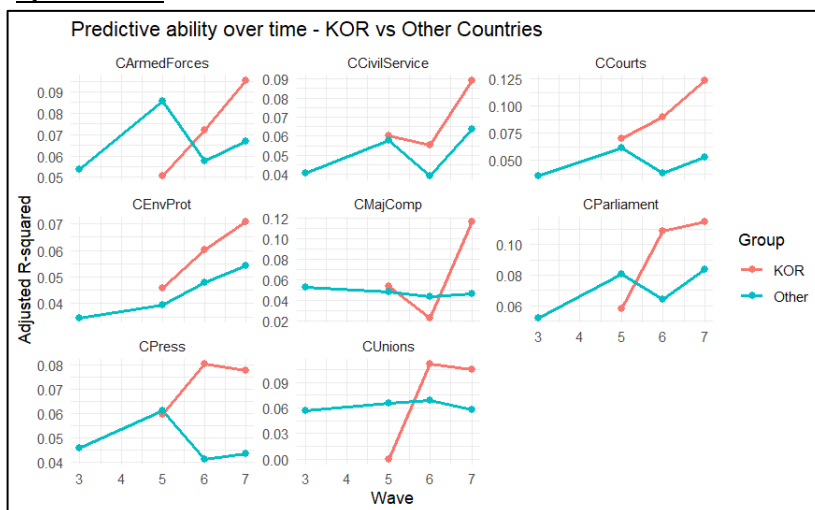
South Korea shows stronger and more variable trends over time, while other countries remain relatively stable. HardWork, ICQDetermination, and ICQImagination increase, while FutureRespect and ACTSport decline. PolInterest rises and converges with other countries. Overall, changes are more dynamic in South Korea, with minimal shifts elsewhere.



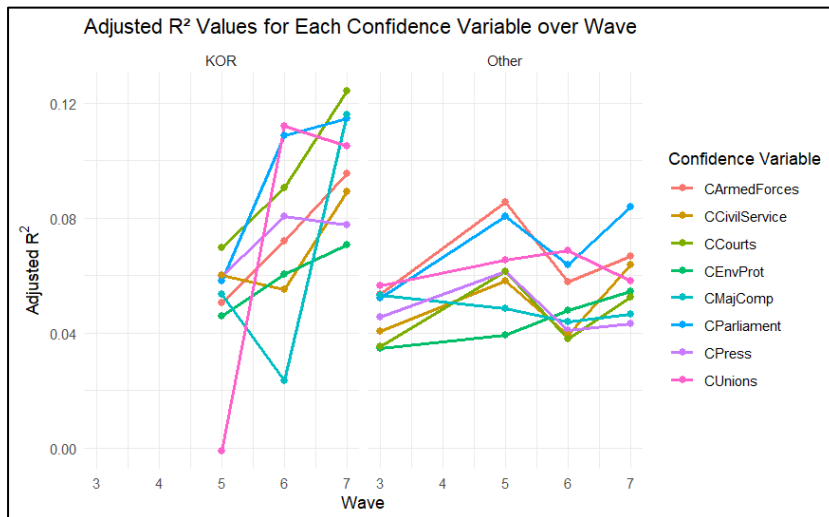
South Korea shows clearer upward shifts over time, especially for HardWork and LifeSatis, with increasing medians. CArmedForces and CParliament also rise gradually. In contrast, other countries remain stable. South Korea starts lower but converges toward, or even

exceeds, other countries, with slightly reduced variability, indicating stronger, more dynamic changes.

Question 3b

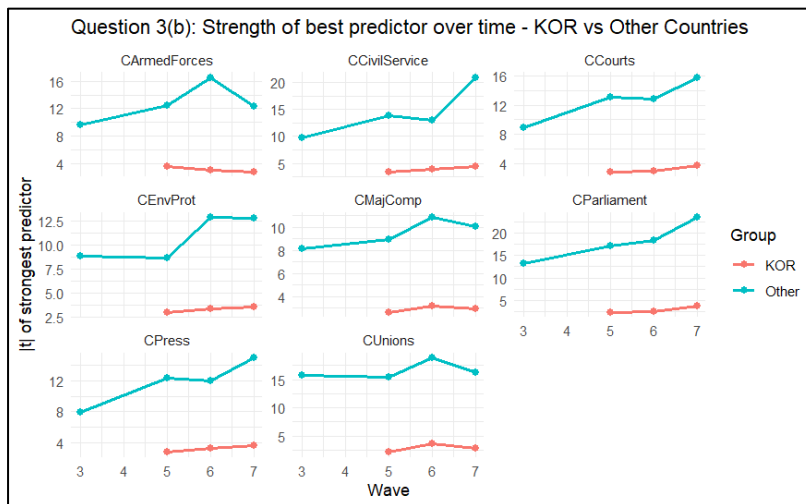


Adjusted R^2 increases over time in South Korea, especially for courts, parliament, and major companies, showing increased predictive ability. Other countries remain relatively stable with lower values. Although South Korea eventually surpasses others in some outcomes, the adjusted R^2 remains low overall, indicating limited explanatory power.



Adjusted R² increases over time in South Korea, especially for courts, parliament, and major companies, revealing improved model effectiveness. In contrast, other countries show lower and more stable values with only minor changes. Despite this, the adjusted R² remains low for both groups, showing limited overall

explanatory power.



The plot shows the strength of the best predictor over time using |t|-values. Other countries have consistently higher and increasing t-values, indicating stronger and more stable relationships. South Korea shows lower t-values that are only slightly increasing, suggesting weaker predictor effects despite some improvement in model fit.

Appendix

33986010 A1 FIT3152

Libraries

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
library(ggplot2)
library(corrplot)
```

Warning: package 'corrplot' was built under R version 4.5.3

corrplot 0.95 loaded

```
library(MASS)
```

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

select

```
library(knitr)
library(stringr)
library(purrr)
```

Dataset

```
set.seed(33986010) # Your Student Number
VCData = read.csv("WVSEExtract.csv")
VC = VCData[sample(1:nrow(VCData),100000, replace=FALSE),]
VC = VC[,c(1:3,sort(sample(4:50,25,replace=FALSE)),
sort(sample(51:65,8,replace=FALSE)))]
#write.csv(VC, "FIT3152A1Data_NgLiXian.csv", row.names = FALSE)
```

Appendix 1

Q1a

```
overview <- data.frame(
  Metric = c("Rows", "Columns", "Missing (%)", "Numeric Vars", "Character Vars"),
  Value = c(
    nrow(VC),
    ncol(VC),
    round(mean(is.na(VC)) * 100, 2),
    sum(sapply(VC, is.numeric)),
    sum(sapply(VC, is.character))
  )
)

knitr::kable(overview)
```

Metric	Value
Rows	100000
Columns	36
Missing (%)	0
Numeric Vars	35
Character Vars	1

```
# TABLE: DATA TYPES
data_dict <- data.frame(
  Variable = names(VC),
  Type = sapply(VC, class),
  Example = sapply(VC, function(x) head(x, 1))
)

knitr::kable(data_dict)
```

	Variable	Type	Example
Wave	Wave	integer	7
Country	Country	character	NLD
Year	Year	integer	2022
ILFam	ILFam	integer	2
ILReligion	ILReligion	integer	3
ICQResonsibility	ICQResonsibility	integer	1
ICQImagination	ICQImagination	integer	1
ICQThrift	ICQThrift	integer	0
ICQDetermination	ICQDetermination	integer	0
ICQObedience	ICQObedience	integer	0
ACTReligion	ACTReligion	integer	0
ACTSport	ACTSport	integer	0
ACTArtsEd	ACTArtsEd	integer	0
ACTUnions	ACTUnions	integer	0
ACTPolParty	ACTPolParty	integer	0
ACTEnvOrg	ACTEnvOrg	integer	0
ACTProfessional	ACTProfessional	integer	0
ACTHumanitarian	ACTHumanitarian	integer	0
LifeSatis	LifeSatis	integer	7
FutureTech	FutureTech	integer	1
FutureRespect	FutureRespect	integer	1
PolInterest	PolInterest	integer	2
PolPetition	PolPetition	integer	2

	Variable	Type	Example
PolBoycott	PolBoycott	integer	2
PolDemons	PolDemons	integer	1
PrivateState	PrivateState	integer	7
HardWork	HardWork	integer	4
PolExperts	PolExperts	integer	2
CArmedForces	CArmedForces	integer	3
CPress	CPress	integer	3
CUnions	CUnions	integer	3
CParliament	CParliament	integer	3
CCivilService	CCivilService	integer	2
CMajComp	CMajComp	integer	3
CEnvProt	CEnvProt	integer	2
CCourts	CCourts	integer	1

```
write.csv(data_dict, "q1a_data_types.csv", row.names = FALSE)
```

```
# IDENTIFIER VARIABLES
identifier_vars <- c("Wave", "Country", "Year")

# OUTCOME VARIABLES
outcome_vars <- setdiff(grep("^C", names(VC), value = TRUE), "Country")

# PREDICTOR VARIABLES
predictor_vars <- setdiff(names(VC), c(identifier_vars, outcome_vars))
# Numeric predictor variables
predictor_num_vars <- predictor_vars[
  sapply(VC[predictor_vars], is.numeric) &
  !(predictor_vars %in% c("Country", "Year", "Wave"))
]
num_cols <- names(VC)[sapply(VC, is.numeric)]

# FIGURE: COUNT OF ZEROS PER NUMERICAL VARIABLE
zero_count <- sapply(VC[num_cols], function(x) sum(x == 0, na.rm = TRUE))

# Create plotting table
zero_df <- data.frame(
  Variable = names(zero_count),
  Zero_Count = as.numeric(zero_count),
  Type = ifelse(names(zero_count) %in% outcome_vars,
    "Confidence (Outcome)",
```

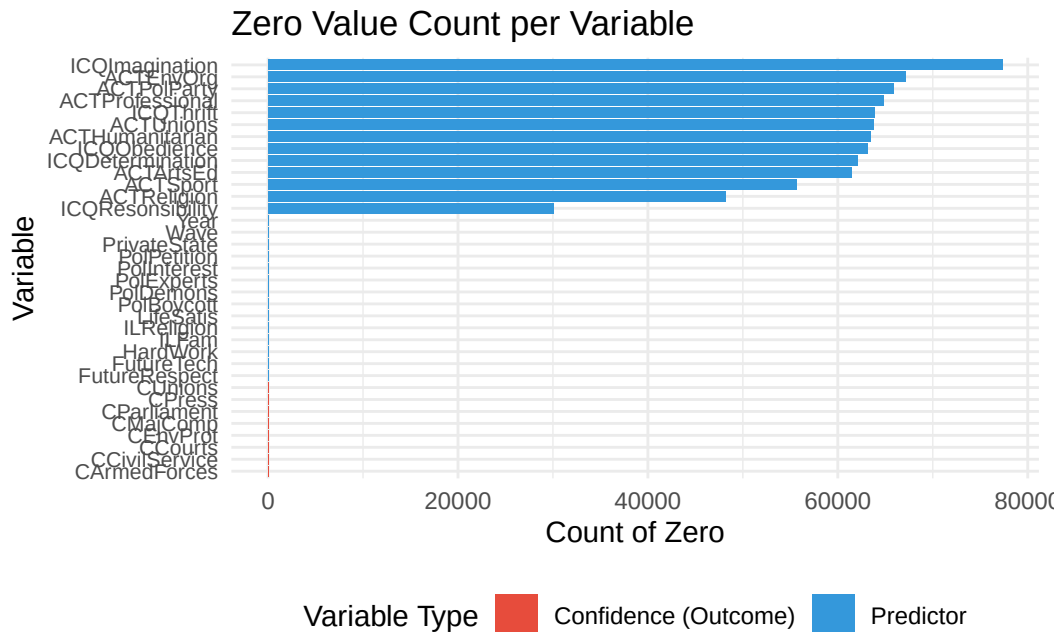
```

        "Predictor")
)

p_zero <- ggplot(zero_df,
               aes(x = reorder(Variable, Zero_Count),
                   y = Zero_Count,
                   fill = Type)) +
  geom_col() +
  coord_flip() +
  scale_fill_manual(values = c(
    "Confidence (Outcome)" = "#E74C3C",
    "Predictor" = "#3498DB"
  )) +
  labs(title = "Zero Value Count per Variable",
       x = "Variable",
       y = "Count of Zero",
       fill = "Variable Type") +
  theme_minimal(base_size = 11) +
  theme(
    legend.position = "bottom",
    axis.text.y = element_text(size = 8)
  )

print(p_zero)

```



```
# numeric_cols <- sapply(VC, is.numeric)
# num_cols <- names(VC)[numeric_cols]

# NEGATIVE -> NA
VC[num_cols] <- lapply(VC[num_cols], function(x) replace(x, x < 0, NA))

# INVALID 0s -> NA
valid_zero <- c(
  grep("^ACT", names(VC), value = TRUE),
  grep("^ICQ", names(VC), value = TRUE)
)
zero_invalid_vars <- setdiff(num_cols, valid_zero)
VC[zero_invalid_vars] <- lapply(VC[zero_invalid_vars], function(x) {
  x[x == 0] <- NA
  x
})
```

```
# FIGURE: MISSING VALUE COUNT PER VARIABLE
# Missing count
missing_count <- colSums(is.na(VC[, num_cols, drop = FALSE]))

# Create plotting table
missing_df <- data.frame(
```

```

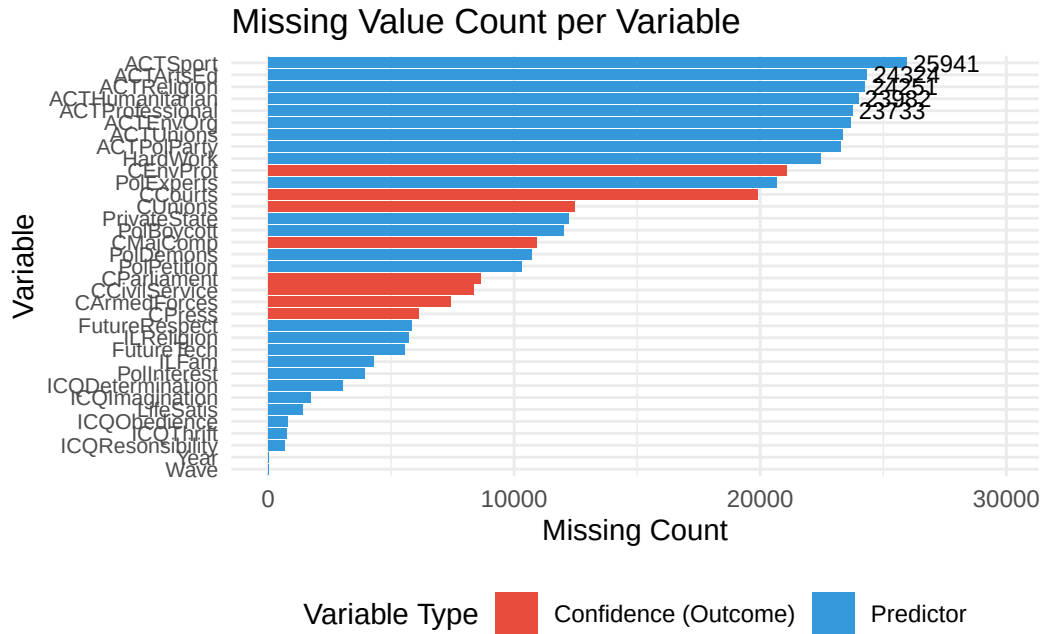
Variable = names(missing_count),
Missing_Count = as.numeric(missing_count),
Type = ifelse(names(missing_count) %in% outcome_vars,
              "Confidence (Outcome)",
              "Predictor")
)

top5_missing <- missing_df %>%
  arrange(desc(Missing_Count)) %>%
  slice(1:5)

p_missing <- ggplot(missing_df,
                   aes(x = reorder(Variable, Missing_Count),
                       y = Missing_Count,
                       fill = Type)) +
  geom_col() +
  geom_text(
    data = top5_missing,
    aes(label = Missing_Count),
    hjust = -0.1,
    size = 3
  ) +
  coord_flip() +
  scale_fill_manual(values = c(
    "Confidence (Outcome)" = "#E74C3C",
    "Predictor" = "#3498DB"
  )) +
  labs(
    title = "Missing Value Count per Variable",
    x = "Variable",
    y = "Missing Count",
    fill = "Variable Type"
  ) +
  theme_minimal(base_size = 11) +
  expand_limits(y = max(missing_df$Missing_Count) * 1.15) +
  theme(
    legend.position = "bottom",
    axis.text.y = element_text(size = 8)
  )

print(p_missing)

```



```
# TABLE: ALL NUMERICAL VARIABLES
summary_table <- data.frame(
  Variable = num_cols,
  Min = sapply(VC[num_cols], function(x) min(x, na.rm = TRUE)),
  Q1 = sapply(VC[num_cols], function(x) quantile(x, 0.25, na.rm = TRUE)),
  Median = sapply(VC[num_cols], function(x) median(x, na.rm = TRUE)),
  Mean = sapply(VC[num_cols], function(x) mean(x, na.rm = TRUE)),
  Q3 = sapply(VC[num_cols], function(x) quantile(x, 0.75, na.rm = TRUE)),
  Max = sapply(VC[num_cols], function(x) max(x, na.rm = TRUE)),
  SD = sapply(VC[num_cols], function(x) sd(x, na.rm = TRUE)),
  MissingPct = sapply(VC[num_cols], function(x) mean(is.na(x)) * 100)
)

summary_table[, -1] <- round(summary_table[, -1], 2)
knitr::kable(summary_table, caption = "Summary of numerical variables")
```

Table 3: Summary of numerical variables

	Variable	Min	Q1	Me- dian	Mean	Q3	Max	SD	Missing- Pct
Wave	Wave	1	3	5	4.78	6	7	1.71	0.00
Year	Year	1981	1998	2006	2005.56	2013	2023	9.93	0.00

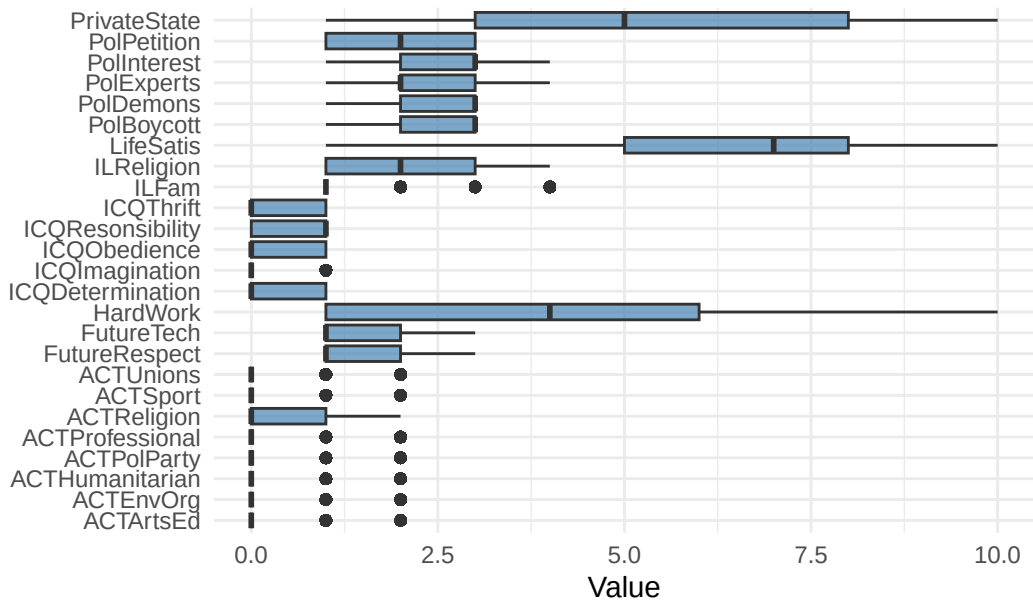
	Variable	Min	Q1	Me- dian	Mean	Q3	Max	SD	Missing- Pct
ILFam	ILFam	1	1	1	1.11	1	4	0.36	4.31
ILReligion	ILReligion	1	1	2	1.97	3	4	1.08	5.70
ICQResonsi- bility	ICQResonsi- bility	0	0	1	0.70	1	1	0.46	0.66
ICQImagina- tion	ICQImagina- tion	0	0	0	0.21	0	1	0.41	1.74
ICQThrift	ICQThrift	0	0	0	0.36	1	1	0.48	0.76
ICQDetermi- nation	ICQDetermi- nation	0	0	0	0.36	1	1	0.48	3.04
ICQObedi- ence	ICQObedi- ence	0	0	0	0.36	1	1	0.48	0.80
ACTReligion	ACTReligion	0	0	0	0.55	1	2	0.79	24.25
ACTSport	ACTSport	0	0	0	0.38	0	2	0.70	25.94
ACTArtsEd	ACTArtsEd	0	0	0	0.28	0	2	0.62	24.32
ACTUnions	ACTUnions	0	0	0	0.22	0	2	0.52	23.34
ACTPolParty	ACTPolParty	0	0	0	0.19	0	2	0.49	23.28
ACTEnvOrg	ACTEnvOrg	0	0	0	0.16	0	2	0.47	23.65
ACTProfes- sional	ACTProfes- sional	0	0	0	0.21	0	2	0.54	23.73
ACTHuman- itarian	ACTHuman- itarian	0	0	0	0.24	0	2	0.57	23.98
LifeSatis	LifeSatis	1	5	7	6.76	8	10	2.39	1.41
FutureTech	FutureTech	1	1	1	1.37	2	3	0.63	5.54
FutureRe- spect	FutureRe- spect	1	1	1	1.54	2	3	0.73	5.85
PolInterest	PolInterest	1	2	3	2.66	3	4	0.96	3.94
PolPetition	PolPetition	1	1	2	2.14	3	3	0.81	10.30
PolBoycott	PolBoycott	1	2	3	2.54	3	3	0.65	12.03
PolDemons	PolDemons	1	2	3	2.37	3	3	0.72	10.70
PrivateState	PrivateState	1	3	5	5.50	8	10	2.85	12.22
HardWork	HardWork	1	1	4	4.31	6	10	2.88	22.45
PolExperts	PolExperts	1	2	2	2.34	3	4	0.92	20.67
CArmed- Forces	CArmed- Forces	1	1	2	2.18	3	4	0.92	7.43
CPress	CPress	1	2	3	2.59	3	4	0.86	6.10
CUnions	CUnions	1	2	3	2.70	3	4	0.89	12.47
CParliament	CParliament	1	2	3	2.72	3	4	0.93	8.63
CCivilService	CCivilService	1	2	3	2.57	3	4	0.87	8.34
CMajComp	CMajComp	1	2	3	2.57	3	4	0.86	10.91
CEnvProt	CEnvProt	1	2	2	2.33	3	4	0.87	21.07

	Variable	Min	Q1	Me- dian	Mean	Q3	Max	SD	Missing- Pct
CCourts	CCourts	1	2	2	2.43	3	4	0.92	19.89

```
# BOX PLOT: ALL NUMERICAL PREDICTORS
numeric_vars <- predictor_num_vars

VC %>%
  dplyr::select(all_of(numeric_vars)) %>%
  tidyr::pivot_longer(
    cols = everything(),
    names_to = "Variable",
    values_to = "Value"
  ) %>%
  dplyr::filter(!is.na(Value)) %>%
  ggplot(aes(x = Variable, y = Value)) +
  geom_boxplot(fill = "steelblue", alpha = 0.7, outlier.alpha = 0.2) +
  coord_flip() +
  labs(
    title = "Boxplot of All Numerical Predictors",
    x = NULL,
    y = "Value"
  ) +
  theme_minimal()
```

Boxplot of All Numerical Predictors



```
IL_vars <- grep("^IL", names(VC), value = TRUE)
ICQ_vars <- grep("^ICQ", names(VC), value = TRUE)
Pol_vars <- grep("^Pol", names(VC), value = TRUE)
ACT_vars <- grep("^ACT", names(VC), value = TRUE)
extra_vars <- c("LifeSatis", "FutureTech", "FutureRespect",
               "PrivateState", "HardWork")
```

```
# NUMBER OF UNIQUE COUNTRY
length(unique(VC$Country))
```

```
[1] 58
```

```
# COUNTRY SAMPLE IMBALANCE
library(patchwork)
```

Warning: package 'patchwork' was built under R version 4.5.3

Attaching package: 'patchwork'

The following object is masked from 'package:MASS':

area

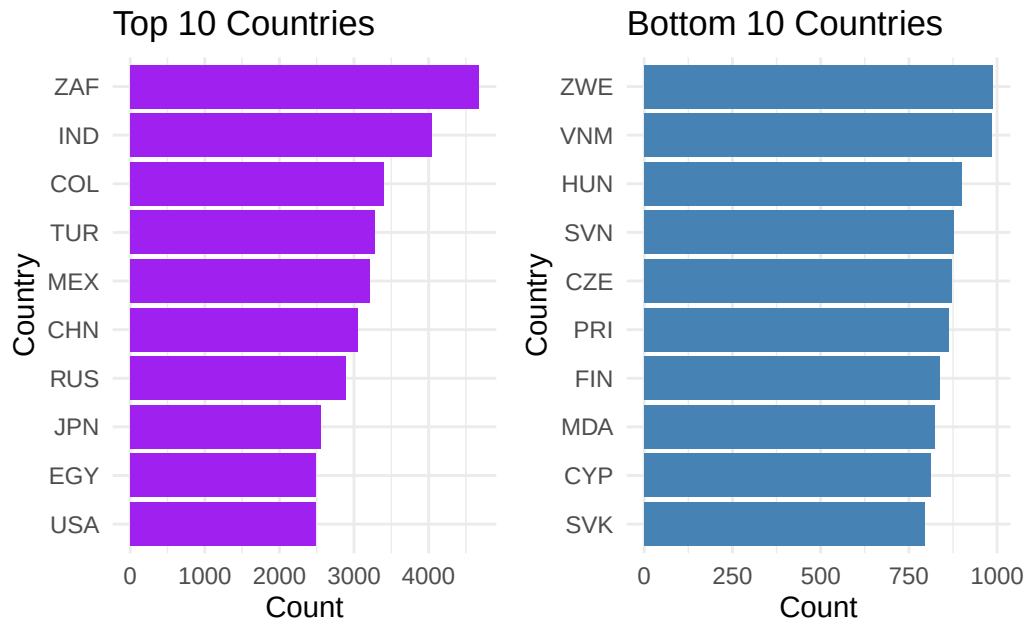
```
country_counts <- VC %>%
  count(Country, sort = TRUE)

top10 <- country_counts %>% slice_head(n = 10)
bottom10 <- country_counts %>% slice_tail(n = 10)

p1 <- ggplot(top10, aes(x = reorder(Country, n), y = n)) +
  geom_col(fill = "purple") +
  coord_flip() +
  labs(
    title = "Top 10 Countries",
    x = "Country",
    y = "Count"
  ) +
  theme_minimal()

p2 <- ggplot(bottom10, aes(x = reorder(Country, n), y = n)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(
    title = "Bottom 10 Countries",
    x = "Country",
    y = "Count"
  ) +
  theme_minimal()

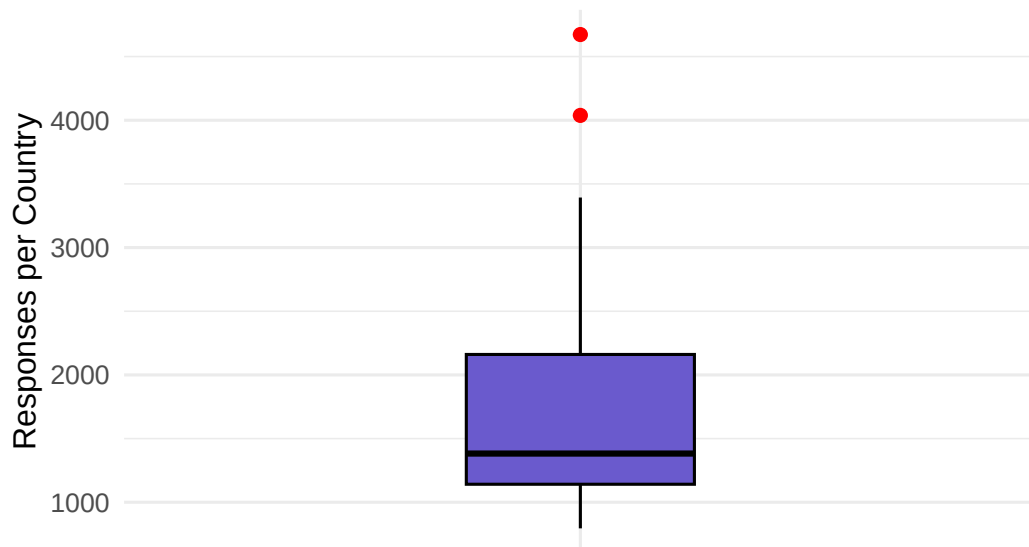
p1 + p2
```



```
# BOX PLOT: DISTRIBUTION OF COUNTRY SAMPLE SIZE
country_counts %>%
  ggplot(aes(x = "", y = n)) +
  geom_boxplot(
    fill = "#6A5ACD",
    color = "black",
    width = 0.3,
    outlier.color = "red",
    outlier.size = 2
  ) +
  labs(
    title = "Distribution of Country Sample Sizes",
    subtitle = "Number of survey responses per country",
    y = "Responses per Country",
    x = NULL
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold"),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank()
  )
)
```

Distribution of Country Sample Sizes

Number of survey responses per country



Appendix 2

Q2a

```
focus_country <- "KOR"

# FOCUS COUNTRY & OTHER SUBSETS
VC_focus <- VC %>% filter(Country == focus_country)
VC_other <- VC %>% filter(Country != focus_country)

cat("Korea nrow:", nrow(VC_focus), "\n")
```

Korea nrow: 2339

```
cat("Other Countries nrow:", nrow(VC_other))
```

Other Countries nrow: 97661

```

# CORRELATION MATRIX
cor_mat <- cor(
  VC %>%
    dplyr::select(all_of(predictor_num_vars)),
  use = "pairwise.complete.obs"
)

# TABLE: HIGHLY CORRELATED PAIRS
high_corr_pairs <- which(
  abs(cor_mat) >= 0.7 & abs(cor_mat) < 1,
  arr.ind = TRUE
)

high_corr_table <- data.frame(
  Var1 = rownames(cor_mat)[high_corr_pairs[,1]],
  Var2 = colnames(cor_mat)[high_corr_pairs[,2]],
  Correlation = cor_mat[high_corr_pairs]
) %>%
  distinct()

knitr::kable(high_corr_table)

```

Var1	Var2	Correlation
------	------	-------------

```

predictor_q2a <- predictor_num_vars

```

```

# TABLE: WELCH T-TESTS
q2a_results <- map_dfr(predictor_q2a, function(var){

  x <- VC_focus[[var]]
  y <- VC_other[[var]]

  test <- t.test(x, y, var.equal = FALSE)

  tibble(
    Variable = var,
    Focus_Mean = mean(x, na.rm = TRUE),
    Focus_SD = sd(x, na.rm = TRUE),
    Other_Mean = mean(y, na.rm = TRUE),
    Other_SD = sd(y, na.rm = TRUE),

```

```

    Mean_Diff = mean(x, na.rm = TRUE) - mean(y, na.rm = TRUE),
    P_Value = test$p.value
  )
}) %>%
  arrange(desc(abs(Mean_Diff)))

knitr::kable(q2a_results, digits = 3)

```

Variable	Fo- cus_Mean	Fo- cus_SD	Other_Mean	Other_SD	Mean_Diff	P_Value
FutureRespect	2.334	0.767	1.522	0.716	0.812	0.000
ILReligion	2.469	0.982	1.959	1.076	0.511	0.000
LifeSatis	6.370	2.003	6.771	2.398	-0.401	0.000
PrivateState	5.149	2.376	5.511	2.857	-0.362	0.000
PolPetition	1.865	0.738	2.146	0.813	-0.282	0.000
PolBoycott	2.286	0.618	2.545	0.651	-0.259	0.000
ICQObedience	0.112	0.315	0.370	0.483	-0.259	0.000
ICQThrift	0.593	0.491	0.351	0.477	0.242	0.000
ICQResonsibility	0.853	0.354	0.694	0.461	0.159	0.000
HardWork	4.175	2.513	4.315	2.889	-0.140	0.022
ACTUnions	0.097	0.357	0.223	0.528	-0.126	0.000
PolInterest	2.548	0.779	2.660	0.969	-0.112	0.000
ACTSport	0.487	0.705	0.376	0.702	0.112	0.000
ACTPolParty	0.083	0.335	0.189	0.496	-0.106	0.000
PolExperts	2.443	0.768	2.339	0.921	0.104	0.000
PolDemons	2.281	0.694	2.368	0.724	-0.086	0.000
ACTProfessional	0.135	0.442	0.215	0.545	-0.080	0.000
ACTHumanitarian	0.178	0.469	0.238	0.573	-0.060	0.000
ICQDetermination	0.413	0.492	0.358	0.479	0.055	0.000
FutureTech	1.325	0.589	1.375	0.626	-0.050	0.000
ICQImagination	0.235	0.424	0.213	0.409	0.022	0.013
ILFam	1.094	0.307	1.112	0.365	-0.017	0.011
ACTArtsEd	0.293	0.589	0.278	0.617	0.015	0.304
ACTReligion	0.543	0.740	0.552	0.789	-0.008	0.667
ACTEnvOrg	0.156	0.452	0.162	0.466	-0.005	0.655

```

write.csv(q2a_results, "q2a_welch_ttest_results.csv", row.names = FALSE)

```

```

# TOP 10 DIFFERENCES (DUMBBELL PLOT)
top10_diff <- q2a_results %>%
  slice_max(abs(Mean_Diff), n = 10)

ggplot(top10_diff,
  aes(y = reorder(Variable, Mean_Diff))) +

  geom_segment(aes(
    x = Other_Mean,
    xend = Focus_Mean,
    yend = Variable
  ),
  color = "grey70",
  linewidth = 1.2) +

  geom_point(
    aes(x = Other_Mean, colour = "Other Countries"),
    size = 3
  ) +

  geom_point(
    aes(x = Focus_Mean, colour = "Focus Country"),
    size = 3
  ) +

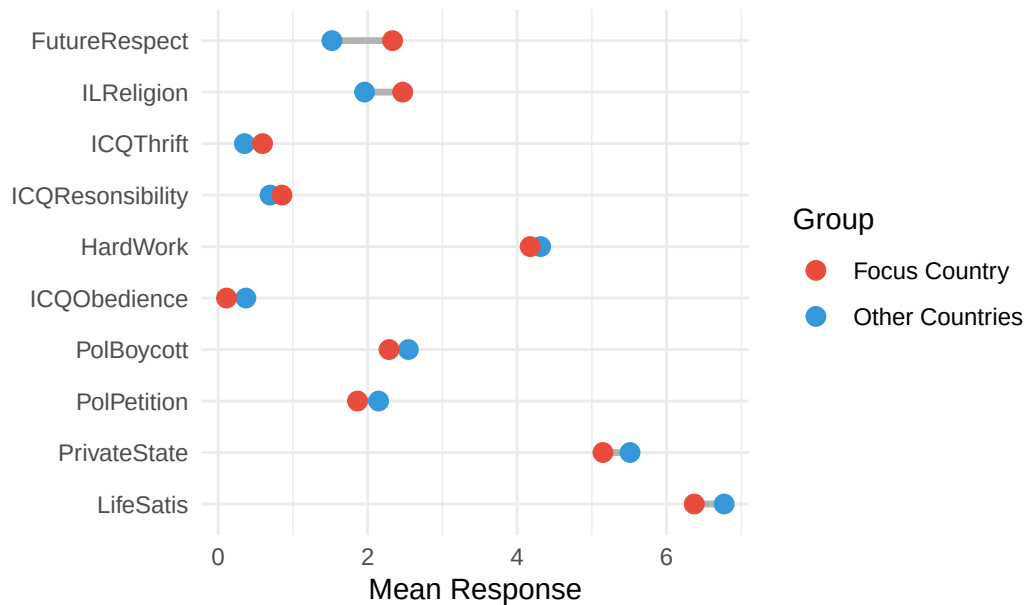
  scale_colour_manual(
    values = c(
      "Other Countries" = "#3498DB",
      "Focus Country" = "#E74C3C"
    ),
    name = "Group"
  ) +

  labs(
    title = paste("Top 10 Attribute Differences:", focus_country, "vs Other Countries"),
    x = "Mean Response",
    y = NULL
  ) +

  theme_minimal()

```

Top 10 Attribute Differences: KOR vs Other Countries



```
# TABLE: Top significant mean differences (clean for report)

q2a_summary <- q2a_results %>%
  mutate(
    Direction = ifelse(Mean_Diff > 0, "Focus Higher", "Focus Lower")
  ) %>%
  filter(P_Value < 0.05) %>%           # keep only significant
  arrange(desc(abs(Mean_Diff))) %>%   # largest differences first
  slice_head(n = 10) %>%             # top 10 only
  mutate(
    Mean_Diff = round(Mean_Diff, 3),
    P_Value = ifelse(P_Value < 0.001, "<0.001", round(P_Value, 3))
  ) %>%
  dplyr::select(
    Variable,
    Mean_Diff,
    P_Value,
    Direction
  )

# Clean table for report
knitr::kable(
  q2a_summary,
```

```
caption = "Top 10 significant differences between Korea and other countries"
)
```

Table 6: Top 10 significant differences between Korea and other countries

Variable	Mean_Diff	P_Value	Direction
FutureRespect	0.812	<0.001	Focus Higher
ILReligion	0.511	<0.001	Focus Higher
LifeSatis	-0.401	<0.001	Focus Lower
PrivateState	-0.362	<0.001	Focus Lower
PolPetition	-0.282	<0.001	Focus Lower
PolBoycott	-0.259	<0.001	Focus Lower
ICQObedience	-0.259	<0.001	Focus Lower
ICQThrift	0.242	<0.001	Focus Higher
ICQResonsibility	0.159	<0.001	Focus Higher
HardWork	-0.140	0.022	Focus Lower

```
write.csv(q2a_summary, "q2a_top10_results.csv", row.names = FALSE)
```

```
write.csv(q2a_results, "q2a_full_results.csv", row.names = FALSE)
```

```
# BOXPLOT: TOP DIFFERING VARIABLES
```

```
var_groups <- list(
  IL = IL_vars,
  ICQ = ICQ_vars,
  Political = Pol_vars,
  ACT = ACT_vars,
  Additional = extra_vars
)
```

```
# Build grouped scores for each respondent
```

```
focus_grouped <- VC_focus %>%
  dplyr::transmute(
    Group = focus_country,
    IL = rowMeans(dplyr::select(., all_of(IL_vars)), na.rm = TRUE),
    ICQ = rowMeans(dplyr::select(., all_of(ICQ_vars)), na.rm = TRUE),
    Political = rowMeans(dplyr::select(., all_of(Pol_vars)), na.rm = TRUE),
    ACT = rowMeans(dplyr::select(., all_of(ACT_vars)), na.rm = TRUE),
    Additional = rowMeans(dplyr::select(., all_of(extra_vars)), na.rm = TRUE)
  )
```

```

other_grouped <- VC_other %>%
  dplyr::transmute(
    Group = "Other Countries",
    IL = rowMeans(dplyr::select(., all_of(IL_vars)), na.rm = TRUE),
    ICQ = rowMeans(dplyr::select(., all_of(ICQ_vars)), na.rm = TRUE),
    Political = rowMeans(dplyr::select(., all_of(Pol_vars)), na.rm = TRUE),
    ACT = rowMeans(dplyr::select(., all_of(ACT_vars)), na.rm = TRUE),
    Additional = rowMeans(dplyr::select(., all_of(extra_vars)), na.rm = TRUE)
  )

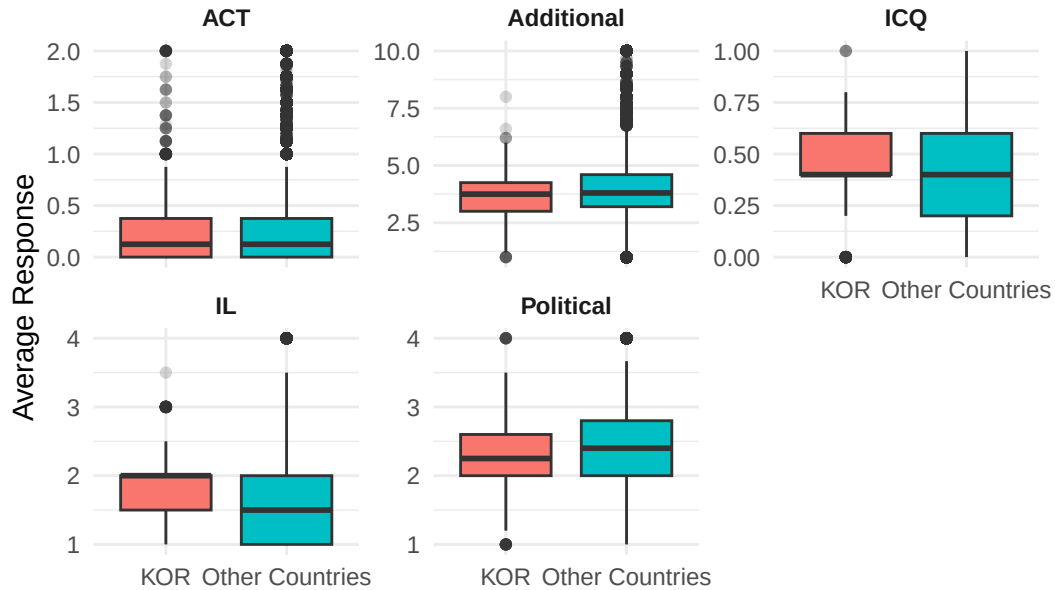
boxplot_group_data <- bind_rows(focus_grouped, other_grouped) %>%
  tidyr::pivot_longer(
    cols = -Group,
    names_to = "VariableGroup",
    values_to = "Value"
  )

ggplot(boxplot_group_data,
       aes(x = Group, y = Value, fill = Group)) +
  geom_boxplot(outlier.alpha = 0.2) +
  facet_wrap(~ VariableGroup, scales = "free_y") +
  labs(
    title = paste("Distribution by Variable Group:", focus_country, "vs Other Countries"),
    x = NULL,
    y = "Average Response"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    strip.text = element_text(face = "bold")
  )

```

Warning: Removed 28924 rows containing non-finite outside the scale range (`stat\_boxplot()`).

Distribution by Variable Group: KOR vs Other Countries



Q2b

```
predictor_q2b <- predictor_q2a
confidence_vars <- outcome_vars
```

```
# Run stepwise and forward models
run_models_selection <- function(data, predictors, outcomes) {

  model_list <- lapply(outcomes, function(out) {

    # Keep only needed columns for this outcome
    model_data <- data %>%
      dplyr::select(all_of(c(out, predictors))) %>%
      na.omit()

    # Standardise variables so coefficients are comparable
    model_data <- model_data %>%
      dplyr::mutate(across(everything(), ~ as.numeric(scale(.))))

    full_formula <- as.formula(
```

```

    paste(out, "~", paste(predictors, collapse = " + "))
  )

  null_formula <- as.formula(
    paste(out, "~ 1")
  )

  full_model <- lm(full_formula, data = model_data)
  null_model <- lm(null_formula, data = model_data)

  step_model <- stepAIC(
    full_model,
    direction = "both",
    trace = FALSE
  )

  forward_model <- stepAIC(
    null_model,
    scope = list(lower = null_formula, upper = full_formula),
    direction = "forward",
    trace = FALSE
  )

  list(
    stepwise = step_model,
    forward = forward_model,
    n = nrow(model_data)
  )
})

names(model_list) <- outcomes
return(model_list)
}

# Model performance table
get_model_perf <- function(model_list) {

  bind_rows(lapply(names(model_list), function(out) {

    step_sum <- summary(model_list[[out]]$stepwise)
    fwd_sum <- summary(model_list[[out]]$forward)
  }

```

```

step_p <- pf(
  step_sum$fstatistic[1],
  step_sum$fstatistic[2],
  step_sum$fstatistic[3],
  lower.tail = FALSE
)

fwd_p <- pf(
  fwd_sum$fstatistic[1],
  fwd_sum$fstatistic[2],
  fwd_sum$fstatistic[3],
  lower.tail = FALSE
)

tibble(
  Outcome = out,
  N = model_list[[out]]$n,
  Stepwise_R2 = step_sum$r.squared,
  Stepwise_Adj_R2 = step_sum$adj.r.squared,
  Stepwise_Model_P = step_p,
  Forward_R2 = fwd_sum$r.squared,
  Forward_Adj_R2 = fwd_sum$adj.r.squared,
  Forward_Model_P = fwd_p
)
))) %>%
  dplyr::mutate(
    Best_Method = ifelse(Stepwise_Adj_R2 >= Forward_Adj_R2, "Stepwise", "Forward"),
    Best_R2 = pmax(Stepwise_R2, Forward_R2),
    Best_Adj_R2 = pmax(Stepwise_Adj_R2, Forward_Adj_R2),
    Best_Model_P = ifelse(Best_Method == "Stepwise", Stepwise_Model_P, Forward_Model_P)
  ) %>%
  dplyr::arrange(desc(Best_Adj_R2))
}

# Extract coefficients from best model
get_coefficients_best <- function(model_list, group_name) {

  bind_rows(lapply(names(model_list), function(out) {

    step_sum <- summary(model_list[[out]]$stepwise)
    fwd_sum <- summary(model_list[[out]]$forward)

```

```

best_model <- if (step_sum$adj.r.squared >= fwd_sum$adj.r.squared) {
  model_list[[out]]$stepwise
} else {
  model_list[[out]]$forward
}

coef_df <- as.data.frame(summary(best_model)$coefficients)
coef_df$Variable <- rownames(coef_df)
coef_df$Outcome <- out
coef_df$Group <- group_name
coef_df$Significant <- coef_df$`Pr(>|t|)` < 0.05
coef_df$Abs_t <- abs(coef_df$t value)
coef_df$Abs_Beta <- abs(coef_df$Estimate)
rownames(coef_df) <- NULL

coef_df
))) %>%
  dplyr::filter(Variable != "(Intercept)")
}

# Run models
models_focus <- run_models_selection(
  data = VC_focus,
  predictors = predictor_q2b,
  outcomes = confidence_vars
)

# Model performance summary
model_perf_focus <- get_model_perf(models_focus)

knitr::kable(
  model_perf_focus,
  digits = 4,
  caption = "Model performance for confidence outcomes in the focus country"
)

```

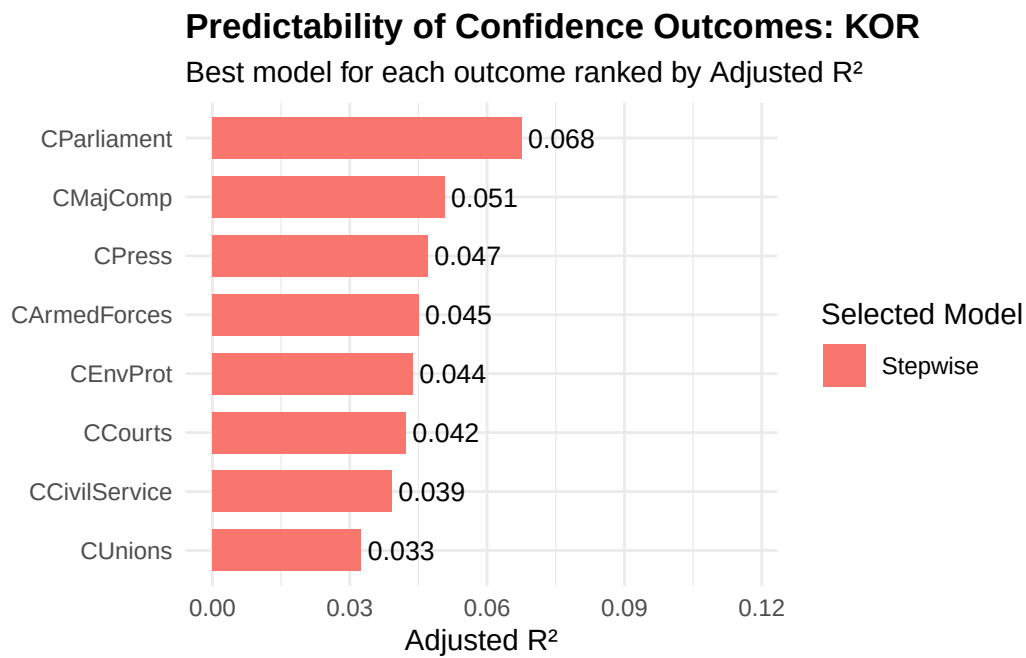
Table 7: Model performance for confidence outcomes in the focus country

Out- come	N	Step- wise_R2	Step- wise_Adj_R2	Step- wise_R2_Model_P	For- ward_R2	For- ward_Adj_R2	For- ward_R2_Model_P	Best_PMB	Best_R2	Best_Adj_R2	Best_R2_Model_P
CPar- lia- ment	945	0.0774	0.0675	0	0.0774	0.0675	0	Step- wise	0.0774	0.0675	0
CMaj- Comp	945	0.0618	0.0507	0	0.0594	0.0493	0	Step- wise	0.0618	0.0507	0
CPress	946	0.0541	0.0471	0	0.0541	0.0471	0	Step- wise	0.0541	0.0471	0
CArmed Forces	946	0.0531	0.0450	0	0.0531	0.0450	0	Step- wise	0.0531	0.0450	0
CEn- vProt	946	0.0519	0.0438	0	0.0483	0.0422	0	Step- wise	0.0519	0.0438	0
CCourts	946	0.0484	0.0423	0	0.0484	0.0423	0	Step- wise	0.0484	0.0423	0
CCivilSe- vice	945	0.0473	0.0392	0	0.0407	0.0356	0	Step- wise	0.0473	0.0392	0
CU- nions	944	0.0407	0.0325	0	0.0407	0.0325	0	Step- wise	0.0407	0.0325	0

```
write.csv(model_perf_focus, "q2b_model_perf_focus.csv", row.names = FALSE)
```

```
# Plot: which confidence outcome is most reliably predicted
model_perf_focus %>%
  ggplot(aes(x = reorder(Outcome, Best_Adj_R2),
                y = Best_Adj_R2,
                fill = Best_Method)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = round(Best_Adj_R2, 3)),
            hjust = -0.1, size = 3.5) +
  coord_flip() +
  scale_y_continuous(limits = c(0, max(model_perf_focus$Best_Adj_R2) + 0.05)) +
  labs(
    title = paste("Predictability of Confidence Outcomes:", focus_country),
    subtitle = "Best model for each outcome ranked by Adjusted R2",
    x = NULL,
    y = "Adjusted R2",
    fill = "Selected Model"
  ) +
  theme_minimal(base_size = 11) +
```

```
theme(
  plot.title = element_text(face = "bold"),
  axis.text.y = element_text(size = 9)
)
```



```
# Extract coefficients from best model for each outcome
coef_focus <- get_coefficients_best(models_focus, focus_country)
```

```
# Best predictor for each confidence outcome
# rank by absolute t-value
best_predictors_focus <- coef_focus %>%
  dplyr::filter(Significant) %>%
  dplyr::group_by(Outcome) %>%
  dplyr::slice_max(order_by = Abs_t, n = 1, with_ties = FALSE) %>%
  dplyr::ungroup()

knitr::kable(
  best_predictors_focus,
  digits = 4,
  caption = "Strongest predictor of each confidence outcome in the focus country"
)
```

Table 8: Strongest predictor of each confidence outcome in the focus country

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
0.1089	0.0320	3.4026	0.0007	FutureRespect	CArmed-Forces	KOR	TRUE	3.4026	0.1089
-	0.0322	-	0.0011	IC-QThrift	CCivilService	KOR	TRUE	3.2748	0.1056
0.1056		3.2748							
0.1305	0.0324	4.0319	0.0001	FutureRespect	CCourts	KOR	TRUE	4.0319	0.1305
0.1404	0.0321	4.3668	0.0000	Hard-Work	CEnvProt	KOR	TRUE	4.3668	0.1404
0.1096	0.0321	3.4169	0.0007	ILReligion	CMajComp	KOR	TRUE	3.4169	0.1096
0.1226	0.0319	3.8414	0.0001	FutureRespect	CParliament	KOR	TRUE	3.8414	0.1226
0.1251	0.0321	3.8963	0.0001	Hard-Work	CPress	KOR	TRUE	3.8963	0.1251
-	0.0321	-	0.0035	PrivateSetate	CUnions	KOR	TRUE	2.9280	0.0940
0.0940		2.9280							

```
write.csv(best_predictors_focus, "q2b_best_predict_focus.csv", row.names = FALSE)
```

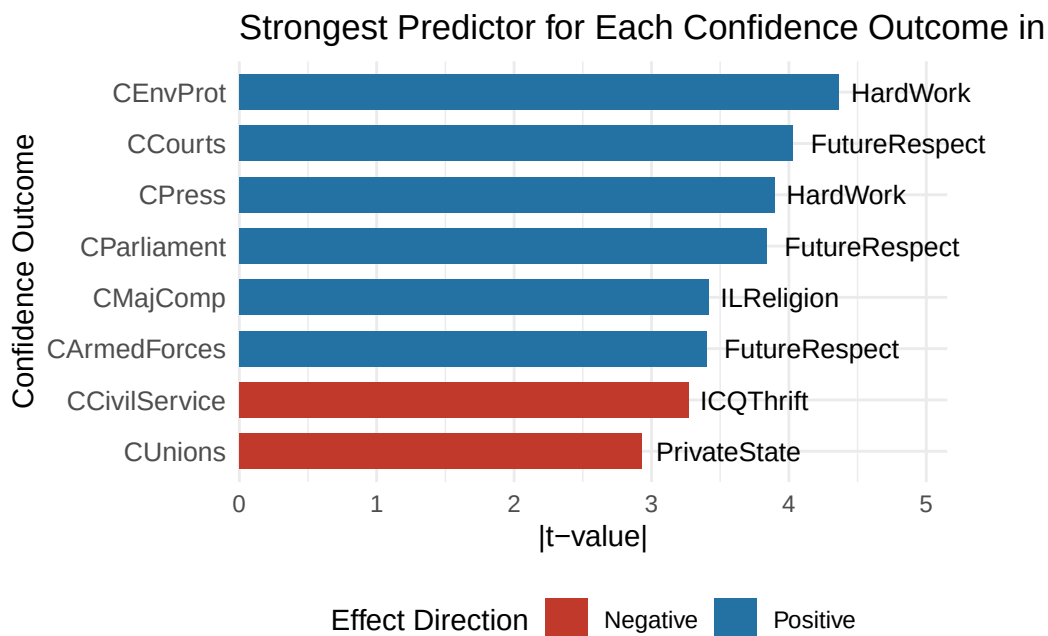
```
best_predictors_plot <- best_predictors_focus %>%
  mutate(
    Outcome_label = stringr::str_wrap(Outcome, width = 12),
    Predictor_label = stringr::str_wrap(Variable, width = 14),
    Direction = ifelse(Estimate > 0, "Positive", "Negative")
  )

ggplot(best_predictors_plot,
  aes(x = reorder(Outcome_label, Abs_t),
    y = Abs_t,
    fill = Direction)) +
  geom_col(width = 0.7) +
  coord_flip(clip = "off") +
  geom_text(
    aes(label = Predictor_label),
    hjust = -0.1,
    size = 3.5
  ) +
```

```

scale_fill_manual(
  values = c("Positive" = "#2471A3", "Negative" = "#C0392B")
) +
scale_y_continuous(
  expand = expansion(mult = c(0, 0.18))
) +
labs(
  title = "Strongest Predictor for Each Confidence Outcome in Korea",
  x = "Confidence Outcome",
  y = "|t-value|",
  fill = "Effect Direction"
) +
theme_minimal(base_size = 11) +
theme(
  legend.position = "bottom",
  plot.margin = margin(5.5, 40, 5.5, 5.5),
  axis.text.y = element_text(size = 10)
)

```



```

# 13. Heatmap of standardised predictors
heatmap_focus <- coef_focus %>%
  dplyr::select(Outcome, Variable, Estimate, Significant) %>%

```

```

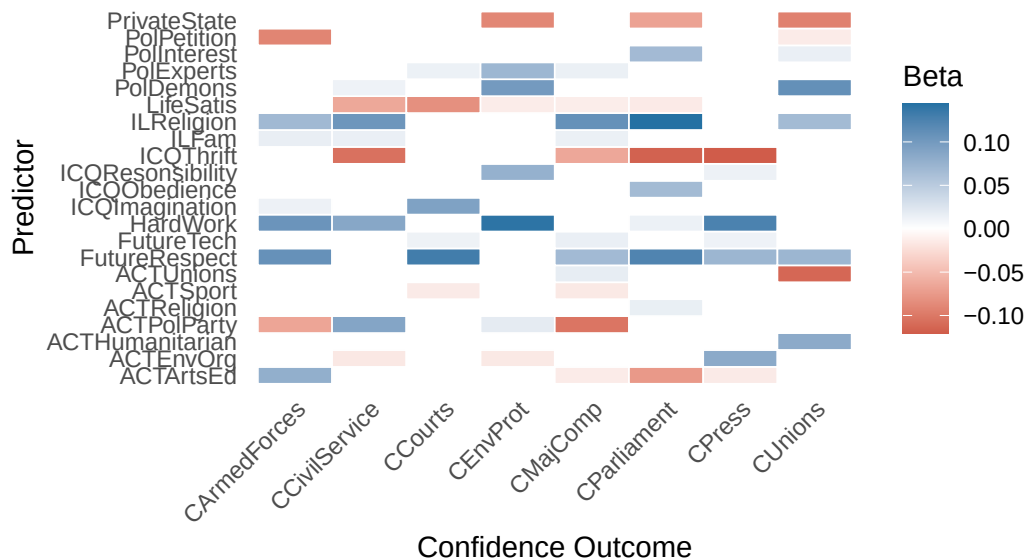
tidyr::complete(Outcome, Variable, fill = list(Estimate = 0, Significant = FALSE))

ggplot(heatmap_focus, aes(x = Outcome, y = Variable, fill = Estimate, alpha = Significant)) +
  geom_tile(color = "white", linewidth = 0.3) +
  scale_fill_gradient2(
    low = "#C0392B",
    mid = "white",
    high = "#2471A3",
    midpoint = 0
  ) +
  scale_alpha_manual(values = c("TRUE" = 1, "FALSE" = 0.25), guide = "none") +
  labs(
    title = paste("Standardised Predictors of Confidence:", focus_country),
    subtitle = "Darker tiles indicate statistically significant effects",
    x = "Confidence Outcome",
    y = "Predictor",
    fill = "Beta"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 9),
    axis.text.y = element_text(size = 9),
    panel.grid = element_blank()
  )

```

Standardised Predictors of Confidence: KOR

Darker tiles indicate statistically significant effects



```
# Assumption checks for the most reliably predicted outcome
best_outcome <- model_perf_focus %>%
  dplyr::slice_max(order_by = Best_Adj_R2, n = 1) %>%
  dplyr::pull(Outcome)

step_sum_best <- summary(models_focus[[best_outcome]]$stepwise)
fwd_sum_best <- summary(models_focus[[best_outcome]]$forward)

best_model <- if (step_sum_best$adj.r.squared >= fwd_sum_best$adj.r.squared) {
  models_focus[[best_outcome]]$stepwise
} else {
  models_focus[[best_outcome]]$forward
}

cat("Most reliably predicted confidence outcome:", best_outcome, "\n")
```

Most reliably predicted confidence outcome: CParliament

```
cat("Selected model:",
  ifelse(step_sum_best$adj.r.squared >= fwd_sum_best$adj.r.squared,
    "Stepwise", "Forward"),
  "\n")
```

Selected model: Stepwise

```
# Save diagnostics to file instead of tiny plotting pane
png("Q2b_assumption_checks.png", width = 1200, height = 1000, res = 120)
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
plot(best_model)
dev.off()
```

pdf
2

Q2c

```
# TABLE: MODEL PERFORMANCE FOR OTHER COUNTRIES
models_other <- run_models_selection(
  data = VC_other,
  predictors = predictor_q2b,
  outcomes = confidence_vars
)

model_perf_other <- get_model_perf(models_other)

knitr::kable(
  model_perf_other,
  digits = 4,
  caption = "Model performance for confidence outcomes in other countries"
)
```

Table 9: Model performance for confidence outcomes in other countries

Out- come	N	Step- wise_R ²	Step- wise_Adj _w R ²	Step- wise_Model_P	For- ward_P	For- ward_Adj _w R ²	For- ward_Model_P	Best_P	Best_Adj _w R ²	Best_Model_P
CPar- lia- ment	4887	0.0628	0.0623	0	0.0628	0.0623	0	Step- wise	0.0628	0.0623
CArmed Forces	4844	0.0589	0.0585	0	0.0589	0.0585	0	For- ward	0.0589	0.0585
CU- nions	4787	0.0550	0.0546	0	0.0550	0.0546	0	For- ward	0.0550	0.0546

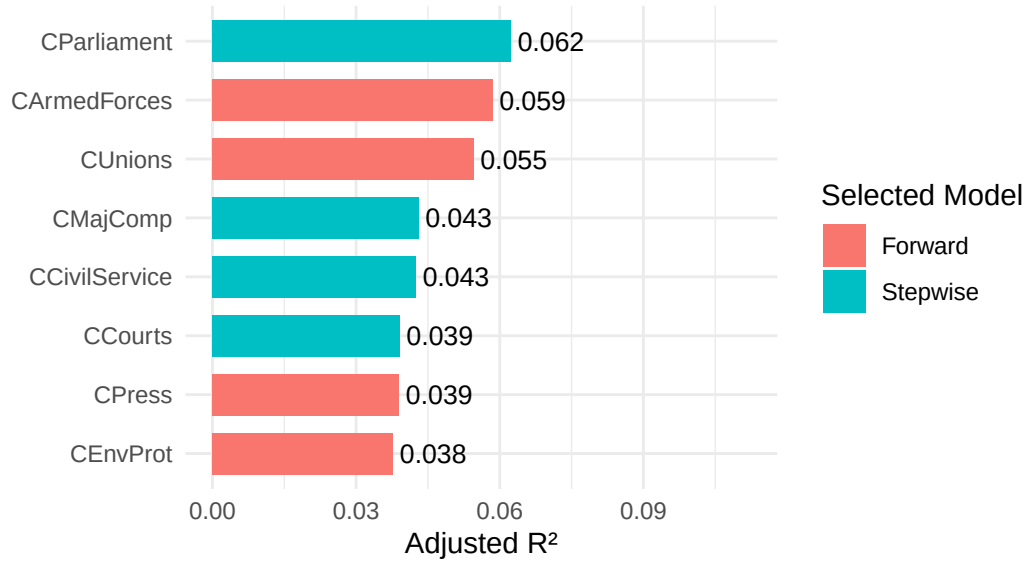
Out- come	N	Step- wise_R2	Step- wise_Adj_R2	Step- wise_R2_Model_P	For- ward_P_R2	For- ward_Adj_R2	For- ward_R2_Model_P	Best_P Method	Best_R2	Best_Adj_R2	Model_P
CMaj- Comp	48798	0.0435	0.0431	0	0.0435	0.0431	0	Step- wise	0.0435	0.0431	0
CCivilSer- vice	49416	0.0429	0.0425	0	0.0429	0.0425	0	Step- wise	0.0429	0.0425	0
CCourts	48914	0.0396	0.0391	0	0.0396	0.0391	0	Step- wise	0.0396	0.0391	0
CPress	50024	0.0394	0.0390	0	0.0394	0.0390	0	For- ward	0.0394	0.0390	0
CEn- vProt	48203	0.0382	0.0378	0	0.0382	0.0378	0	For- ward	0.0382	0.0378	0

```
write.csv(model_perf_other, "q2c_model_perf.csv", row.names = FALSE)
```

```
# PLOT: PREDICTABILITY OF OTHER COUNTRIES
model_perf_other %>%
  ggplot(aes(x = reorder(Outcome, Best_Adj_R2),
               y = Best_Adj_R2,
               fill = Best_Method)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = round(Best_Adj_R2, 3)),
            hjust = -0.1, size = 3.5) +
  coord_flip() +
  scale_y_continuous(
    limits = c(0, max(model_perf_other$Best_Adj_R2) + 0.05)
  ) +
  labs(
    title = "Predictability of Confidence Outcomes: Other Countries",
    subtitle = "Best model for each outcome ranked by Adjusted R2",
    x = NULL,
    y = "Adjusted R2",
    fill = "Selected Model"
  ) +
  theme_minimal()
```

Predictability of Confidence Outcomes: Other Countries

Best model for each outcome ranked by Adjusted R²



```
# Extract coefficients
coef_other <- get_coefficients_best(models_other, "Other Countries")

knitr::kable(
  coef_other,
  digits = 4,
  caption = "Standardised coefficients from best models for other countries"
)
```

Table 10: Standardised coefficients from best models for other countries

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
0.1272	0.0050	25.4297	0.0000	ILReligion	CArmed-Forces	Other Countries	TRUE	25.4297	0.1272
0.0809	0.0045	18.0015	0.0000	Hard-Work	CArmed-Forces	Other Countries	TRUE	18.0015	0.0809
0.0900	0.0047	19.1654	0.0000	PolInterest	CArmed-Forces	Other Countries	TRUE	19.1654	0.0900

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
0.0672	0.0046	14.4476	0.0000	FutureRespect	CArmed-Forces	Other Countries	TRUE	14.4476	0.0672
- 0.0748	0.0053	- 14.1420	0.0000	PolDemons	CArmed-Forces	Other Countries	TRUE	14.1420	0.0748
- 0.0569	0.0045	- 12.7443	0.0000	LifeSatis	CArmed-Forces	Other Countries	TRUE	12.7443	0.0569
0.0432	0.0045	9.5150	0.0000	FutureTech	CArmed-Forces	Other Countries	TRUE	9.5150	0.0432
0.0523	0.0051	10.1693	0.0000	ACTReligion	CArmed-Forces	Other Countries	TRUE	10.1693	0.0523
0.0401	0.0045	8.9153	0.0000	ILFam	CArmed-Forces	Other Countries	TRUE	8.9153	0.0401
- 0.0314	0.0051	- 6.2004	0.0000	ACTPolParty	CArmed-Forces	Other Countries	TRUE	6.2004	0.0314
- 0.0350	0.0044	- 7.8852	0.0000	IC-QThrift	CArmed-Forces	Other Countries	TRUE	7.8852	0.0350
- 0.0330	0.0050	- 6.5805	0.0000	ACTHumanitarian	CArmed-Forces	Other Countries	TRUE	6.5805	0.0330
0.0318	0.0050	6.3916	0.0000	ACTArtsEd	CArmed-Forces	Other Countries	TRUE	6.3916	0.0318
- 0.0154	0.0049	- 3.1765	0.0015	ACTUnions	CArmed-Forces	Other Countries	TRUE	3.1765	0.0154
- 0.0137	0.0045	- 3.0450	0.0023	ICQImagination	CArmed-Forces	Other Countries	TRUE	3.0450	0.0137
0.0082	0.0045	1.7963	0.0725	ICQObedience	CArmed-Forces	Other Countries	FALSE	1.7963	0.0082

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
-0.0084	0.0045	-1.8852	0.0594	PrivateS-tate	CArmed-Forces	Other Countries	FALSE	1.8852	0.0084
0.0088	0.0054	1.6334	0.1024	PolBoycott	CArmed-Forces	Other Countries	FALSE	1.6334	0.0088
0.1085	0.0047	23.2155	0.0000	PolInterest	CPress	Other Countries	TRUE	23.2155	0.1085
-0.0925	0.0054	-17.2287	0.0000	PolPetition	CPress	Other Countries	TRUE	17.2287	0.0925
0.0553	0.0045	12.2080	0.0000	FutureTech	CPress	Other Countries	TRUE	12.2080	0.0553
-0.0400	0.0054	-7.3582	0.0000	ACTEnvOrg	CPress	Other Countries	TRUE	7.3582	0.0400
0.0400	0.0050	7.9550	0.0000	ILReligion	CPress	Other Countries	TRUE	7.9550	0.0400
-0.0392	0.0044	-8.8196	0.0000	LifeSatis	CPress	Other Countries	TRUE	8.8196	0.0392
0.0372	0.0045	8.3212	0.0000	Hard-Work	CPress	Other Countries	TRUE	8.3212	0.0372
-0.0327	0.0044	-7.3806	0.0000	IC-QThrift	CPress	Other Countries	TRUE	7.3806	0.0327
0.0304	0.0044	6.8657	0.0000	PolExperts	CPress	Other Countries	TRUE	6.8657	0.0304
-0.0312	0.0051	-6.0630	0.0000	ACTPol-Party	CPress	Other Countries	TRUE	6.0630	0.0312
-0.0228	0.0045	-5.0896	0.0000	PrivateS-tate	CPress	Other Countries	TRUE	5.0896	0.0228

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
- 0.0246	0.0045	- 5.4733	0.0000	ICQImagination	CPress	Other Countries	TRUE	5.4733	0.0246
- 0.0233	0.0052	- 4.5235	0.0000	PolDemons	CPress	Other Countries	TRUE	4.5235	0.0233
- 0.0187	0.0045	- 4.1082	0.0000	ICQObedience	CPress	Other Countries	TRUE	4.1082	0.0187
- 0.0207	0.0045	- 4.6318	0.0000	ICQDetermination	CPress	Other Countries	TRUE	4.6318	0.0207
- 0.0218	0.0049	- 4.4082	0.0000	ACTUnions	CPress	Other Countries	TRUE	4.4082	0.0218
0.0156	0.0052	2.9876	0.0028	ACTProfessional	CPress	Other Countries	TRUE	2.9876	0.0156
0.0128	0.0046	2.7611	0.0058	FutureRespect	CPress	Other Countries	TRUE	2.7611	0.0128
- 0.0118	0.0051	- 2.3125	0.0208	AC-TArtsEd	CPress	Other Countries	TRUE	2.3125	0.0118
0.0148	0.0053	2.7669	0.0057	ACTHumanitarian	CPress	Other Countries	TRUE	2.7669	0.0148
- 0.0090	0.0051	- 1.7536	0.0795	ACTReligion	CPress	Other Countries	FALSE	1.7536	0.0090
- 0.1660	0.0049	- 33.6780	0.0000	ACTUnions	CUnions	Other Countries	TRUE	33.6780	0.1660
0.1157	0.0047	24.8210	0.0000	PolInterest	CUnions	Other Countries	TRUE	24.8210	0.1157
- 0.0539	0.0045	- 11.8949	0.0000	PrivateState	CUnions	Other Countries	TRUE	11.8949	0.0539

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
-0.0558	0.0045	-12.3289	0.0000	LifeSatis	CUnions	Other Countries	TRUE	12.3289	0.0558
-0.0579	0.0055	-10.4956	0.0000	PolPetition	CUnions	Other Countries	TRUE	10.4956	0.0579
0.0433	0.0054	8.0456	0.0000	PolBoycott	CUnions	Other Countries	TRUE	8.0456	0.0433
0.0289	0.0046	6.2751	0.0000	FutureTech	CUnions	Other Countries	TRUE	6.2751	0.0289
0.0279	0.0045	6.1548	0.0000	HardWork	CUnions	Other Countries	TRUE	6.1548	0.0279
0.0310	0.0052	5.9654	0.0000	ACTProfessional	CUnions	Other Countries	TRUE	5.9654	0.0310
-0.0280	0.0045	-6.2441	0.0000	ICQThrift	CUnions	Other Countries	TRUE	6.2441	0.0280
-0.0237	0.0046	-5.1937	0.0000	ICQImagination	CUnions	Other Countries	TRUE	5.1937	0.0237
0.0298	0.0052	5.7713	0.0000	ILReligion	CUnions	Other Countries	TRUE	5.7713	0.0298
0.0174	0.0052	3.3723	0.0007	ACTReligion	CUnions	Other Countries	TRUE	3.3723	0.0174
-0.0273	0.0052	-5.2072	0.0000	ACTEnvOrg	CUnions	Other Countries	TRUE	5.2072	0.0273
0.0152	0.0051	2.9677	0.0030	ACTSport	CUnions	Other Countries	TRUE	2.9677	0.0152
0.0159	0.0045	3.5474	0.0004	PolExperts	CUnions	Other Countries	TRUE	3.5474	0.0159

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
- 0.0126	0.0045	- 2.7827	0.0054	ICQDetermination	CU-nions	Other Countries	TRUE	2.7827	0.0126
0.0138	0.0045	3.0420	0.0024	ILFam	CU-nions	Other Countries	TRUE	3.0420	0.0138
- 0.0115	0.0047	- 2.4532	0.0142	FutureRespect	CU-nions	Other Countries	TRUE	2.4532	0.0115
0.0110	0.0053	2.0699	0.0385	AC-TArtsEd	CU-nions	Other Countries	TRUE	2.0699	0.0110
0.0087	0.0045	1.9468	0.0516	IC-QResonsibility	CU-nions	Other Countries	FALSE	1.9468	0.0087
0.0172	0.0045	3.8532	0.0001	ILFam	CParliament	Other Countries	TRUE	3.8532	0.0172
0.0092	0.0048	1.9305	0.0535	ILReligion	CParliament	Other Countries	FALSE	1.9305	0.0092
0.0096	0.0045	2.1567	0.0310	IC-QResonsibility	CParliament	Other Countries	TRUE	2.1567	0.0096
- 0.0271	0.0045	- 6.0378	0.0000	ICQImagination	CParliament	Other Countries	TRUE	6.0378	0.0271
- 0.0299	0.0044	- 6.7568	0.0000	IC-QThrift	CParliament	Other Countries	TRUE	6.7568	0.0299
- 0.0200	0.0045	- 4.4659	0.0000	ICQDetermination	CParliament	Other Countries	TRUE	4.4659	0.0200
0.0261	0.0046	5.7173	0.0000	ICQObedience	CParliament	Other Countries	TRUE	5.7173	0.0261
- 0.0266	0.0049	- 5.4244	0.0000	ACTU-nions	CParliament	Other Countries	TRUE	5.4244	0.0266

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
- 0.0513	0.0051	- 10.1167	0.0000	ACTPol-Party	CParliament	Other Countries	TRUE	10.1167	0.0513
- 0.0328	0.0051	- 6.4477	0.0000	ACTEnvOrg	CParliament	Other Countries	TRUE	6.4477	0.0328
0.0220	0.0050	4.3681	0.0000	ACTProfessional	CParliament	Other Countries	TRUE	4.3681	0.0220
- 0.0877	0.0044	- 19.7582	0.0000	LifeSatis	CParliament	Other Countries	TRUE	19.7582	0.0877
0.0301	0.0045	6.6611	0.0000	FutureTech	CParliament	Other Countries	TRUE	6.6611	0.0301
0.0342	0.0046	7.3715	0.0000	FutureRespect	CParliament	Other Countries	TRUE	7.3715	0.0342
0.1697	0.0047	36.2409	0.0000	PolInterest	CParliament	Other Countries	TRUE	36.2409	0.1697
- 0.0735	0.0056	- 13.0705	0.0000	PolPetition	CParliament	Other Countries	TRUE	13.0705	0.0735
0.0181	0.0056	3.2044	0.0014	PolBoycott	CParliament	Other Countries	TRUE	3.2044	0.0181
- 0.0434	0.0055	- 7.9216	0.0000	PolDemons	CParliament	Other Countries	TRUE	7.9216	0.0434
- 0.0351	0.0045	- 7.8671	0.0000	PrivateS-tate	CParliament	Other Countries	TRUE	7.8671	0.0351
0.0634	0.0045	14.1954	0.0000	Hard-Work	CParliament	Other Countries	TRUE	14.1954	0.0634
- 0.0395	0.0044	- 8.9343	0.0000	PolExperts	CParliament	Other Countries	TRUE	8.9343	0.0395

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
0.0207	0.0045	4.6115	0.0000	ILFam	CCivilService	Other Countries	TRUE	4.6115	0.0207
0.0287	0.0048	5.9665	0.0000	ILReligion	CCivilService	Other Countries	TRUE	5.9665	0.0287
0.0122	0.0045	2.7306	0.0063	IC-QResonsibility	CCivilService	Other Countries	TRUE	2.7306	0.0122
- 0.0227	0.0044	- 5.1045	0.0000	IC-QThrift	CCivilService	Other Countries	TRUE	5.1045	0.0227
- 0.0126	0.0045	- 2.8016	0.0051	ICQDetermination	CCivilService	Other Countries	TRUE	2.8016	0.0126
0.0342	0.0046	7.4508	0.0000	ICQObedience	CCivilService	Other Countries	TRUE	7.4508	0.0342
- 0.0312	0.0049	- 6.4035	0.0000	ACTUnions	CCivilService	Other Countries	TRUE	6.4035	0.0312
- 0.0146	0.0051	- 2.8576	0.0043	ACTPolParty	CCivilService	Other Countries	TRUE	2.8576	0.0146
- 0.0349	0.0053	- 6.6226	0.0000	ACTEnvOrg	CCivilService	Other Countries	TRUE	6.6226	0.0349
- 0.0104	0.0051	- 2.0139	0.0440	ACTHumanitarian	CCivilService	Other Countries	TRUE	2.0139	0.0104
- 0.0674	0.0045	- 15.0981	0.0000	LifeSatis	CCivilService	Other Countries	TRUE	15.0981	0.0674
0.0366	0.0045	8.0484	0.0000	FutureTech	CCivilService	Other Countries	TRUE	8.0484	0.0366
0.0194	0.0046	4.1641	0.0000	FutureRespect	CCivilService	Other Countries	TRUE	4.1641	0.0194

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
0.1373	0.0047	29.1567	0.0000	PolInterest	CCivilService	Other Countries	TRUE	29.1567	0.1373
- 0.0444	0.0057	- 7.8450	0.0000	PolPetition	CCivilService	Other Countries	TRUE	7.8450	0.0444
0.0194	0.0057	3.4238	0.0006	PolBoycott	CCivilService	Other Countries	TRUE	3.4238	0.0194
- 0.0455	0.0055	- 8.2545	0.0000	PolDemos	CCivilService	Other Countries	TRUE	8.2545	0.0455
- 0.0381	0.0045	- 8.4811	0.0000	PrivateState	CCivilService	Other Countries	TRUE	8.4811	0.0381
0.0576	0.0045	12.8241	0.0000	HardWork	CCivilService	Other Countries	TRUE	12.8241	0.0576
- 0.0083	0.0044	- 1.8591	0.0630	PolExperts	CCivilService	Other Countries	FALSE	1.8591	0.0083
0.0251	0.0045	5.5480	0.0000	ILFam	CMajComp	Other Countries	TRUE	5.5480	0.0251
0.0382	0.0051	7.4520	0.0000	ILReligion	CMajComp	Other Countries	TRUE	7.4520	0.0382
- 0.0102	0.0045	- 2.2833	0.0224	ICQThrift	CMajComp	Other Countries	TRUE	2.2833	0.0102
- 0.0103	0.0045	- 2.2939	0.0218	ICQDetermination	CMajComp	Other Countries	TRUE	2.2939	0.0103
- 0.0422	0.0052	- 8.1090	0.0000	ACTReligion	CMajComp	Other Countries	TRUE	8.1090	0.0422
- 0.0248	0.0051	- 4.8534	0.0000	ACTSport	CMajComp	Other Countries	TRUE	4.8534	0.0248

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
0.0115	0.0053	2.1456	0.0319	ACTArtsEd	CMaj-Comp	Other Countries	TRUE	2.1456	0.0115
- 0.0170	0.0050	- 3.3920	0.0007	ACTUnions	CMaj-Comp	Other Countries	TRUE	3.3920	0.0170
- 0.0183	0.0052	- 3.5182	0.0004	ACTPolParty	CMaj-Comp	Other Countries	TRUE	3.5182	0.0183
- 0.0236	0.0055	- 4.2754	0.0000	ACTEnvOrg	CMaj-Comp	Other Countries	TRUE	4.2754	0.0236
- 0.0154	0.0053	- 2.9033	0.0037	ACTProfessional	CMaj-Comp	Other Countries	TRUE	2.9033	0.0154
0.0121	0.0054	2.2481	0.0246	ACTHumanitarian	CMaj-Comp	Other Countries	TRUE	2.2481	0.0121
- 0.0809	0.0045	- 17.9456	0.0000	LifeSatis	CMaj-Comp	Other Countries	TRUE	17.9456	0.0809
0.0597	0.0046	13.0438	0.0000	FutureTech	CMaj-Comp	Other Countries	TRUE	13.0438	0.0597
0.0516	0.0047	11.0595	0.0000	FutureRespect	CMaj-Comp	Other Countries	TRUE	11.0595	0.0516
0.0656	0.0047	13.8501	0.0000	PolInterest	CMaj-Comp	Other Countries	TRUE	13.8501	0.0656
- 0.0353	0.0057	- 6.1700	0.0000	PolPetition	CMaj-Comp	Other Countries	TRUE	6.1700	0.0353
- 0.0126	0.0057	- 2.2034	0.0276	PolBoycott	CMaj-Comp	Other Countries	TRUE	2.2034	0.0126
- 0.0330	0.0056	- 5.9365	0.0000	PolDemons	CMaj-Comp	Other Countries	TRUE	5.9365	0.0330

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
0.0323	0.0045	7.1495	0.0000	PrivateS-tate	CMaj-Comp	Other Countries	TRUE	7.1495	0.0323
0.0585	0.0045	12.9456	0.0000	Hard-Work	CMaj-Comp	Other Countries	TRUE	12.9456	0.0585
0.0346	0.0045	7.7481	0.0000	PolExperts	CMaj-Comp	Other Countries	TRUE	7.7481	0.0346
- 0.0952	0.0055	- 17.4571	0.0000	ACTEnvOrg	CEnvProt	Other Countries	TRUE	17.4571	0.0952
- 0.0919	0.0045	- 20.2902	0.0000	LifeSatis	CEnvProt	Other Countries	TRUE	20.2902	0.0919
0.0552	0.0046	11.9451	0.0000	FutureTech	CEnvProt	Other Countries	TRUE	11.9451	0.0552
0.0606	0.0056	10.8233	0.0000	PolDemons	CEnvProt	Other Countries	TRUE	10.8233	0.0606
0.0407	0.0048	8.5221	0.0000	PolInterest	CEnvProt	Other Countries	TRUE	8.5221	0.0407
0.0249	0.0046	5.4557	0.0000	ILFam	CEnvProt	Other Countries	TRUE	5.4557	0.0249
0.0267	0.0045	5.9183	0.0000	PolExperts	CEnvProt	Other Countries	TRUE	5.9183	0.0267
0.0299	0.0046	6.5515	0.0000	Hard-Work	CEnvProt	Other Countries	TRUE	6.5515	0.0299
- 0.0290	0.0046	- 6.3513	0.0000	PrivateS-tate	CEnvProt	Other Countries	TRUE	6.3513	0.0290
- 0.0306	0.0046	- 6.6791	0.0000	ICQImagination	CEnvProt	Other Countries	TRUE	6.6791	0.0306

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
0.0260	0.0047	5.5044	0.0000	FutureRespect	CEnvProt	Other Countries	TRUE	5.5044	0.0260
- 0.0313	0.0053	- 5.8608	0.0000	ACTHumanitarian	CEnvProt	Other Countries	TRUE	5.8608	0.0313
0.0254	0.0053	4.8355	0.0000	ACTProfessional	CEnvProt	Other Countries	TRUE	4.8355	0.0254
- 0.0192	0.0045	- 4.2369	0.0000	ICQResponsibility	CEnvProt	Other Countries	TRUE	4.2369	0.0192
0.0247	0.0052	4.7521	0.0000	ACTPolParty	CEnvProt	Other Countries	TRUE	4.7521	0.0247
0.0210	0.0049	4.2873	0.0000	ILReligion	CEnvProt	Other Countries	TRUE	4.2873	0.0210
0.0133	0.0047	2.8518	0.0043	ICQObedience	CEnvProt	Other Countries	TRUE	2.8518	0.0133
0.0137	0.0058	2.3883	0.0169	PolPetition	CEnvProt	Other Countries	TRUE	2.3883	0.0137
- 0.0112	0.0058	- 1.9508	0.0511	PolBoycott	CEnvProt	Other Countries	FALSE	1.9508	0.0112
- 0.0091	0.0050	- 1.8118	0.0700	ACTUnions	CEnvProt	Other Countries	FALSE	1.8118	0.0091
- 0.0077	0.0046	- 1.6954	0.0900	ICQDetermination	CEnvProt	Other Countries	FALSE	1.6954	0.0077
0.0199	0.0045	4.4071	0.0000	ILFam	CCourts	Other Countries	TRUE	4.4071	0.0199
0.0266	0.0052	5.1415	0.0000	ILReligion	CCourts	Other Countries	TRUE	5.1415	0.0266

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
-0.0184	0.0045	-4.0528	0.0001	ICQImagination	CCourts	Other Countries	TRUE	4.0528	0.0184
-0.0219	0.0045	-4.9029	0.0000	IC-QThrift	CCourts	Other Countries	TRUE	4.9029	0.0219
-0.0169	0.0045	-3.7307	0.0002	ICQDetermination	CCourts	Other Countries	TRUE	3.7307	0.0169
0.0169	0.0046	3.6850	0.0002	ICQObedience	CCourts	Other Countries	TRUE	3.6850	0.0169
0.0174	0.0052	3.3480	0.0008	ACTReligion	CCourts	Other Countries	TRUE	3.3480	0.0174
-0.0079	0.0049	-1.6099	0.1074	ACT-Sport	CCourts	Other Countries	FALSE	1.6099	0.0079
-0.0180	0.0049	-3.6462	0.0003	ACTUnions	CCourts	Other Countries	TRUE	3.6462	0.0180
-0.0142	0.0052	-2.7499	0.0060	ACTPol-Party	CCourts	Other Countries	TRUE	2.7499	0.0142
-0.0202	0.0054	-3.7546	0.0002	ACTEnvOrg	CCourts	Other Countries	TRUE	3.7546	0.0202
-0.0089	0.0053	-1.6836	0.0923	ACTHumanitarian	CCourts	Other Countries	FALSE	1.6836	0.0089
-0.0893	0.0045	-19.7828	0.0000	LifeSatis	CCourts	Other Countries	TRUE	19.7828	0.0893
0.0345	0.0046	7.5219	0.0000	FutureTech	CCourts	Other Countries	TRUE	7.5219	0.0345
0.0137	0.0047	2.9221	0.0035	FutureRespect	CCourts	Other Countries	TRUE	2.9221	0.0137

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
0.1181	0.0047	24.9313	0.0000	PolInterest	CCourts	Other Countries	TRUE	24.9313	0.1181
-0.0214	0.0057	-3.7472	0.0002	PolPetition	CCourts	Other Countries	TRUE	3.7472	0.0214
0.0267	0.0057	4.6778	0.0000	PolBoycott	CCourts	Other Countries	TRUE	4.6778	0.0267
-0.0671	0.0055	-12.1293	0.0000	PolDemos	CCourts	Other Countries	TRUE	12.1293	0.0671
-0.0218	0.0045	-4.8216	0.0000	PrivateState	CCourts	Other Countries	TRUE	4.8216	0.0218
0.0732	0.0045	16.1786	0.0000	HardWork	CCourts	Other Countries	TRUE	16.1786	0.0732
-0.0153	0.0045	-3.4177	0.0006	PolExperts	CCourts	Other Countries	TRUE	3.4177	0.0153

```
# BEST PREDICTORS FOR OTHER COUNTRIES
best_predictors_other <- coef_other %>%
  dplyr::filter(Significant) %>%
  dplyr::group_by(Outcome) %>%
  dplyr::slice_max(order_by = Abs_t, n = 1, with_ties = FALSE) %>%
  dplyr::ungroup()

knitr::kable(
  best_predictors_other,
  digits = 4,
  caption = "Strongest predictor of each confidence outcome in other countries"
)
```

Table 11: Strongest predictor of each confidence outcome in other countries

Estimate	Std. Error	t value	Pr(> t)	Variable	Outcome	Group	Significant	Abs_t	Abs_Beta
0.1272	0.0050	25.4297	0	ILReligion	CArmed-Forces	Other Countries	TRUE	25.4297	0.1272
0.1373	0.0047	29.1567	0	PolInterest	CCivilService	Other Countries	TRUE	29.1567	0.1373
0.1181	0.0047	24.9313	0	PolInterest	CCourts	Other Countries	TRUE	24.9313	0.1181
-	0.0045	-	0	Life-Satis	CEnvProt	Other Countries	TRUE	20.2902	0.0919
0.0919		20.2902							
-	0.0045	-	0	Life-Satis	CMajComp	Other Countries	TRUE	17.9456	0.0809
0.0809		17.9456							
0.1697	0.0047	36.2409	0	PolInterest	CParliament	Other Countries	TRUE	36.2409	0.1697
0.1085	0.0047	23.2155	0	PolInterest	CPress	Other Countries	TRUE	23.2155	0.1085
-	0.0049	-	0	ACTUnions	CU-nions	Other Countries	TRUE	33.6780	0.1660
0.1660		33.6780							

```
write.csv(best_predictors_other, "q2c_strongest_predictor.csv", row.names = FALSE)
```

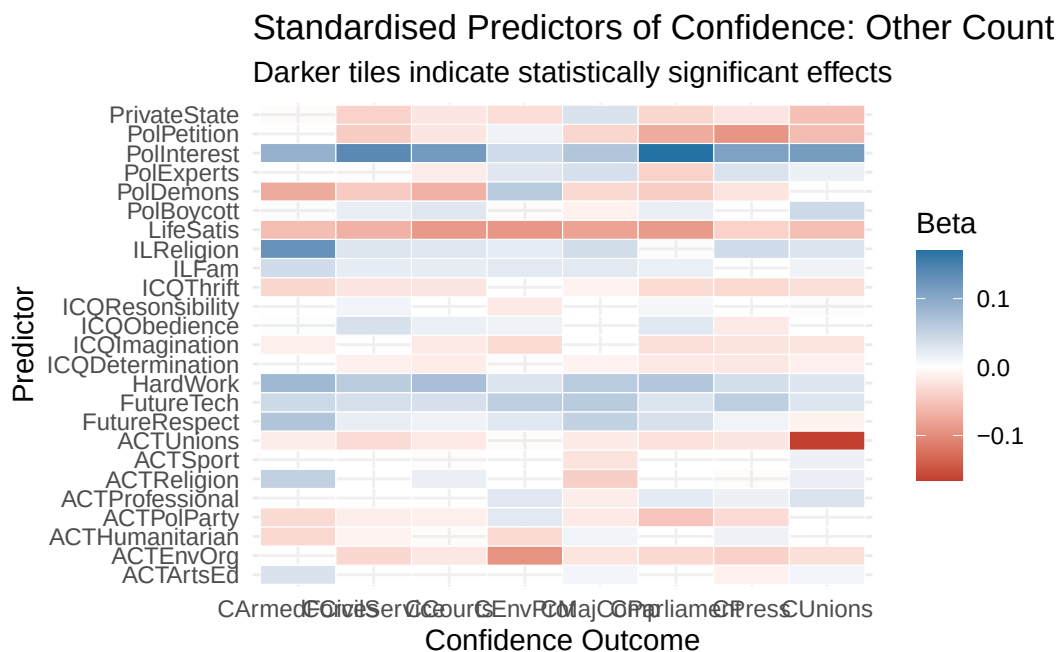
```
# HEATMAP FOR OTHER COUNTRIES
heatmap_other <- coef_other %>%
  dplyr::select(Outcome, Variable, Estimate, Significant) %>%
  tidyr::complete(
    Outcome,
    Variable,
    fill = list(Estimate = 0, Significant = FALSE)
  )

ggplot(heatmap_other,
  aes(x = Outcome, y = Variable,
    fill = Estimate, alpha = Significant)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(
    low = "#C0392B",
    mid = "white",
    high = "#2471A3",
    midpoint = 0
  ) +
```

```

scale_alpha_manual(
  values = c("TRUE" = 1, "FALSE" = 0.25),
  guide = "none"
) +
labs(
  title = "Standardised Predictors of Confidence: Other Countries",
  subtitle = "Darker tiles indicate statistically significant effects",
  x = "Confidence Outcome",
  y = "Predictor",
  fill = "Beta"
) +
theme_minimal()

```



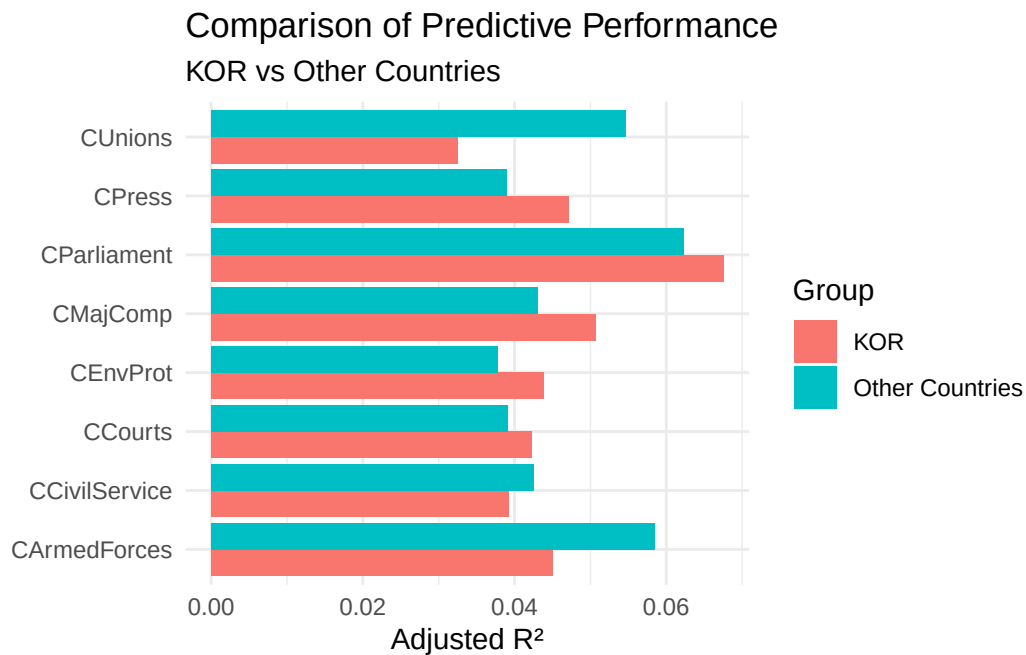
```

compare_perf <- bind_rows(
  model_perf_focus %>% mutate(Group = focus_country),
  model_perf_other %>% mutate(Group = "Other Countries")
)

ggplot(compare_perf,
  aes(x = Outcome, y = Best_Adj_R2, fill = Group)) +
  geom_col(position = "dodge") +

```

```
coord_flip() +
labs(
  title = "Comparison of Predictive Performance",
  subtitle = paste(focus_country, "vs Other Countries"),
  x = NULL,
  y = "Adjusted R2"
) +
theme_minimal()
```



```
# BEST PREDICTORS
compare_predictors <- best_predictors_focus %>%
  dplyr::select(Outcome, Focus_Predictor = Variable) %>%
  left_join(
    best_predictors_other %>%
      dplyr::select(Outcome, Other_Predictor = Variable),
    by = "Outcome"
  )

knitr::kable(
  compare_predictors,
  caption = "Comparison of strongest predictors: Focus Country vs Other Countries"
)
```

Table 12: Comparison of strongest predictors: Focus Country vs Other Countries

Outcome	Focus_Predictor	Other_Predictor
CArmedForces	FutureRespect	ILReligion
CCivilService	ICQThrift	PolInterest
CCourts	FutureRespect	PolInterest
CEnvProt	HardWork	LifeSatis
CMajComp	ILReligion	LifeSatis
CParliament	FutureRespect	PolInterest
CPress	HardWork	PolInterest
CUnions	PrivateState	ACTUnions

```
write.csv(compare_predictors, "q2c_strongest_focus_others.csv", row.names = FALSE)
```

Appendix 3

Q3a

```
time_var <- "Wave"

VC_focus <- VC_focus %>%
  mutate(Wave = as.numeric(as.character(Wave)))

VC_other <- VC_other %>%
  mutate(Wave = as.numeric(as.character(Wave)))

# Create combined dataset
combined <- dplyr::bind_rows(
  VC_focus %>% dplyr::mutate(Group = focus_country),
  VC_other %>% dplyr::mutate(Group = "Others")
)

# Correlation with Time (Wave) (Selection rule: |r| >= 0.05)
all_vars <- intersect(names(VC_focus), names(VC_other))
all_vars <- setdiff(all_vars, c("Country", "Wave", "Year"))

all_vars <- all_vars[
  sapply(VC_focus[, all_vars, drop = FALSE], is.numeric) &
```

```

  supply(VC_other[, all_vars, drop = FALSE], is.numeric)
]

all_vars <- as.character(all_vars)

compute_time_cor <- function(data, vars, time_col) {
  supply(vars, function(v) {
    if (!(v %in% names(data))) return(NA_real_)
    if (!(time_col %in% names(data))) return(NA_real_)

    d <- data[!is.na(data[[v]]) & !is.na(data[[time_col]]), , drop = FALSE]

    if (nrow(d) < 10) return(NA_real_)
    if (!is.numeric(d[[v]]) || !is.numeric(d[[time_col]])) return(NA_real_)

    cor(d[[time_col]], d[[v]], method = "pearson")
  })
}

focus_time_cor <- compute_time_cor(VC_focus, all_vars, time_var)
others_time_cor <- compute_time_cor(VC_other, all_vars, time_var)

time_cor_df <- data.frame(
  Variable      = all_vars,
  Focus_r       = round(focus_time_cor, 3),
  Others_r      = round(others_time_cor, 3),
  Focus_select  = abs(focus_time_cor) >= 0.05,
  Others_select = abs(others_time_cor) >= 0.05
)

cat("\n=== Correlation screening with Wave ===\n")

```

=== Correlation screening with Wave ===

```
print(time_cor_df, row.names = FALSE)
```

Variable	Focus_r	Others_r	Focus_select	Others_select
ILFam	0.031	-0.032	FALSE	FALSE
ILReligion	0.132	0.016	TRUE	FALSE
ICQResonsibility	0.145	0.029	TRUE	FALSE

ICQImagination	0.313	0.046	TRUE	FALSE
ICQThrift	0.058	0.004	TRUE	FALSE
ICQDetermination	0.170	0.027	TRUE	FALSE
ICQObedience	-0.094	-0.047	TRUE	FALSE
ACTReligion	0.040	0.006	FALSE	FALSE
ACTSport	-0.176	-0.023	TRUE	FALSE
ACTArtsEd	0.040	0.019	FALSE	FALSE
ACTUnions	0.000	-0.024	FALSE	FALSE
ACTPolParty	0.004	0.002	FALSE	FALSE
ACTEnvOrg	-0.050	0.026	FALSE	FALSE
ACTProfessional	-0.018	0.022	FALSE	FALSE
ACTHumanitarian	-0.097	0.032	TRUE	FALSE
LifeSatis	0.135	0.067	TRUE	TRUE
FutureTech	0.160	0.044	TRUE	FALSE
FutureRespect	-0.169	0.042	TRUE	FALSE
PolInterest	0.179	0.002	TRUE	FALSE
PolPetition	0.116	0.069	TRUE	TRUE
PolBoycott	0.092	0.012	TRUE	FALSE
PolDemons	-0.052	0.022	TRUE	FALSE
PrivateState	0.130	0.048	TRUE	FALSE
HardWork	0.246	0.006	TRUE	FALSE
PolExperts	0.052	-0.016	TRUE	FALSE
CArmedForces	0.322	-0.082	TRUE	TRUE
CPress	0.125	0.056	TRUE	TRUE
CUnions	0.207	-0.017	TRUE	FALSE
CParliament	0.241	0.054	TRUE	TRUE
CCivilService	0.261	-0.013	TRUE	FALSE
CMajComp	-0.065	0.026	TRUE	FALSE
CEnvProt	0.168	0.030	TRUE	FALSE
CCourts	0.165	-0.016	TRUE	FALSE

```
selected_vars <- time_cor_df$Variable[
  time_cor_df$Focus_select | time_cor_df$Others_select
]

cat("\nVariables selected for further analysis:\n")
```

Variables selected for further analysis:

```
cat(paste(selected_vars, collapse = ", "), "\n")
```

ILReligion, ICQResonsibility, ICQImagination, ICQThrift, ICQDetermination, ICQObedience, ACTReligion, ACTSport, ACTArtsEd, ACTUnions, ACTPolParty, ACTEnvOrg, ACTProfessional, ACTHumanitarian, LifeSatis, FutureTech, FutureRespect, PolInterest, PolPetition, PolBoycott, PolDemons, PrivateState, HardWork

```
# Optional neat table
knitr::kable(
  time_cor_df,
  digits = 3,
  caption = "Screening variables by Pearson correlation with Wave"
)
```

Table 13: Screening variables by Pearson correlation with Wave

	Variable	Focus_r	Others_r	Focus_select	Others_select
ILFam	ILFam	0.031	-0.032	FALSE	FALSE
ILReligion	ILReligion	0.132	0.016	TRUE	FALSE
ICQResonsibility	ICQResonsibility	0.145	0.029	TRUE	FALSE
ICQImagination	ICQImagination	0.313	0.046	TRUE	FALSE
ICQThrift	ICQThrift	0.058	0.004	TRUE	FALSE
ICQDetermination	ICQDetermination	0.170	0.027	TRUE	FALSE
ICQObedience	ICQObedience	-0.094	-0.047	TRUE	FALSE
ACTReligion	ACTReligion	0.040	0.006	FALSE	FALSE
ACTSport	ACTSport	-0.176	-0.023	TRUE	FALSE
ACTArtsEd	ACTArtsEd	0.040	0.019	FALSE	FALSE
ACTUnions	ACTUnions	0.000	-0.024	FALSE	FALSE
ACTPolParty	ACTPolParty	0.004	0.002	FALSE	FALSE
ACTEnvOrg	ACTEnvOrg	-0.050	0.026	FALSE	FALSE
ACTProfessional	ACTProfessional	-0.018	0.022	FALSE	FALSE
ACTHumanitarian	ACTHumanitarian	-0.097	0.032	TRUE	FALSE
LifeSatis	LifeSatis	0.135	0.067	TRUE	TRUE
FutureTech	FutureTech	0.160	0.044	TRUE	FALSE
FutureRespect	FutureRespect	-0.169	0.042	TRUE	FALSE
PolInterest	PolInterest	0.179	0.002	TRUE	FALSE
PolPetition	PolPetition	0.116	0.069	TRUE	TRUE
PolBoycott	PolBoycott	0.092	0.012	TRUE	FALSE
PolDemons	PolDemons	-0.052	0.022	TRUE	FALSE
PrivateState	PrivateState	0.130	0.048	TRUE	FALSE
HardWork	HardWork	0.246	0.006	TRUE	FALSE

	Variable	Focus_r	Others_r	Focus_select	Others_select
PolExperts	PolExperts	0.052	-0.016	TRUE	FALSE
CArmedForces	CArmedForces	0.322	-0.082	TRUE	TRUE
CPress	CPress	0.125	0.056	TRUE	TRUE
CUnions	CUnions	0.207	-0.017	TRUE	FALSE
CParliament	CParliament	0.241	0.054	TRUE	TRUE
CCivilService	CCivilService	0.261	-0.013	TRUE	FALSE
CMajComp	CMajComp	-0.065	0.026	TRUE	FALSE
CEnvProt	CEnvProt	0.168	0.030	TRUE	FALSE
CCourts	CCourts	0.165	-0.016	TRUE	FALSE

```
write.csv(time_cor_df, "q3a_pearson_correlation_wave.csv", row.names = FALSE)
```

```
# Lollipop chart (Focus vs Others)
cor_long <- time_cor_df %>%
  dplyr::select(Variable, Focus_r, Others_r) %>%
  tidyr::pivot_longer(
    cols = c(Focus_r, Others_r),
    names_to = "Group",
    values_to = "Correlation"
  ) %>%
  dplyr::mutate(
    Group = ifelse(Group == "Focus_r", focus_country, "Others")
  )

p_lollipop <- ggplot(cor_long, aes(x = Correlation, y = Variable)) +

  # reference lines
  geom_vline(
    xintercept = c(-0.05, 0.05),
    linetype = "dashed",
    colour = "grey50",
    linewidth = 0.5
  ) +
  geom_vline(
    xintercept = 0,
    colour = "black",
    linewidth = 0.4
  ) +

  # connecting line (focus vs others)
```

```

geom_line(
  aes(group = Variable),
  colour = "grey70",
  linewidth = 0.7
) +

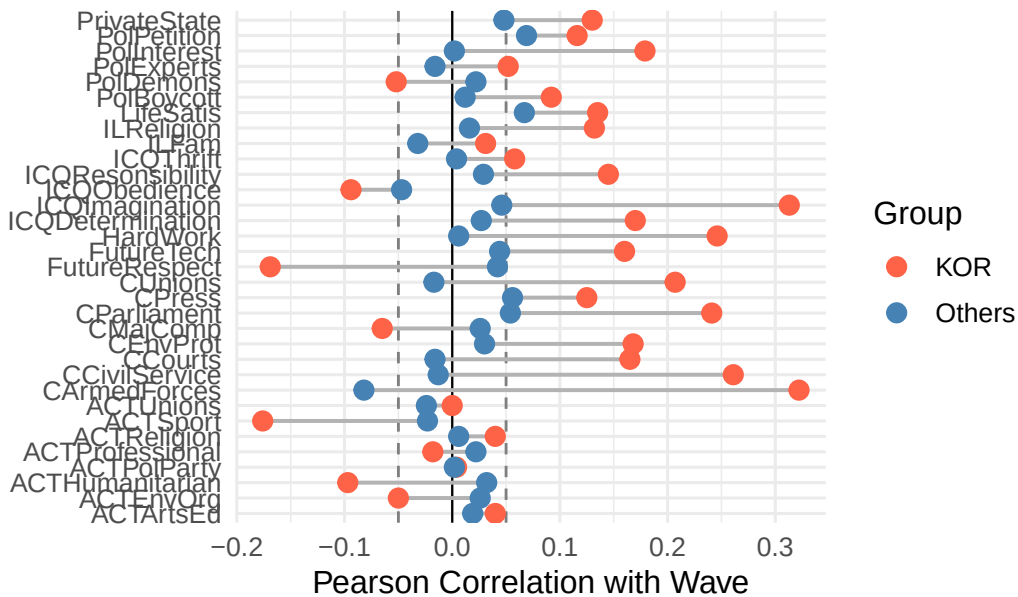
# points
geom_point(
  aes(colour = Group),
  size = 3
) +

theme_minimal(base_size = 12) +
labs(
  title = paste("Correlation with Wave:", focus_country, "vs Others"),
  x = "Pearson Correlation with Wave",
  y = NULL,
  colour = "Group"
) +
scale_colour_manual(
  values = setNames(
    c("steelblue", "tomato"),
    c("Others", focus_country)
  )
)

print(p_lollipop)

```

Correlation with Wave: KOR vs Others



```
# VISUALISE MEAN RESPONSE OVER TIME
selected_pred <- intersect(selected_vars, predictor_vars)
selected_conf <- intersect(selected_vars, outcome_vars)

# Top 6 predictors by strongest time correlation
top6_predictors <- time_cor_df %>%
  dplyr::filter(Variable %in% selected_pred) %>%
  dplyr::mutate(abs_r = pmax(abs(Focus_r), abs(Others_r), na.rm = TRUE)) %>%
  dplyr::arrange(desc(abs_r)) %>%
  dplyr::slice_head(n = 6) %>%
  dplyr::pull(Variable)

plot_line_trends <- function(vars_to_plot, title_suffix) {

  if (length(vars_to_plot) == 0) return(NULL)

  plot_data <- combined %>%
    dplyr::select(Group, Wave, all_of(vars_to_plot)) %>%
    tidyr::pivot_longer(
      ~c(Group, Wave),
      names_to = "Variable",
      values_to = "Value"
    ) %>%
```

```

dplyr::group_by(Group, Wave, Variable) %>%
dplyr::summarise(
  Mean = mean(Value, na.rm = TRUE),
  .groups = "drop"
)

ggplot(
  plot_data,
  aes(x = Wave, y = Mean, colour = Group, group = Group)
) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  facet_wrap(~Variable, scales = "free_y", ncol = 3) +
  theme_minimal(base_size = 11) +
  labs(
    title = paste("Mean Response by Time:", title_suffix),
    subtitle = paste(focus_country, "vs Others"),
    x = "Wave",
    y = "Mean Score",
    colour = "Group"
  ) +
  scale_colour_manual(
    values = setNames(
      c("steelblue", "tomato"),
      c("Others", focus_country)
    )
  )
}

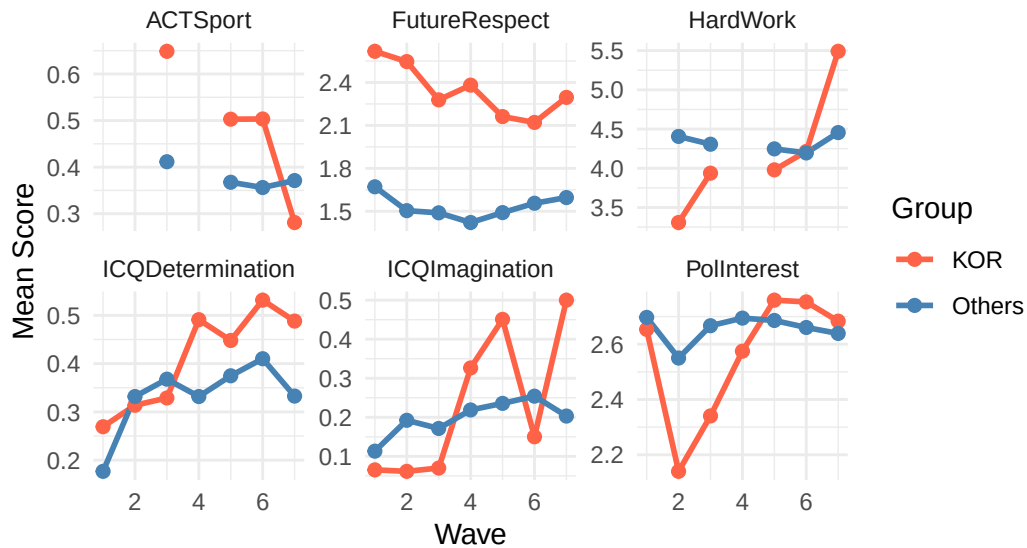
# First plot = top 6 predictors only
plot_line_trends(top6_predictors, "Top 6 Predictor Variables")

```

Warning: Removed 6 rows containing missing values or values outside the scale range (``geom_line()``).

Warning: Removed 10 rows containing missing values or values outside the scale range (``geom_point()``).

Mean Response by Time: Top 6 Predictor Variables KOR vs Others

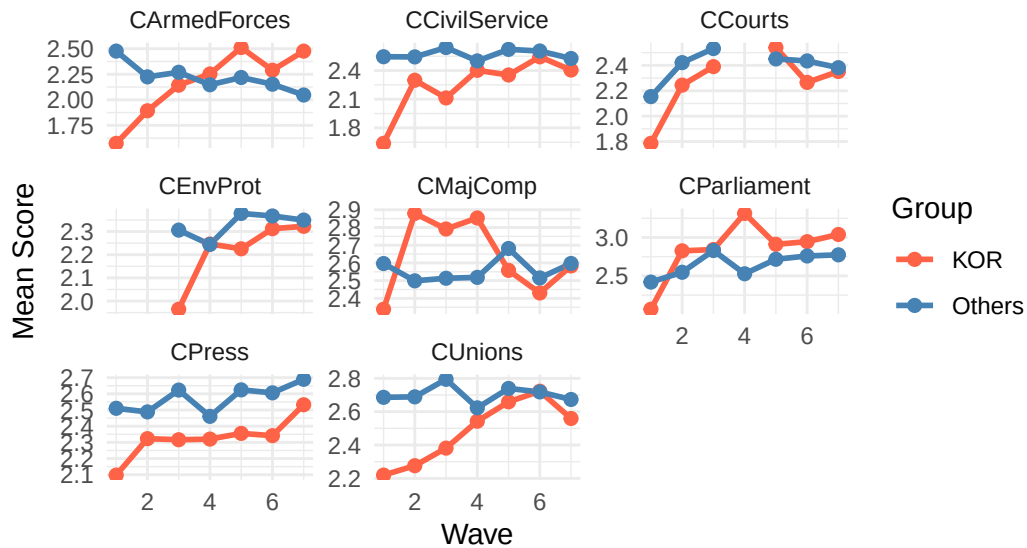


```
# Second plot = all selected confidence variables
plot_line_trends(selected_conf, "Confidence Variables")
```

Warning: Removed 4 rows containing missing values or values outside the scale range (``geom_line()``).

Warning: Removed 6 rows containing missing values or values outside the scale range (``geom_point()``).

Mean Response by Time: Confidence Variables KOR vs Others



```
# TEST CHANGE OVER TIME WITHIN EACH GROUP
# If residual normality looks acceptable -> ANOVA
# Else -> Kruskal-Wallis
run_method1_test <- function(data, vars, group_name) {

  results <- lapply(vars, function(v) {

    d <- data %>%
      dplyr::select(Wave, all_of(v)) %>%
      filter(!is.na(Wave), !is.na(.data[[v]]))

    # Must have enough observations and at least 2 waves
    if (nrow(d) < 10 || n_distinct(d$Wave) < 2) {
      return(data.frame(
        Group = group_name,
        Variable = v,
        Test = NA,
        Statistic = NA,
        P_Value = NA,
        Significant = NA
      ))
    }
  })
}
```

```

d$WaveFactor <- factor(d$Wave)

# fit model for residual normality check
lm_fit <- try(lm(as.formula(paste(v, "~ WaveFactor")), data = d), silent = TRUE)

use_anova <- FALSE

if (!inherits(lm_fit, "try-error")) {
  res <- residuals(lm_fit)

  # Shapiro only when feasible
  if (length(res) >= 3 && length(res) <= 5000) {
    shp <- try(shapiro.test(res), silent = TRUE)

    if (!inherits(shp, "try-error") && shp$p.value > 0.05) {
      use_anova <- TRUE
    }
  }
}

if (use_anova) {
  fit <- aov(as.formula(paste(v, "~ WaveFactor")), data = d)
  tab <- summary(fit)[[1]]

  data.frame(
    Group = group_name,
    Variable = v,
    Test = "ANOVA",
    Statistic = unname(tab$`F value`[1]),
    P_Value = unname(tab$`Pr(>F)`[1]),
    Significant = unname(tab$`Pr(>F)`[1]) < 0.05
  )
} else {
  kw <- kruskal.test(as.formula(paste(v, "~ WaveFactor")), data = d)

  data.frame(
    Group = group_name,
    Variable = v,
    Test = "Kruskal-Wallis",
    Statistic = unname(kw$statistic),
    P_Value = unname(kw$p.value),
  )
}

```

```

        Significant = unname(kw$p.value) < 0.05
    )
  }
})

bind_rows(results)
}

method1_focus <- run_method1_test(VC_focus, selected_vars, focus_country)
method1_other <- run_method1_test(VC_other, selected_vars, "Others")

method1_results <- bind_rows(method1_focus, method1_other) %>%
  arrange(Group, P_Value)

knitr::kable(
  method1_results,
  digits = 4,
  caption = "Method 1: ANOVA or Kruskal-Wallis across waves"
)

```

Table 14: Method 1: ANOVA or Kruskal-Wallis across waves

Group	Variable	Test	Statistic	P_Value	Significant
KOR	ICQImagination	Kruskal-Wallis	408.1506	0e+00	TRUE
KOR	CArmedForces	Kruskal-Wallis	323.2363	0e+00	TRUE
KOR	CParliament	Kruskal-Wallis	323.0574	0e+00	TRUE
KOR	CCivilService	Kruskal-Wallis	276.5679	0e+00	TRUE
KOR	ICQResonsibility	Kruskal-Wallis	208.5895	0e+00	TRUE
KOR	PolInterest	Kruskal-Wallis	189.2874	0e+00	TRUE
KOR	ACTHumanitarian	Kruskal-Wallis	177.6530	0e+00	TRUE
KOR	HardWork	Kruskal-Wallis	168.7214	0e+00	TRUE
KOR	CCourts	Kruskal-Wallis	166.4641	0e+00	TRUE
KOR	ICQThrift	Kruskal-Wallis	158.8688	0e+00	TRUE
KOR	FutureTech	Kruskal-Wallis	153.2802	0e+00	TRUE
KOR	CMajComp	Kruskal-Wallis	145.3579	0e+00	TRUE
KOR	PolPetition	Kruskal-Wallis	143.2054	0e+00	TRUE
KOR	CUnions	Kruskal-Wallis	127.0160	0e+00	TRUE
KOR	FutureRespect	Kruskal-Wallis	115.0674	0e+00	TRUE
KOR	PolBoycott	Kruskal-Wallis	112.2389	0e+00	TRUE
KOR	LifeSatis	Kruskal-Wallis	84.6412	0e+00	TRUE
KOR	ICQDetermination	Kruskal-Wallis	84.8152	0e+00	TRUE

Group	Variable	Test	Statistic	P_Value	Significant
KOR	CEnvProt	Kruskal-Wallis	67.5611	0e+00	TRUE
KOR	ACTSport	Kruskal-Wallis	57.7724	0e+00	TRUE
KOR	PrivateState	Kruskal-Wallis	58.5054	0e+00	TRUE
KOR	PolDemons	Kruskal-Wallis	58.0192	0e+00	TRUE
KOR	CPress	Kruskal-Wallis	54.0476	0e+00	TRUE
KOR	ILReligion	Kruskal-Wallis	48.3974	0e+00	TRUE
KOR	PolExperts	Kruskal-Wallis	27.6329	0e+00	TRUE
KOR	ICQObedience	Kruskal-Wallis	27.9618	1e-04	TRUE
Others	ILReligion	Kruskal-Wallis	2411.1290	0e+00	TRUE
Others	ICQObedience	Kruskal-Wallis	1293.1504	0e+00	TRUE
Others	CParliament	Kruskal-Wallis	1181.0147	0e+00	TRUE
Others	LifeSatis	Kruskal-Wallis	1126.1156	0e+00	TRUE
Others	ICQResonsibility	Kruskal-Wallis	1081.1543	0e+00	TRUE
Others	PolPetition	Kruskal-Wallis	1042.3947	0e+00	TRUE
Others	ICQThrift	Kruskal-Wallis	891.1705	0e+00	TRUE
Others	CArmedForces	Kruskal-Wallis	870.1613	0e+00	TRUE
Others	ICQDetermination	Kruskal-Wallis	688.7308	0e+00	TRUE
Others	FutureRespect	Kruskal-Wallis	648.4186	0e+00	TRUE
Others	CPress	Kruskal-Wallis	638.0968	0e+00	TRUE
Others	FutureTech	Kruskal-Wallis	625.6949	0e+00	TRUE
Others	ICQImagination	Kruskal-Wallis	587.9075	0e+00	TRUE
Others	PolDemons	Kruskal-Wallis	547.1602	0e+00	TRUE
Others	CCourts	Kruskal-Wallis	529.1389	0e+00	TRUE
Others	CMajComp	Kruskal-Wallis	524.0804	0e+00	TRUE
Others	CCivilService	Kruskal-Wallis	313.6259	0e+00	TRUE
Others	PolBoycott	Kruskal-Wallis	302.5310	0e+00	TRUE
Others	PrivateState	Kruskal-Wallis	290.1668	0e+00	TRUE
Others	CUnions	Kruskal-Wallis	269.0408	0e+00	TRUE
Others	ACTHumanitarian	Kruskal-Wallis	225.7875	0e+00	TRUE
Others	CEnvProt	Kruskal-Wallis	217.0532	0e+00	TRUE
Others	PolInterest	Kruskal-Wallis	119.7060	0e+00	TRUE
Others	ACTSport	Kruskal-Wallis	73.1465	0e+00	TRUE
Others	HardWork	Kruskal-Wallis	66.1831	0e+00	TRUE
Others	PolExperts	Kruskal-Wallis	37.6018	0e+00	TRUE

```
write.csv(method1_results, "q3a_anova_kruskal.csv", row.names = FALSE)
```

```
# Significant only
method1_sig <- method1_results %>%
```

```

filter(Significant == TRUE) %>%
  arrange(Group, P_Value)

knitr::kable(
  method1_sig,
  digits = 4,
  caption = "Significant attributes changing across waves"
)

```

Table 15: Significant attributes changing across waves

Group	Variable	Test	Statistic	P_Value	Significant
KOR	ICQImagination	Kruskal-Wallis	408.1506	0e+00	TRUE
KOR	CArmedForces	Kruskal-Wallis	323.2363	0e+00	TRUE
KOR	CParliament	Kruskal-Wallis	323.0574	0e+00	TRUE
KOR	CCivilService	Kruskal-Wallis	276.5679	0e+00	TRUE
KOR	ICQResonsibility	Kruskal-Wallis	208.5895	0e+00	TRUE
KOR	PolInterest	Kruskal-Wallis	189.2874	0e+00	TRUE
KOR	ACTHumanitarian	Kruskal-Wallis	177.6530	0e+00	TRUE
KOR	HardWork	Kruskal-Wallis	168.7214	0e+00	TRUE
KOR	CCourts	Kruskal-Wallis	166.4641	0e+00	TRUE
KOR	ICQThrift	Kruskal-Wallis	158.8688	0e+00	TRUE
KOR	FutureTech	Kruskal-Wallis	153.2802	0e+00	TRUE
KOR	CMajComp	Kruskal-Wallis	145.3579	0e+00	TRUE
KOR	PolPetition	Kruskal-Wallis	143.2054	0e+00	TRUE
KOR	CUnions	Kruskal-Wallis	127.0160	0e+00	TRUE
KOR	FutureRespect	Kruskal-Wallis	115.0674	0e+00	TRUE
KOR	PolBoycott	Kruskal-Wallis	112.2389	0e+00	TRUE
KOR	LifeSatis	Kruskal-Wallis	84.6412	0e+00	TRUE
KOR	ICQDetermination	Kruskal-Wallis	84.8152	0e+00	TRUE
KOR	CEnvProt	Kruskal-Wallis	67.5611	0e+00	TRUE
KOR	ACTSport	Kruskal-Wallis	57.7724	0e+00	TRUE
KOR	PrivateState	Kruskal-Wallis	58.5054	0e+00	TRUE
KOR	PolDemons	Kruskal-Wallis	58.0192	0e+00	TRUE
KOR	CPress	Kruskal-Wallis	54.0476	0e+00	TRUE
KOR	ILReligion	Kruskal-Wallis	48.3974	0e+00	TRUE
KOR	PolExperts	Kruskal-Wallis	27.6329	0e+00	TRUE
KOR	ICQObedience	Kruskal-Wallis	27.9618	1e-04	TRUE
Others	ILReligion	Kruskal-Wallis	2411.1290	0e+00	TRUE
Others	ICQObedience	Kruskal-Wallis	1293.1504	0e+00	TRUE
Others	CParliament	Kruskal-Wallis	1181.0147	0e+00	TRUE

Group	Variable	Test	Statistic	P_Value	Significant
Others	LifeSatis	Kruskal-Wallis	1126.1156	0e+00	TRUE
Others	ICQResonsibility	Kruskal-Wallis	1081.1543	0e+00	TRUE
Others	PolPetition	Kruskal-Wallis	1042.3947	0e+00	TRUE
Others	ICQThrift	Kruskal-Wallis	891.1705	0e+00	TRUE
Others	CArmedForces	Kruskal-Wallis	870.1613	0e+00	TRUE
Others	ICQDetermination	Kruskal-Wallis	688.7308	0e+00	TRUE
Others	FutureRespect	Kruskal-Wallis	648.4186	0e+00	TRUE
Others	CPress	Kruskal-Wallis	638.0968	0e+00	TRUE
Others	FutureTech	Kruskal-Wallis	625.6949	0e+00	TRUE
Others	ICQImagination	Kruskal-Wallis	587.9075	0e+00	TRUE
Others	PolDemons	Kruskal-Wallis	547.1602	0e+00	TRUE
Others	CCourts	Kruskal-Wallis	529.1389	0e+00	TRUE
Others	CMajComp	Kruskal-Wallis	524.0804	0e+00	TRUE
Others	CCivilService	Kruskal-Wallis	313.6259	0e+00	TRUE
Others	PolBoycott	Kruskal-Wallis	302.5310	0e+00	TRUE
Others	PrivateState	Kruskal-Wallis	290.1668	0e+00	TRUE
Others	CUnions	Kruskal-Wallis	269.0408	0e+00	TRUE
Others	ACTHumanitarian	Kruskal-Wallis	225.7875	0e+00	TRUE
Others	CEnvProt	Kruskal-Wallis	217.0532	0e+00	TRUE
Others	PolInterest	Kruskal-Wallis	119.7060	0e+00	TRUE
Others	ACTSport	Kruskal-Wallis	73.1465	0e+00	TRUE
Others	HardWork	Kruskal-Wallis	66.1831	0e+00	TRUE
Others	PolExperts	Kruskal-Wallis	37.6018	0e+00	TRUE

```
write.csv(method1_sig, "q3a_significant_changing.csv", row.names = FALSE)
```

```
# SUMMARY
get_direction_summary <- function(data, vars, group_name) {

  out <- lapply(vars, function(v) {
    d <- data %>%
      dplyr::select(Wave, all_of(v)) %>%
      filter(!is.na(Wave), !is.na(.data[[v]])) %>%
      group_by(Wave) %>%
      summarise(Mean = mean(.data[[v]], na.rm = TRUE), .groups = "drop") %>%
      arrange(Wave)

    if (nrow(d) < 2) {
      return(data.frame(
        Group = group_name,
```

```

    Variable = v,
    First_Wave_Mean = NA,
    Last_Wave_Mean = NA,
    Mean_Change = NA,
    Direction = NA
  ))
}

change <- d$Mean[nrow(d)] - d$Mean[1]

direction <- ifelse(
  abs(change) < 0.05, "Stable",
  ifelse(change > 0, "Increasing", "Decreasing")
)

data.frame(
  Group = group_name,
  Variable = v,
  First_Wave_Mean = d$Mean[1],
  Last_Wave_Mean = d$Mean[nrow(d)],
  Mean_Change = change,
  Direction = direction
)
})

bind_rows(out)
}

direction_focus <- get_direction_summary(VC_focus, selected_vars, focus_country)
direction_other <- get_direction_summary(VC_other, selected_vars, "Others")
direction_summary <- bind_rows(direction_focus, direction_other)

# INTERACTION REGRESSION:
# Test whether trend over time differs between focus country and Others

combined2 <- combined %>%
  mutate(Group_dummy = ifelse(Group == focus_country, 1, 0))

interaction_results <- combined2 %>%
  dplyr::select(Wave, Group_dummy, all_of(selected_vars)) %>%
  tidyr::pivot_longer(
    cols = all_of(selected_vars),

```

```

    names_to = "Variable",
    values_to = "Value"
  ) %>%
  tidyr::drop_na() %>%
  dplyr::group_by(Variable) %>%
  dplyr::group_modify(~{
    fit <- lm(Value ~ Wave + Group_dummy + Wave:Group_dummy, data = .x)
    s <- summary(fit)

    coef_tab <- as.data.frame(s$coefficients)
    coef_tab$term <- rownames(coef_tab)
    rownames(coef_tab) <- NULL

    names(coef_tab) <- c("estimate", "std.error", "statistic", "p.value", "term")

    coef_tab %>%
      dplyr::filter(term == "Wave:Group_dummy") %>%
      dplyr::transmute(
        Interaction_Effect = estimate,
        Interaction_P = p.value,
        Interaction_Significant = ifelse(p.value < 0.05, "Yes", "No"),
        Trend_Comparison = case_when(
          p.value >= 0.05 ~ "No significant difference in trend",
          estimate > 0 ~ paste(focus_country, "changes more positively over time"),
          estimate < 0 ~ paste(focus_country, "changes more negatively over time")
        )
      )
  }) %>%
  dplyr::ungroup()

knitr::kable(
  interaction_results,
  digits = 4,
  caption = "Interaction regression: does the trend over time differ between groups?"
)

```

Table 16: Interaction regression: does the trend over time differ between groups?

Variable	Interac- tion_Effect	Interac- tion_P	Interaction_Sig- nificant	Trend_Comparison
ACTHuman- itarian	-0.0332	0.0000	Yes	KOR changes more negatively over time
ACTSport	-0.0711	0.0000	Yes	KOR changes more negatively over time
CArmed- Forces	0.1780	0.0000	Yes	KOR changes more positively over time
CCivilSer- vice	0.1056	0.0000	Yes	KOR changes more positively over time
CCourts	0.0672	0.0000	Yes	KOR changes more positively over time
CEnvProt	0.0625	0.0000	Yes	KOR changes more positively over time
CMajComp	-0.0388	0.0000	Yes	KOR changes more negatively over time
CParliament	0.0726	0.0000	Yes	KOR changes more positively over time
CPress	0.0184	0.0476	Yes	KOR changes more positively over time
CUnions	0.0865	0.0000	Yes	KOR changes more positively over time
FutureRe- spect	-0.0841	0.0000	Yes	KOR changes more negatively over time
FutureTech	0.0322	0.0000	Yes	KOR changes more positively over time
HardWork	0.3235	0.0000	Yes	KOR changes more positively over time
ICQDetermi- nation	0.0353	0.0000	Yes	KOR changes more positively over time
ICQImagina- tion	0.0572	0.0000	Yes	KOR changes more positively over time
ICQObedi- ence	-0.0016	0.7504	No	No significant difference in trend
ICQResonsi- bility	0.0185	0.0002	Yes	KOR changes more positively over time
ICQThrift	0.0135	0.0090	Yes	KOR changes more positively over time
ILReligion	0.0650	0.0000	Yes	KOR changes more positively over time

Variable	Interaction_Effect	Interaction_P	Interaction_Significant	Trend_Comparison
LifeSatis	0.0361	0.1763	No	No significant difference in trend
PolBoycott	0.0249	0.0007	Yes	KOR changes more positively over time
PolDemons	-0.0281	0.0006	Yes	KOR changes more negatively over time
PolExperts	0.0382	0.0158	Yes	KOR changes more positively over time
PolInterest	0.0709	0.0000	Yes	KOR changes more positively over time
PolPetition	0.0115	0.2061	No	No significant difference in trend
PrivateState	0.0955	0.0101	Yes	KOR changes more positively over time

```
write.csv(interaction_results, "q3a_interaction_regression.csv", row.names = FALSE)
```

```
interaction_results_small <- interaction_results %>%
  dplyr::filter(Interaction_Significant == "Yes") %>%
  dplyr::arrange(desc(abs(Interaction_Effect))) %>%
  dplyr::slice_head(n = 10)

knitr::kable(
  interaction_results_small,
  digits = 3,
  caption = "Top 10 significant interaction effects (Korea vs Others over time)"
)
```

Table 17: Top 10 significant interaction effects (Korea vs Others over time)

Variable	Interaction_Effect	Interaction_P	Interaction_Significant	Trend_Comparison
HardWork	0.324	0.00	Yes	KOR changes more positively over time
CArmed-Forces	0.178	0.00	Yes	KOR changes more positively over time
CCivilService	0.106	0.00	Yes	KOR changes more positively over time

Variable	Interaction_Effect	Interaction_P	Interaction_Significant	Trend_Comparison
PrivateS-tate	0.096	0.01	Yes	KOR changes more positively over time
CUnions	0.086	0.00	Yes	KOR changes more positively over time
FutureRe-spect	-0.084	0.00	Yes	KOR changes more negatively over time
CParliament	0.073	0.00	Yes	KOR changes more positively over time
ACTSport	-0.071	0.00	Yes	KOR changes more negatively over time
PolInter-est	0.071	0.00	Yes	KOR changes more positively over time
CCourts	0.067	0.00	Yes	KOR changes more positively over time

```
write.csv(interaction_results_small, "q3a_top10_interaction_regression.csv", row.names = FALSE)
```

```
# FINAL SUMMARY
q3a_summary <- method1_results %>%
  left_join(direction_summary, by = c("Group", "Variable")) %>%
  left_join(interaction_results, by = "Variable") %>%
  arrange(Group, P_Value)

knitr::kable(
  q3a_summary,
  digits = 4,
  caption = "Q3(a) summary: within-group change and between-group trend comparison"
)
```

Table 18: Q3(a) summary: within-group change and between-group trend comparison

Group	Variable	Test	Statistic	P_Value	Sig-nif-i-	First_Wave_Mean	Second_Wave_Mean	Change	Inter- rec- tion_Ef- fect	In- ter- ac- tion_Pant	Inter- ac- tion_Sig- nifi-	Trend_Com- parison
KORC- QImag- ina- tion	Kruskal- Wallis	108.1566	0.0000	TRUE	0.0654	0.5000	0.4346	In- creas- ing	0.0572	0.0000	Yes	KOR changes more positively over time
KORC Armed Forces	Kruskal- Wallis	323.2363	0.0000	TRUE	1.5742	2.4763	0.9021	In- creas- ing	0.1780	0.0000	Yes	KOR changes more positively over time
KORC Par- lia- ment	Kruskal- Wallis	323.0574	0.0000	TRUE	2.0661	3.0385	0.9723	In- creas- ing	0.0726	0.0000	Yes	KOR changes more positively over time
KORC Civil vice	Kruskal- Wallis	276.5679	0.0000	TRUE	1.6381	2.4053	0.7672	In- creas- ing	0.1056	0.0000	Yes	KOR changes more positively over time
KORC- QReson- sibil- ity	Kruskal- Wallis	208.5895	0.0000	TRUE	0.5577	0.8580	0.3003	In- creas- ing	0.0185	0.0002	Yes	KOR changes more positively over time
KORC PolIn- ter- est	Kruskal- Wallis	189.2874	0.0000	TRUE	2.6537	2.6834	0.0297	Sta- ble	0.0709	0.0000	Yes	KOR changes more positively over time
KORC- THu- man- itar- ian	Kruskal- Wallis	177.6530	0.0000	TRUE	0.0615	0.0621	0.0006	Sta- ble	- 0.0332	0.0000	Yes	KOR changes more negatively over time

Group	Vari- able	Test	Statis- tic	Sig- nif- i- P_Value	First_Wave_Mean	Second_Wave_Mean	Change	Inter- ac- tion_Effect	In- ter- ac- tion_Pant	Inter- ac- tion_Sig-	Trend_Com- parison		
KOR	Hard- Work	Kruskal- Wallis	168.724	0.0000	TRUE	3.3086	5.4911	2.1826	In- creas- ing	0.3235	0.0000	Yes	KOR changes more positively over time
KOR	Court- Wallis	Kruskal- Wallis	166.464	0.0000	TRUE	1.7860	2.3521	0.5661	In- creas- ing	0.0672	0.0000	Yes	KOR changes more positively over time
KOR	HC- QThrift	Kruskal- Wallis	158.868	0.0000	TRUE	0.3269	0.4586	0.1317	In- creas- ing	0.0135	0.0090	Yes	KOR changes more positively over time
KOR	Fu- tureTech	Kruskal- Wallis	153.892	0.0000	TRUE	1.1892	1.5562	0.3670	In- creas- ing	0.0322	0.0000	Yes	KOR changes more positively over time
KOR	CMa- j- Comp	Kruskal- Wallis	145.357	0.0000	TRUE	2.3386	2.5828	0.2443	In- creas- ing	- 0.0388	0.0000	Yes	KOR changes more negatively over time
KOR	PolPe- ti- tion	Kruskal- Wallis	143.205	0.0000	TRUE	2.0588	2.1834	0.1246	In- creas- ing	0.0115	0.2061	No	No significant difference in trend
KOR	CU- nions	Kruskal- Wallis	127.066	0.0000	TRUE	2.2196	2.5592	0.3396	In- creas- ing	0.0865	0.0000	Yes	KOR changes more positively over time

Variable Group	Test	Statistic	P_Value	Sig-nif-i-cant	First_Wave_Mean	Second_Wave_Mean	Change	Inter-action_Effect	Inter-action_Effect_Pant	Inter-action_Sig-nif-icant	Trend_Comparison	
KORFutureRespect	Kruskal Wallis	15.0674	0.0000	TRUE	2.6178	2.2959	-0.3219	Decreasing	-0.0841	0.0000	Yes	KOR changes more negatively over time
KORPol-Boycott	Kruskal Wallis	12.2389	0.0000	TRUE	2.4667	2.4497	-0.0170	Stable	0.0249	0.0007	Yes	KOR changes more positively over time
KORLifeSatis	Kruskal Wallis	4.6402	0.0310	TRUE	5.4229	6.6982	1.2753	Increasing	0.0361	0.1763	No	No significant difference in trend
KORHC-QDetrimination	Kruskal Wallis	4.8152	0.0290	TRUE	0.2692	0.4882	0.2189	Increasing	0.0353	0.0000	Yes	KOR changes more positively over time
KORCEnvProt	Kruskal Wallis	7.5601	0.0060	TRUE	1.9648	2.3225	0.3577	Increasing	0.0625	0.0000	Yes	KOR changes more positively over time
KORACT-Sport	Kruskal Wallis	17.7724	0.0000	TRUE	0.6486	0.2811	-0.3676	Decreasing	-0.0711	0.0000	Yes	KOR changes more negatively over time
KORPrivateS-tate	Kruskal Wallis	18.5054	0.0000	TRUE	4.9026	5.6568	0.7542	Increasing	0.0955	0.0101	Yes	KOR changes more positively over time

Vari-Group	Test	Statis- tic	P_Val	Sig-nif- i-	First_Wave	Wave_Mean	Change	Inter-rec- tion_Ef- fect	In-ter- ac-tion_Pant	Inter-ac- tion_Sig-nifi-	Trend_Com- parison
KORPolDem-Kruska-Wallis	18.0192+00	TRUE	2.4975	2.3373	-	Decreasing	0.1602	-0.0281	0.0006	Yes	KOR changes more negatively over time
KORCPress-Kruska-Wallis	14.0476+00	TRUE	2.0977	2.5325	0.4349	Increasing	0.0184	0.0476	Yes	KOR changes more positively over time	
KORIL-Religion	Kruska-Wallis	18.3974+00	TRUE	2.2416	2.7219	0.4803	Increasing	0.0650	0.0000	Yes	KOR changes more positively over time
KORPol-Experts	Kruska-Wallis	27.6329+00	TRUE	2.2705	2.4349	0.1644	Increasing	0.0382	0.0158	Yes	KOR changes more positively over time
KORIC-QObedi-ence	Kruska-Wallis	27.9618-04	TRUE	0.1192	0.0651	-	Decreasing	0.0541	-0.0016	0.7504	No significant difference in trend
OthIL-ers Religion	Kruska-Wallis	2411.029000	TRUE	2.1641	2.0567	-	Decreasing	0.0650	0.0000	Yes	KOR changes more positively over time
OthIC-ers QObedi-ence	Kruska-Wallis	1293.059400	TRUE	0.3182	0.2814	-	Stable	-0.0016	0.7504	No	No significant difference in trend

Group	Vari- able	Test	Statis- tic	Sig- nif- i- P_Value	First_Wave	Wave_Win- dow	Mean_Change	Inter- rec- tion_Ef- fect	In- ter- ac- tion_Pant	Inter- ac- tion_Sig- nifi- cant	Trend_Com- parison
Oth-CPar- ers lia- ment		Kruskal- Wallis	1181.064700	TRUE	2.4191	2.7738	0.3547	In- creas- ing	0.0726	0.0000	Yes KOR changes more positively over time
Oth-Life- ers Satis		Kruskal- Wallis	1126.065600	TRUE	7.3289	7.1096	- 0.2193	De- creas- ing	0.0361	0.1763	No No significant difference in trend
Oth-IC- ers QReson- sibil- ity		Kruskal- Wallis	1081.054300	TRUE	0.4665	0.6575	0.1911	In- creas- ing	0.0185	0.0002	Yes KOR changes more positively over time
Oth-PolPe- ers ti- tion		Kruskal- Wallis	1042.304700	TRUE	1.9477	2.1341	0.1863	In- creas- ing	0.0115	0.2061	No No significant difference in trend
Oth-IC- ers QThrif		Kruskal- Wallis	891.170500	TRUE	0.2036	0.2937	0.0901	In- creas- ing	0.0135	0.0090	Yes KOR changes more positively over time
Oth-CArmed ers Forces		Kruskal- Wallis	870.166300	TRUE	2.4768	2.0470	- 0.4298	De- creas- ing	0.1780	0.0000	Yes KOR changes more positively over time
Oth-IC- ers QDe- ter- mi- na- tion		Kruskal- Wallis	688.730800	TRUE	0.1768	0.3327	0.1559	In- creas- ing	0.0353	0.0000	Yes KOR changes more positively over time

Group	Variable	Test	Statistic	P_Value	Signif- i- cant	First_Wave_Mean	Second_Wave_Mean	Change	Inter- rec- tion_Ef- fect	In- ter- ac- tion_Pant	Inter- ac- tion_Sig- nifi- cant	Trend_Com- parison
Oth-Fu- ers	tur- eRe- spect	Kruskal- Wallis	648.4086	0.0000	TRUE	1.6711	1.5959	-0.0752	Decreasing	-0.0841	0.0000	Yes KOR changes more negatively over time
Oth-CPress ers		Kruskal- Wallis	638.0068	0.0000	TRUE	2.5105	2.6897	0.1792	Increasing	0.0184	0.0476	Yes KOR changes more positively over time
Oth-Fu- ers	tureTech	Kruskal- Wallis	625.6049	0.0000	TRUE	1.4568	1.4097	-0.0471	Stable	0.0322	0.0000	Yes KOR changes more positively over time
Oth-IC- ers	QImag- ina- tion	Kruskal- Wallis	587.9075	0.0000	TRUE	0.1133	0.2032	0.0899	Increasing	0.0572	0.0000	Yes KOR changes more positively over time
Oth-PolDem ers		Kruskal- Wallis	547.1662	0.0000	TRUE	2.4337	2.3514	-0.0823	Decreasing	-0.0281	0.0006	Yes KOR changes more negatively over time
Oth-CCourt ers		Kruskal- Wallis	529.1889	0.0000	TRUE	2.1542	2.3818	0.2276	Increasing	0.0672	0.0000	Yes KOR changes more positively over time
Oth-CMa- ers	j- Comp	Kruskal- Wallis	524.0804	0.0000	TRUE	2.5963	2.5968	0.0005	Stable	-0.0388	0.0000	Yes KOR changes more negatively over time

Vari- Group	able	Test	Statis- tic	Sig- nif- i- P_Value	First_Wave_Mean	Second_Wave_Mean	Change	Inter- rec- tion_Ef- fect	In- ter- ac- tion_Pant	Inter- ac- tion_Sig- nifi- cant	Trend_Com- parison
Oth-CCiv- ers vice	Kruskal- Wallis	313.6259	0.0000	TRUE	2.5447	2.5264	-0.0183	Stable	0.1056	0.0000	Yes KOR changes more positively over time
Oth-Pol- ers Boy- cott	Kruskal- Wallis	302.5310	0.0000	TRUE	2.5852	2.5111	-0.0742	Decreasing	0.0249	0.0007	Yes KOR changes more positively over time
Oth-Pri- ers vat- eS- tate	Kruskal- Wallis	290.1668	0.0000	TRUE	5.2208	5.6060	0.3853	Increasing	0.0955	0.0101	Yes KOR changes more positively over time
Oth-CU- ers nions	Kruskal- Wallis	269.0408	0.0000	TRUE	2.6860	2.6731	-0.0129	Stable	0.0865	0.0000	Yes KOR changes more positively over time
Oth-AC- ers THU- man- itar- ian	Kruskal- Wallis	225.7875	0.0000	TRUE	0.1442	0.2792	0.1349	Increasing	-0.0332	0.0000	Yes KOR changes more negatively over time
Oth-CEN- ers vProt	Kruskal- Wallis	217.0532	0.0000	TRUE	2.3061	2.3493	0.0432	Stable	0.0625	0.0000	Yes KOR changes more positively over time
Oth-PolIn- ers ter- est	Kruskal- Wallis	119.7060	0.0000	TRUE	2.6970	2.6389	-0.0581	Decreasing	0.0709	0.0000	Yes KOR changes more positively over time

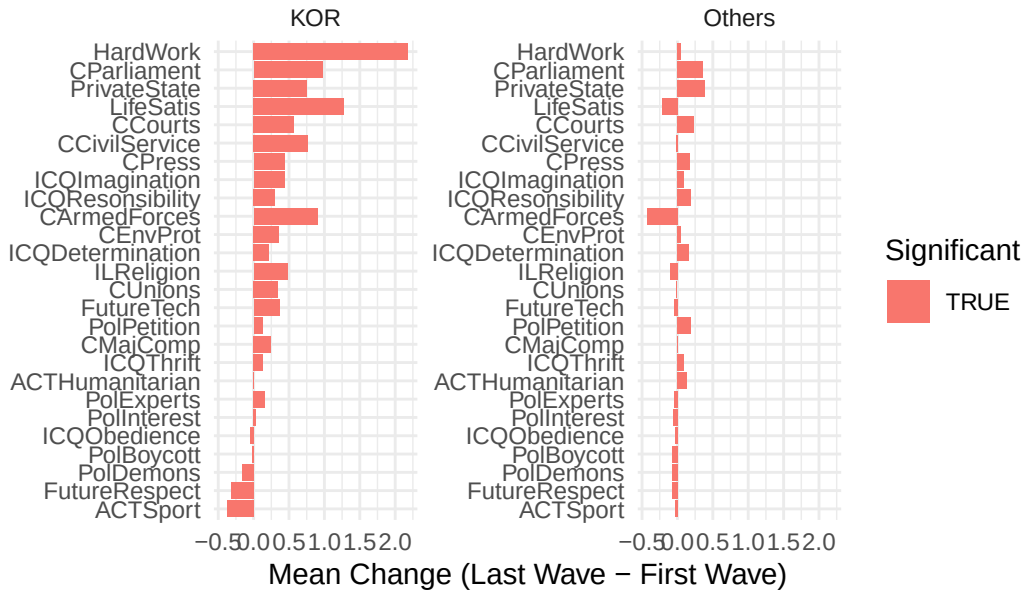
Group	Variable	Test	Statistic	P_Value	Signif- i- cant	First_Wave_Mean	Last_Wave_Mean	Mean_Change	Direction	Effect_Size	Interaction	Signif- icant	Trend_Com- parison
Oth- ers	ACT- Sport	Kruska- Wallis	73.1465	+0.0000	TRUE	0.4116	0.3711	-0.0404	Stable	-0.0711	0.0000	Yes	KOR changes more negatively over time
Oth- ers	Hard- Work	Kruska- Wallis	66.1834	+0.0000	TRUE	4.4061	4.4553	0.0492	Stable	0.3235	0.0000	Yes	KOR changes more positively over time
Oth- ers	Pol- Ex- perts	Kruska- Wallis	37.6008	+0.0000	TRUE	2.3678	2.3215	-0.0463	Stable	0.0382	0.0158	Yes	KOR changes more positively over time

```
write.csv(q3a_summary, "q3a_summary.csv", row.names = FALSE)
```

```
change_plot_data <- q3a_summary %>%
  filter(!is.na(Mean_Change)) %>%
  mutate(Variable = reorder(Variable, Mean_Change))

ggplot(change_plot_data,
  aes(x = Mean_Change, y = Variable, fill = Significant)) +
  geom_col() +
  facet_wrap(~ Group, scales = "free_y") +
  theme_minimal(base_size = 11) +
  labs(
    title = "Direction and Magnitude of Change",
    x = "Mean Change (Last Wave - First Wave)",
    y = NULL,
    fill = "Significant"
  )
```

Direction and Magnitude of Change



```
# BOXPLOT BY WAVE: Top 4 most changing variables
top4_vars <- direction_summary %>%
  filter(Group == focus_country) %>%
  mutate(abs_change = abs(Mean_Change)) %>%
  arrange(desc(abs_change)) %>%
  slice_head(n = 4) %>%
  pull(Variable)

box_data <- combined %>%
  dplyr::select(Group, Wave, all_of(top4_vars)) %>%
  tidyr::pivot_longer(
    ~c(Group, Wave),
    names_to = "Variable",
    values_to = "Value"
  )

p_box <- ggplot(box_data, aes(x = factor(Wave), y = Value, fill = Group)) +
  geom_boxplot(outlier.size = 0.3, alpha = 0.7) +
  facet_wrap(~Variable, scales = "free_y") +
  theme_minimal(base_size = 11) +
  labs(
    title = "Distribution by Wave: Top 4 Changing Variables",
    subtitle = paste(focus_country, "vs Others"),
  )
```

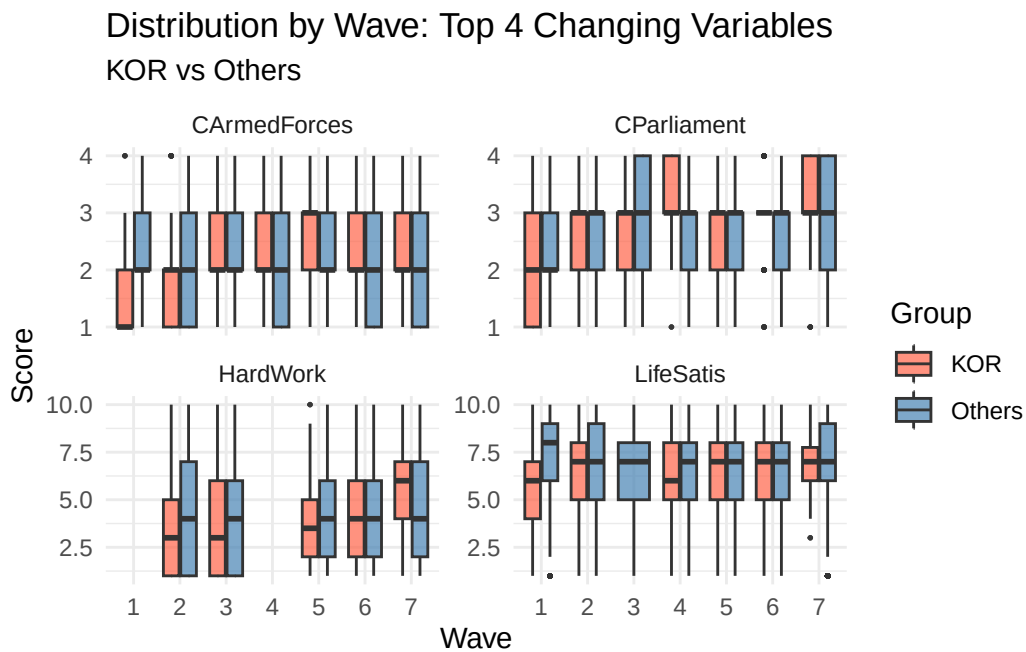
```

    x = "Wave",
    y = "Score",
    fill = "Group"
  ) +
  scale_fill_manual(
    values = setNames(
      c("steelblue", "tomato"),
      c("Others", focus_country)
    )
  )
)

print(p_box)

```

Warning: Removed 39928 rows containing non-finite outside the scale range (`stat\_boxplot()`).



Q3b

```

# For each wave and each outcome:
# fit outcome ~ all predictors
# record adj R2
# identify strongest predictor by largest absolute t value

fit_models_by_wave <- function(data, predictors, outcomes, min_n = 30) {

  results_perf <- list()
  results_coef <- list()

  waves <- sort(unique(data$Wave))

  for (w in waves) {
    data_w <- data %>% dplyr::filter(Wave == w)

    for (outcome in outcomes) {

      needed_vars <- c(outcome, predictors)
      needed_vars <- needed_vars[needed_vars %in% names(data_w)]

      d <- data_w %>%
        dplyr::select(all_of(needed_vars)) %>%
        tidyr::drop_na()

      if (nrow(d) < min_n) next

      formula_txt <- paste(outcome, "~", paste(predictors, collapse = " + "))
      fit <- try(lm(as.formula(formula_txt), data = d), silent = TRUE)

      if (inherits(fit, "try-error")) next

      s <- summary(fit)

      # ----- model performance -----
      fstat <- s$fstatistic
      model_p <- if (!is.null(fstat)) {
        pf(fstat[1], fstat[2], fstat[3], lower.tail = FALSE)
      } else {
        NA
      }

      results_perf[[length(results_perf) + 1]] <- tibble(

```

```

      Wave = w,
      Outcome = outcome,
      N = nrow(d),
      R2 = s$r.squared,
      Adj_R2 = s$adj.r.squared,
      Model_P = model_p
    )

    # ----- coefficient table -----
    coef_tab <- as.data.frame(s$coefficients)
    coef_tab$term <- rownames(coef_tab)
    rownames(coef_tab) <- NULL

    names(coef_tab) <- c("estimate", "std.error", "statistic", "p.value", "term")

    coef_tab <- coef_tab %>%
      dplyr::filter(term != "(Intercept)") %>%
      dplyr::mutate(
        Wave = w,
        Outcome = outcome,
        Abs_t = abs(statistic),
        Significant = p.value < 0.05
      )

    results_coef[[length(results_coef) + 1]] <- coef_tab
  }
}

list(
  performance = bind_rows(results_perf),
  coefficients = bind_rows(results_coef)
)
}

models_focus <- fit_models_by_wave(VC_focus, predictor_vars, outcome_vars, min_n = 30)
models_other <- fit_models_by_wave(VC_other, predictor_vars, outcome_vars, min_n = 30)

model_perf_focus <- models_focus$performance %>% mutate(Group = focus_country)
model_perf_other <- models_other$performance %>% mutate(Group = "Other")

coef_focus <- models_focus$coefficients %>% mutate(Group = focus_country)
coef_other <- models_other$coefficients %>% mutate(Group = "Other")

```

```
# Predictive ability over time

model_perf_all <- bind_rows(model_perf_focus, model_perf_other)

kable(model_perf_all, digits = 4,
      caption = "Model performance over time")
```

Table 19: Model performance over time

Wave	Outcome	N	R2	Adj_R2	Model_P	Group
5	CArmedForces	336	0.1214	0.0506	0.0199	KOR
5	CPress	336	0.1299	0.0597	0.0091	KOR
5	CUnions	336	0.0738	-0.0009	0.4836	KOR
5	CParliament	336	0.1283	0.0580	0.0106	KOR
5	CCivilService	335	0.1306	0.0602	0.0088	KOR
5	CMajComp	336	0.1242	0.0536	0.0154	KOR
5	CEnvProt	336	0.1170	0.0458	0.0294	KOR
5	CCourts	336	0.1392	0.0698	0.0036	KOR
6	CArmedForces	272	0.1576	0.0720	0.0105	KOR
6	CPress	272	0.1654	0.0805	0.0056	KOR
6	CUnions	270	0.1945	0.1119	0.0005	KOR
6	CParliament	271	0.1912	0.1087	0.0006	KOR
6	CCivilService	272	0.1423	0.0552	0.0328	KOR
6	CMajComp	271	0.1137	0.0233	0.1912	KOR
6	CEnvProt	272	0.1470	0.0603	0.0235	KOR
6	CCourts	272	0.1743	0.0904	0.0026	KOR
7	CArmedForces	338	0.1626	0.0955	0.0002	KOR
7	CPress	338	0.1462	0.0777	0.0016	KOR
7	CUnions	338	0.1714	0.1050	0.0001	KOR
7	CParliament	338	0.1804	0.1148	0.0000	KOR
7	CCivilService	338	0.1569	0.0893	0.0005	KOR
7	CMajComp	338	0.1815	0.1159	0.0000	KOR
7	CEnvProt	338	0.1396	0.0707	0.0031	KOR
7	CCourts	338	0.1891	0.1241	0.0000	KOR
3	CArmedForces	10139	0.0558	0.0535	0.0000	Other
3	CPress	10189	0.0479	0.0455	0.0000	Other
3	CUnions	9768	0.0587	0.0563	0.0000	Other
3	CParliament	10066	0.0544	0.0521	0.0000	Other
3	CCivilService	9979	0.0429	0.0405	0.0000	Other
3	CMajComp	9794	0.0557	0.0533	0.0000	Other
3	CEnvProt	9540	0.0371	0.0346	0.0000	Other

Wave	Outcome	N	R2	Adj_R2	Model_P	Group
3	CCourts	10175	0.0378	0.0354	0.0000	Other
5	CArmedForces	9415	0.0881	0.0857	0.0000	Other
5	CPress	10231	0.0636	0.0613	0.0000	Other
5	CUnions	9774	0.0677	0.0654	0.0000	Other
5	CParliament	9403	0.0831	0.0806	0.0000	Other
5	CCivilService	10072	0.0603	0.0580	0.0000	Other
5	CMajComp	9999	0.0510	0.0486	0.0000	Other
5	CEnvProt	9956	0.0418	0.0393	0.0000	Other
5	CCourts	9443	0.0639	0.0614	0.0000	Other
6	CArmedForces	12791	0.0596	0.0578	0.0000	Other
6	CPress	13258	0.0427	0.0408	0.0000	Other
6	CUnions	12632	0.0706	0.0688	0.0000	Other
6	CParliament	13143	0.0656	0.0639	0.0000	Other
6	CCivilService	13091	0.0409	0.0391	0.0000	Other
6	CMajComp	12977	0.0456	0.0437	0.0000	Other
6	CEnvProt	12865	0.0499	0.0480	0.0000	Other
6	CCourts	13198	0.0396	0.0378	0.0000	Other
7	CArmedForces	16102	0.0682	0.0667	0.0000	Other
7	CPress	16346	0.0446	0.0432	0.0000	Other
7	CUnions	15699	0.0596	0.0581	0.0000	Other
7	CParliament	16259	0.0852	0.0838	0.0000	Other
7	CCivilService	16274	0.0651	0.0637	0.0000	Other
7	CMajComp	16028	0.0479	0.0464	0.0000	Other
7	CEnvProt	15842	0.0558	0.0544	0.0000	Other
7	CCourts	16098	0.0541	0.0526	0.0000	Other

```
# strongest outcomes by average adjusted R2
top_outcomes <- model_perf_all %>%
  group_by(Outcome) %>%
  summarise(Mean_Adj_R2 = mean(Adj_R2, na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(Mean_Adj_R2)) %>%
  pull(Outcome)

top_outcomes
```

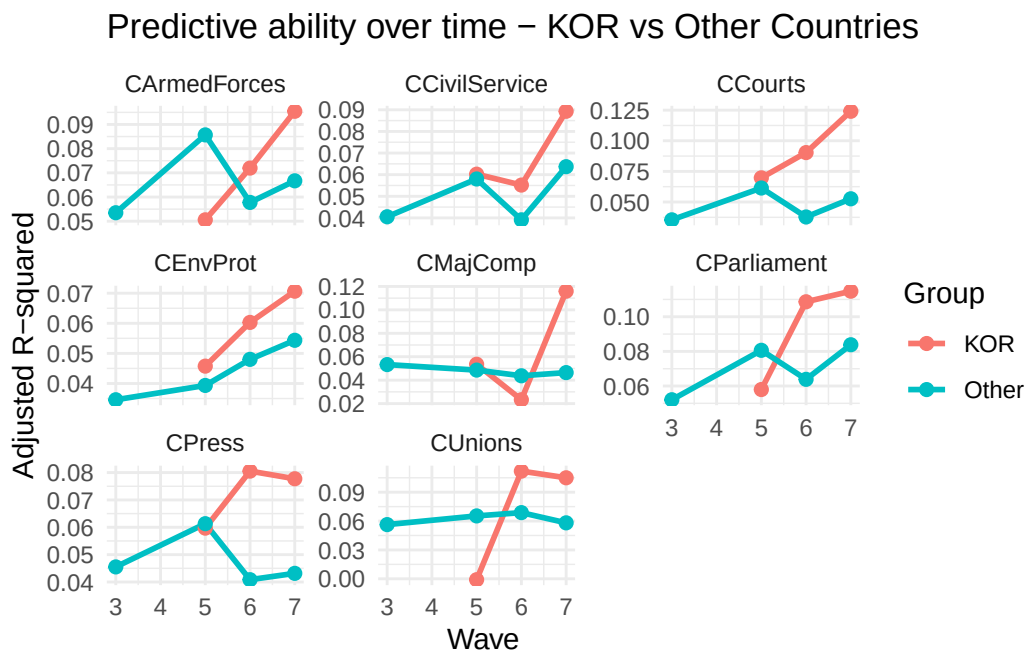
```
[1] "CParliament" "CArmedForces" "CCourts" "CUnions"
[5] "CPress" "CCivilService" "CMajComp" "CEnvProt"
```

```

p_perf <- model_perf_all %>%
  filter(Outcome %in% top_outcomes) %>%
  ggplot(aes(x = Wave, y = Adj_R2, color = Group, group = Group)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  facet_wrap(~ Outcome, scales = "free_y") +
  labs(
    title = paste("Predictive ability over time -", focus_country, "vs Other Countries"),
    x = "Wave",
    y = "Adjusted R-squared"
  ) +
  theme_minimal()

print(p_perf)

```



```

# Strongest predictor in each wave

coef_all <- bind_rows(coef_focus, coef_other)

best_predictor_by_wave <- coef_all %>%
  group_by(Group, Wave, Outcome) %>%
  slice_max(order_by = Abs_t, n = 1, with_ties = FALSE) %>%

```

```

ungroup() %>%
  dplyr::select(Group, Wave, Outcome, term, estimate, statistic, p.value, Abs_t)

kable(best_predictor_by_wave, digits = 4,
      caption = "Strongest predictor by wave and outcome")

```

Table 20: Strongest predictor by wave and outcome

Group	Wave	Outcome	term	estimate	statistic	p.value	Abs_t
KOR	5	CArmed- Forces	FutureRespect	0.1743	3.5763	0.0004	3.5763
KOR	5	CCivilService	ACTPolParty	0.6160	3.3586	0.0009	3.3586
KOR	5	CCourts	FutureRespect	0.1213	2.7937	0.0055	2.7937
KOR	5	CEnvProt	ACTPolParty	0.5290	2.9804	0.0031	2.9804
KOR	5	CMajComp	ACTSport	-0.1273	-2.6245	0.0091	2.6245
KOR	5	CParliament	ICQThrift	-0.2075	-2.3117	0.0214	2.3117
KOR	5	CPress	FutureRespect	0.1068	2.7348	0.0066	2.7348
KOR	5	CUnions	FutureRespect	0.0946	2.0593	0.0403	2.0593
KOR	6	CArmed- Forces	HardWork	0.0563	2.9230	0.0038	2.9230
KOR	6	CCivilService	HardWork	0.0670	3.8531	0.0001	3.8531
KOR	6	CCourts	FutureTech	0.2335	2.9550	0.0034	2.9550
KOR	6	CEnvProt	HardWork	0.0608	3.3381	0.0010	3.3381
KOR	6	CMajComp	HardWork	0.0542	3.1290	0.0020	3.1290
KOR	6	CParliament	HardWork	0.0430	2.5382	0.0118	2.5382
KOR	6	CPress	FutureTech	0.2754	3.2376	0.0014	3.2376
KOR	6	CUnions	PolDemons	0.3239	3.6443	0.0003	3.6443
KOR	7	CArmed- Forces	PolPetition	-0.1961	-2.6951	0.0074	2.6951
KOR	7	CCivilService	ILReligion	0.2121	4.4243	0.0000	4.4243
KOR	7	CCourts	ILReligion	0.1931	3.6547	0.0003	3.6547
KOR	7	CEnvProt	ICQThrift	-0.2386	-3.4971	0.0005	3.4971
KOR	7	CMajComp	ILReligion	0.1464	2.9430	0.0035	2.9430
KOR	7	CParliament	ILReligion	0.2168	3.7894	0.0002	3.7894
KOR	7	CPress	ICQThrift	-0.2814	-3.6472	0.0003	3.6472
KOR	7	CUnions	ILReligion	0.1370	2.8225	0.0051	2.8225
Other	3	CArmed- Forces	PolInterest	0.0975	9.6409	0.0000	9.6409
Other	3	CCivilService	PolInterest	0.0914	9.7388	0.0000	9.7388
Other	3	CCourts	HardWork	0.0269	8.9317	0.0000	8.9317
Other	3	CEnvProt	ILReligion	0.0856	8.8050	0.0000	8.8050

Group	Wave	Outcome	term	estimate	statistic	p.value	Abs_t
Other	3	CMajComp	FutureRespect	0.1037	8.1315	0.0000	8.1315
Other	3	CParliament	PolInterest	0.1281	13.2176	0.0000	13.2176
Other	3	CPress	ILReligion	0.0706	7.9258	0.0000	7.9258
Other	3	CUnions	ACTUnions	-0.2436	-15.8082	0.0000	15.8082
Other	5	CArmed- Forces	ILReligion	0.1229	12.5142	0.0000	12.5142
Other	5	CCivilService	PolInterest	0.1283	13.9123	0.0000	13.9123
Other	5	CCourts	PolInterest	0.1349	13.0773	0.0000	13.0773
Other	5	CEnvProt	LifeSatis	-0.0338	-8.6052	0.0000	8.6052
Other	5	CMajComp	PolInterest	0.0808	8.9478	0.0000	8.9478
Other	5	CParliament	PolInterest	0.1708	17.0739	0.0000	17.0739
Other	5	CPress	PolInterest	0.1128	12.3936	0.0000	12.3936
Other	5	CUnions	ACTUnions	-0.2736	-15.4786	0.0000	15.4786
Other	6	CArmed- Forces	ILReligion	0.1386	16.5882	0.0000	16.5882
Other	6	CCivilService	PolInterest	0.1066	12.9712	0.0000	12.9712
Other	6	CCourts	PolInterest	0.1126	12.8359	0.0000	12.8359
Other	6	CEnvProt	LifeSatis	-0.0437	-12.8489	0.0000	12.8489
Other	6	CMajComp	LifeSatis	-0.0367	-10.8361	0.0000	10.8361
Other	6	CParliament	PolInterest	0.1525	18.2651	0.0000	18.2651
Other	6	CPress	PolInterest	0.0967	12.0022	0.0000	12.0022
Other	6	CUnions	ACTUnions	-0.3157	-18.8885	0.0000	18.8885
Other	7	CArmed- Forces	ILReligion	0.0895	12.3786	0.0000	12.3786
Other	7	CCivilService	PolInterest	0.1554	20.9830	0.0000	20.9830
Other	7	CCourts	PolInterest	0.1266	15.8099	0.0000	15.8099
Other	7	CEnvProt	ACTEnvOrg	-0.2020	-12.8035	0.0000	12.8035
Other	7	CMajComp	LifeSatis	-0.0299	-10.0834	0.0000	10.0834
Other	7	CParliament	PolInterest	0.1836	23.4721	0.0000	23.4721
Other	7	CPress	PolInterest	0.1123	14.9922	0.0000	14.9922
Other	7	CUnions	ACTUnions	-0.2276	-16.1854	0.0000	16.1854

```
write.csv(best_predictor_by_wave, "q3b_best_predictor.csv", row.names = FALSE)

# choose strongest outcomes for predictor-comparison graph
predictor_plot_data <- best_predictor_by_wave %>%
  filter(Outcome %in% top_outcomes)

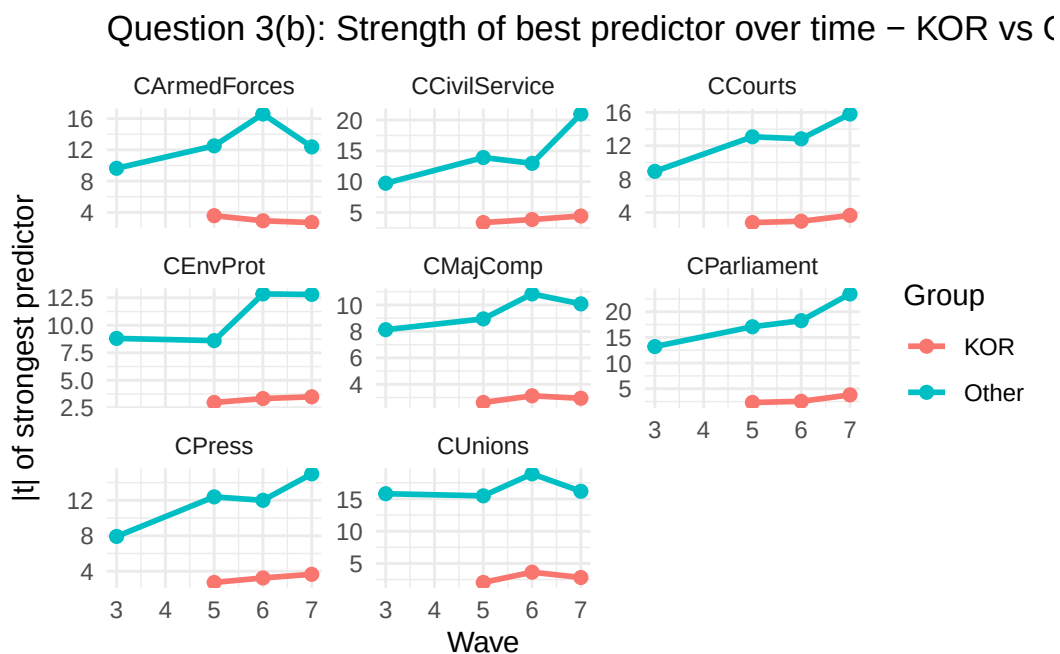
p_best_pred <- ggplot(predictor_plot_data,
```

```

aes(x = Wave, y = Abs_t, color = Group, group = Group)) +
geom_line(linewidth = 1) +
geom_point(size = 2) +
facet_wrap(~ Outcome, scales = "free_y") +
labs(
  title = paste("Question 3(b): Strength of best predictor over time -", focus_country, "v"),
  x = "Wave",
  y = "|t| of strongest predictor"
) +
theme_minimal()

print(p_best_pred)

```



```

# Which predictors appear most often as strongest?
predictor_frequency <- best_predictor_by_wave %>%
  count(Group, Outcome, term, sort = TRUE)

kable(predictor_frequency, digits = 4,
  caption = "Frequency of strongest predictors over time")

```

Table 21: Frequency of strongest predictors over time

Group	Outcome	term	n
Other	CCivilService	PolInterest	4
Other	CParliament	PolInterest	4
Other	CUnions	ACTUnions	4
Other	CArmedForces	ILReligion	3
Other	CCourts	PolInterest	3
Other	CPress	PolInterest	3
Other	CEnvProt	LifeSatis	2
Other	CMajComp	LifeSatis	2
KOR	CArmedForces	FutureRespect	1
KOR	CArmedForces	HardWork	1
KOR	CArmedForces	PolPetition	1
KOR	CCivilService	ACTPolParty	1
KOR	CCivilService	HardWork	1
KOR	CCivilService	ILReligion	1
KOR	CCourts	FutureRespect	1
KOR	CCourts	FutureTech	1
KOR	CCourts	ILReligion	1
KOR	CEnvProt	ACTPolParty	1
KOR	CEnvProt	HardWork	1
KOR	CEnvProt	ICQThrift	1
KOR	CMajComp	ACTSport	1
KOR	CMajComp	HardWork	1
KOR	CMajComp	ILReligion	1
KOR	CParliament	HardWork	1
KOR	CParliament	ICQThrift	1
KOR	CParliament	ILReligion	1
KOR	CPress	FutureRespect	1
KOR	CPress	FutureTech	1
KOR	CPress	ICQThrift	1
KOR	CUnions	FutureRespect	1
KOR	CUnions	ILReligion	1
KOR	CUnions	PolDemons	1
Other	CArmedForces	PolInterest	1
Other	CCourts	HardWork	1
Other	CEnvProt	ACTEnvOrg	1
Other	CEnvProt	ILReligion	1
Other	CMajComp	FutureRespect	1
Other	CMajComp	PolInterest	1
Other	CPress	ILReligion	1

```

write.csv(predictor_frequency, "q3b_frequency_strongest.csv", row.names = FALSE)

# Summary Tables
trend_test_results <- combined %>%
  dplyr::select(Group, Wave, all_of(selected_vars)) %>%
  tidyr::pivot_longer(
    cols = all_of(selected_vars),
    names_to = "Variable",
    values_to = "Value"
  ) %>%
  tidyr::drop_na() %>%
  group_by(Group, Variable) %>%
  group_modify(~{
    fit <- lm(Value ~ Wave, data = .x)
    s <- summary(fit)

    coef_tab <- as.data.frame(s$coefficients)
    coef_tab$term <- rownames(coef_tab)
    rownames(coef_tab) <- NULL

    names(coef_tab) <- c("estimate", "std.error", "statistic", "p.value", "term")

    coef_tab %>%
      dplyr::filter(term == "Wave") %>%
      dplyr::transmute(
        Slope = estimate,
        Std_Error = std.error,
        T_Value = statistic,
        P_Value = p.value,
        Significant = ifelse(p.value < 0.05, "Yes", "No")
      )
  }) %>%
  ungroup()

trend_summary_report <- trend_test_results %>%
  mutate(
    Direction = case_when(
      Slope > 0 & Significant == "Yes" ~ "Increasing",
      Slope < 0 & Significant == "Yes" ~ "Decreasing",
      TRUE ~ "Stable / weak trend"
    )
  )

```

```
)  
  
kable(trend_summary_report, digits = 4,  
      caption = "Report-ready summary of time trends")
```

Table 22: Report-ready summary of time trends

Group	Variable	Slope	Std_Er- ror	T_Value	P_Value	Signifi- cant	Direction
KOR	ACTHumanitarian	-0.0220	0.0056	-3.9013	0.0001	Yes	Decreasing
KOR	ACTSport	-0.0822	0.0125	-6.5805	0.0000	Yes	Decreasing
KOR	CArmed-Forces	0.1331	0.0081	16.3578	0.0000	Yes	Increasing
KOR	CCivilService	0.0991	0.0076	13.0082	0.0000	Yes	Increasing
KOR	CCourts	0.0589	0.0079	7.4938	0.0000	Yes	Increasing
KOR	CEnvProt	0.0804	0.0114	7.0245	0.0000	Yes	Increasing
KOR	CMajComp	-0.0254	0.0081	-3.1263	0.0018	Yes	Decreasing
KOR	CParliament	0.1018	0.0085	11.9275	0.0000	Yes	Increasing
KOR	CPress	0.0465	0.0076	6.0948	0.0000	Yes	Increasing
KOR	CUnions	0.0776	0.0076	10.1468	0.0000	Yes	Increasing
KOR	FutureRespect	-0.0667	0.0081	-8.2771	0.0000	Yes	Decreasing
KOR	FutureTech	0.0483	0.0062	7.7891	0.0000	Yes	Increasing
KOR	HardWork	0.3336	0.0316	10.5711	0.0000	Yes	Increasing
KOR	ICQDetermination	0.0431	0.0052	8.3588	0.0000	Yes	Increasing
KOR	ICQImagination	0.0683	0.0043	15.9531	0.0000	Yes	Increasing
KOR	ICQObedience	-0.0151	0.0033	-4.5413	0.0000	Yes	Decreasing
KOR	ICQResonsibility	0.0264	0.0037	7.0861	0.0000	Yes	Increasing
KOR	ICQThrift	0.0146	0.0052	2.8055	0.0051	Yes	Increasing
KOR	ILReligion	0.0755	0.0125	6.0419	0.0000	Yes	Increasing
KOR	LifeSatis	0.1311	0.0219	5.9823	0.0000	Yes	Increasing
KOR	PolBoycott	0.0294	0.0068	4.2937	0.0000	Yes	Increasing
KOR	PolDemons	-0.0186	0.0077	-2.4208	0.0156	Yes	Decreasing

Group	Variable	Slope	Std_Er- ror	T_Value	P_Value	Signifi- cant	Direction
KOR	PolExperts	0.0280	0.0131	2.1387	0.0326	Yes	Increasing
KOR	PolInterest	0.0718	0.0082	8.7794	0.0000	Yes	Increasing
KOR	PolPetition	0.0446	0.0081	5.5191	0.0000	Yes	Increasing
KOR	PrivateState	0.1812	0.0304	5.9700	0.0000	Yes	Increasing
Oth- ers	ACTHumanitarian	0.0112	0.0013	8.7178	0.0000	Yes	Increasing
Oth- ers	ACTSport	- 0.0111	0.0018	-6.2611	0.0000	Yes	Decreasing
Oth- ers	CArmed- Forces	- 0.0449	0.0018	- 24.8202	0.0000	Yes	Decreasing
Oth- ers	CCivilService	- 0.0065	0.0017	-3.8077	0.0001	Yes	Decreasing
Oth- ers	CCourts	- 0.0083	0.0018	-4.5654	0.0000	Yes	Decreasing
Oth- ers	CEnvProt	0.0179	0.0022	8.2424	0.0000	Yes	Increasing
Oth- ers	CMajComp	0.0134	0.0017	7.8015	0.0000	Yes	Increasing
Oth- ers	CParliament	0.0292	0.0018	16.1673	0.0000	Yes	Increasing
Oth- ers	CPress	0.0282	0.0017	16.9898	0.0000	Yes	Increasing
Oth- ers	CUnions	- 0.0089	0.0018	-4.9945	0.0000	Yes	Decreasing
Oth- ers	FutureRe- spect	0.0174	0.0014	12.6011	0.0000	Yes	Increasing
Oth- ers	FutureTech	0.0161	0.0012	13.3573	0.0000	Yes	Increasing
Oth- ers	HardWork	0.0101	0.0063	1.6092	0.1076	No	Stable / weak trend
Oth- ers	ICQDetermi- nation	0.0078	0.0009	8.4416	0.0000	Yes	Increasing
Oth- ers	ICQImagina- tion	0.0111	0.0008	14.1986	0.0000	Yes	Increasing
Oth- ers	ICQObedi- ence	- 0.0135	0.0009	- 14.6552	0.0000	Yes	Decreasing
Oth- ers	ICQResonsi- bility	0.0079	0.0009	9.0038	0.0000	Yes	Increasing
Oth- ers	ICQThrift	0.0012	0.0009	1.2685	0.2046	No	Stable / weak trend

Group	Variable	Slope	Std_Error	T_Value	P_Value	Significant	Direction
Others	ILReligion	0.0105	0.0022	4.7546	0.0000	Yes	Increasing
Others	LifeSatis	0.0950	0.0045	20.9675	0.0000	Yes	Increasing
Others	PolBoycott	0.0045	0.0013	3.4243	0.0006	Yes	Increasing
Others	PolDemons	0.0095	0.0014	6.5788	0.0000	Yes	Increasing
Others	PolExperts	-0.0102	0.0023	-4.4343	0.0000	Yes	Decreasing
Others	PolInterest	0.0009	0.0019	0.4612	0.6447	No	Stable / weak trend
Others	PolPetition	0.0332	0.0016	20.4514	0.0000	Yes	Increasing
Others	PrivateState	0.0857	0.0061	14.1299	0.0000	Yes	Increasing

```
write.csv(trend_summary_report, "q3b_summary_time_trend.csv", row.names = FALSE)

predictive_summary_report <- model_perf_all %>%
  group_by(Group, Outcome) %>%
  summarise(
    Mean_Adj_R2 = mean(Adj_R2, na.rm = TRUE),
    Max_Adj_R2 = max(Adj_R2, na.rm = TRUE),
    Min_Adj_R2 = min(Adj_R2, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(Group, desc(Mean_Adj_R2))

kable(predictive_summary_report, digits = 4,
       caption = "Report-ready summary of predictive ability over time")
```

Table 23: Report-ready summary of predictive ability over time

Group	Outcome	Mean_Adj_R2	Max_Adj_R2	Min_Adj_R2
KOR	CCourts	0.0947	0.1241	0.0698
KOR	CParliament	0.0938	0.1148	0.0580

Group	Outcome	Mean_Adj_R2	Max_Adj_R2	Min_Adj_R2
KOR	CArmedForces	0.0727	0.0955	0.0506
KOR	CPress	0.0727	0.0805	0.0597
KOR	CUnions	0.0720	0.1119	-0.0009
KOR	CCivilService	0.0682	0.0893	0.0552
KOR	CMajComp	0.0643	0.1159	0.0233
KOR	CEnvProt	0.0589	0.0707	0.0458
Other	CParliament	0.0701	0.0838	0.0521
Other	CArmedForces	0.0659	0.0857	0.0535
Other	CUnions	0.0621	0.0688	0.0563
Other	CCivilService	0.0503	0.0637	0.0391
Other	CMajComp	0.0480	0.0533	0.0437
Other	CPress	0.0477	0.0613	0.0408
Other	CCourts	0.0468	0.0614	0.0354
Other	CEnvProt	0.0441	0.0544	0.0346

```
write.csv(predictive_summary_report, "q3b_summary_predictive.csv", row.names = FALSE)
```

```
# model_perf_focus and model_perf_other should already exist
model_perf_all <- bind_rows(
  model_perf_focus %>% mutate(Group = focus_country),
  model_perf_other %>% mutate(Group = "Other")
)

# optional: keep only outcomes with enough data
plot_perf_data <- model_perf_all %>%
  filter(!is.na(Adj_R2))

ggplot(plot_perf_data,
  aes(x = Wave,
    y = Adj_R2,
    color = Outcome,
    group = Outcome)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  facet_wrap(~ Group) +
  labs(
    title = "Adjusted R2 Values for Each Confidence Variable over Wave",
    x = "Wave",
    y = expression(Adjusted~R2),
    color = "Confidence Variable"
```

```
) +  
theme_minimal()
```

